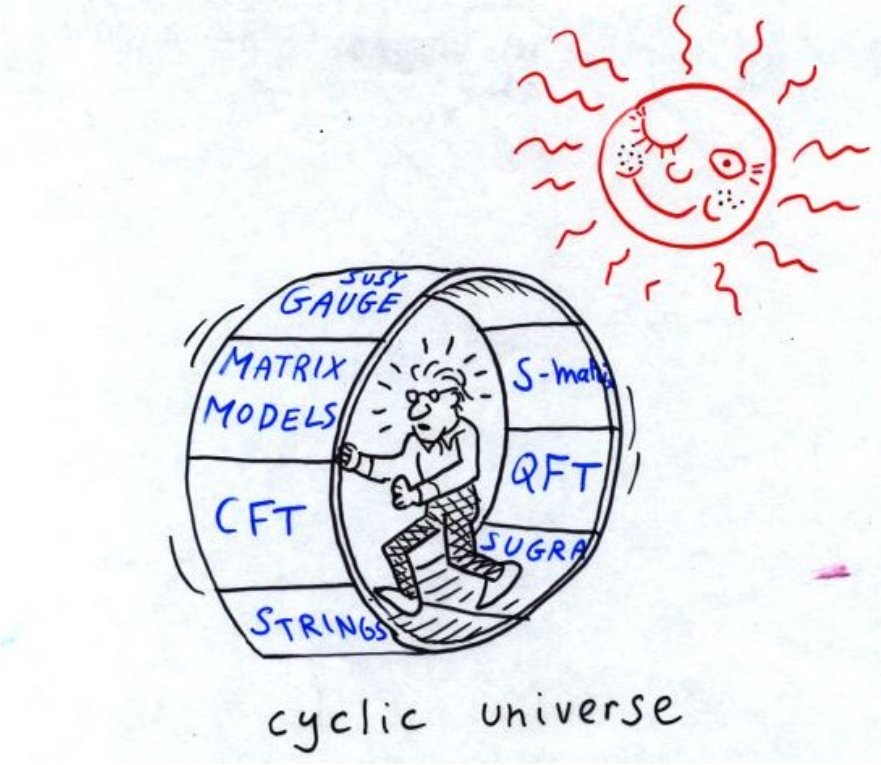
Aspects of Topological Quantum Fields, Strings & Dualities

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I dedicate this thesis to my parents who have endlessly supported me through this journey and have helped make my dream a reality. I am forever grateful.

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Introduction

Background

Topological Field Theories (TFT) are a class of field theories which have no local propagating degrees of freedom and are completely solvable. Because of this they are of great interest as the observables of a quantum theory which is deemed topological can be computed in the semi-classical limit. To better understand how to categorize the topological nature of a theory we must identify what makes a theory topological. A quantum field theory (QFT) is roughly specified by the following data

- A manifold M equipped with a metric h, where M may or may not have boundaries.
- Fields over M: ϕ,ψ, A_μ etc.
- Operators of fields and their associated Hilbert space $\mathcal{H}$.

Generally one starts with a classical description of the Lagrangian of the theory in which the fields are described by sections of bundles over M in some appropriate representation. For a principle bundle we have sections corresponding to gauge fields A while matter fields such as the scalar bosonic field ϕ are sections of vector bundles. Of particular interest are fields $\phi : M \to N$ which are maps between manifolds that act as the embedding maps. These types of theories are called sigma models and are also used to describe the embedding of the classical bosonic string and its worldsheet into a higher-dimensional space-time. After this description is made one can quantize the theory by either canonical quantization or by the path integral formalism in promoting the fields to operators which act on a Hilbert space of states under representations of the Poincare group. To obtain physical information from these theories we use the path integral formalism to integrate over all possible field configurations. Physical states are defined by subsets of the Hilbert space which satisfy a certain set of properties and often depend on the theory of interest. In most cases these theories are described by *local* data of the manifold where the global nature is lost in quantization. This is quite apparent from considering the nature in which we build *local* charts on the manifold. Therefore it is natural to wonder if it is possible to describe the global properties of a QFT that describe the topological aspects of the underlying manifold.

Given this, a QFT can be thought of as topological if we can compute topological invariants of the underlying manifold M from it. Topological invariants describe the global nature of the topological space as they are a class of properties which are invariant, at the weakest case, under

homeomorphisms. By its very definition this would imply that the invariants should not depend on the higher structure which we build on this topology, including the manifold structure and the metric we put on it. Therefore it is common for one to define a topological quantum field theory as a theory which does not depend on the metric and indeed this is one of the most common definitions found in the literature. It is important to stress that inherently topological QFT's contain a metric but as we shall see the physical observables of the theory, the correlation functions of physical operators, do not depend on the choice of metric. From Einstein's theory of general relativity it is almost ingrained in us that the metric is naturally part of the manifold we consider our QFT on but this is in fact a choice we have made. String theory, among other things, resolves this issue by integrating over all possible metrics a manifold can admit in the path integral prescription.

In [29] Witten showed that the three dimensional Chern-Simons $U(N)$ gauge theory could be used to calculate topological invariants of links and knots embedded into the base three-manifold. These were found to be related to the Wilson loop observables of the theory. Furthermore he found that the partition function of this theory was connected to a well known Conformal Field Theory (CFT), the WZW model, which he used to calculate a new topological invariant of three manifolds. Because of the intimate relationship between string theory and conformal field theory it was natural to expect a branch of string theory could be used to obtain a topological theory.

When the first five superstrings were discovered their consistent physical descriptions were found to be anomalie free in ten-dimensions when considering low energies. In an effort to make contact with four dimensional theories the idea of compactification was introduced. The ansatz for this is given by $M^{10} = \mathbb{M}^4 \times M^6$ where the manifold M^6 satisfies certain physical requirements and $\mathbb{M}^4$ is a four-dimensonal space generally taken to me a Minkowski space-time. Of the many requirements found, many were found to lead to M^6 being a Calabi-Yau manifold. Inspired by these ideas Witten showed that there is infact a topological sector of string theories which can be used to compute topological information of certain target space manifolds which were compactified. This was done by considering Kahler supersymmetric sigma models whose Lorentz group is altered by one of the internal fermionic R-symmetrys the theory possesses. The two topological models which were obtained by Witten were later found to be related by a duality in string theory known as mirror symmetry. The mathematical implications of this discovery were surprising and led to new and profound relations in enumerative geometry.

Along with mirror symmetry many other string dualities were discovered during the mid to late 90's. These were known as open-closed string dulaities and gauge-gravity dualities. They, like mirror symmetry did for mathematics, gave new connections between physical theories which were thought to once be unrelated. The discovery of the AdS/CFT correspondence between gauge theory and gravity was one of the motivating reasons to study open-closed string dualities. It was a step closer to finding an intamate relationship between QFT and gravity. Indeed in [11] Gopakumar and Vafa showed that Chern-Simons gauge theory is dual to the topological string on a Calabi-

Yau manifold. Because of what was known about Chern-Simons and knots invariants it seems the natural next step was to consider how to calculate knot invariants in this duality. This is precisely what Ooguri and Vafa did in [21]. This will be the main result which we obtain and work towards in this thesis.

In the first part of this thesis we consider the mathematical foundations needed to describe topological string theories. We begin by looking at bundles and connections on them. We then describe the basics of complex geometry which will underpin much of the manifolds considered. Calabi-Yau manifolds are introduced as Kahler manifolds which $SU(N)$ holonomy. Their modlui are found to be deeply related to the field content of compactified string theories. In the second part we present the basics of string compactification on Calabi-Yau threefolds and conformal field theory needed to understand the topological sigma models and string theories. In the third section we present the formal definition of a TQFT and use it to study knot invariants obtained from Chern-Simons theory. We then present the $N = (2, 2)$ supersymmetric sigma model and its topological twists. Following this is the section of topological string theory where we couple the theories to gravity. In the final section we discuss the open-closed string duality between the Chern-Simons open string field theory and the topological A-model string theory.

Part I

Fundamentals I: Mathematics

1 Complex Geometry & Bundles

Fundamental to the understanding of the mathematical structures used in topological string theory are complex manifolds and bundles defined over them. In particular we are interested in Calabi-Yau manifolds of complex dimension 3 as the target space to our sigma model. The reason for this is that such spaces admit vanishing first Chern-class and allow us to resolve anomalies which arise from the quantization of the axial symmetry of the theory. To get to this point requires a great amount of mathematical machinery which we will state without proof. To keep the background from getting too extensive we assume familiarity with the analysis of real manifolds, Riemannian geometry, differential forms and representation theory. The following is based off of [19], [5], [16], [15] and [3].

1.1 Bundles:

Every aspects of physics, from classical to quantum mechanics, can be formulated in the language of bundles. Therefore it should not be surprising that the same is true for string theory. Aside from allowing us to understand how to define objects such as vector fields on a manifold we can use bundles to understand topological properties of the underlying manifold. Let us now define the basics of bundle theory.

Definition 1. *A bundle (of topological manifolds) is a triple* (E, π, M) *where* E *is called the total space and* M *the base space with both* E *and* M *manifolds.* $\pi : E \to M$ *is called the (bundle) projective map which is surjective. This information is often represented by* $E \xrightarrow{\pi} M$ *or simply* $E \to M$.

A map $s : M \to E$ is called a section of the bundle if $\pi \circ s = id_M$. The space of sections of E over M is denoted $\Gamma(E)$ or $\Gamma(M, E)$ if we would like to make the base manifold apparent. The pre-image of the projection map π over a point is called a fiber of M over p denoted $F_p := \{\pi^{-1}(p) | p \in M\}$. Given two bundles $E_1 \xrightarrow{\pi_1} M_1$ and $E_2 \xrightarrow{\pi_2} M_2$ with maps $f : M_1 \to M_2$ and $g : E_1 \to E_2$ then the pair (f, g) is said to be a bundle morphism if the following diagram commutes

$$\begin{array}{ccc} E_1 & \xrightarrow{g} & E_2 \\ \downarrow{\scriptstyle \pi_1} & & \downarrow{\scriptstyle \pi_2} \\ M_1 & \xrightarrow{f} & M_2 \end{array}$$

i.e. $\pi_2 \circ g = f \circ \pi_1$. Two bundles are isomorphic (at the level of bundles) if both (f, g) and (f^{-1}, g^{-1}) are both bundle morphisms. This is of course only defined if the inverses of (f, g) exists. In the case that the fibers at all points of M are equivalent $F_p = F$ the bundle is given by $E = M \times F$ with $\pi : (m, f) \mapsto m$ which admits the usual product topology. Any bundle that of this form or isomorphic to one is said to be trivial. A bundle which is not trivial is called twisted.

Let G be a lie group and M a smooth manifold. Then a left-action of G on M is a (smooth) map $\sigma : G \times M \to M$ satisfying $\sigma(e, p) = p$ for the identity element $e \in G$ and $\sigma(g_1, \sigma(g_2, p)) = \sigma(g_1 \cdot_G g_2, p)$. A representation $\rho : G \to GL(V)$ induces a left action on V. A right action is defined similarly. We can construct the quotient space $M/\sim := M/G$ defined by the equivalence relation $p \sim q :\Leftrightarrow \exists g \in G$ such that $q = \sigma(g, p)$. The little group is defined by $H_p := \{p \in G | \sigma(g, p) = p\}$ and is said to be free if $H_p = \{e\}$.

Definition 2. *A fiber bundle*[1] *consists of* (E, π, M, F, G) *where* F *is a fiber of* M *and* G *is the structure group which acts from the left on* F*. If* F *is a vector space then the fiber bundle is a vector bundle. For a local chart* $U_i \subset M$ *we define local trivializations* $\varphi : U_i \times F \to \pi^{-1}(U_i)$ *such that* $(\pi \circ \varphi)(p, f) = p$ *which make a fiber bundle locally trivial.*

The tangent bundle TM is simply the disjoint union of all tangent spaces of M, $TM := \sqcup_{p \in M} T_p M$. Locally this has coordinates $(x^1, \cdots, x^m, p^1, \cdots, p^m)$ where $(x^1, \cdots, x^m)$ are coordinates on M and $(p^1, \cdots, p^m)$ coordinates on $T_p M$, the tangent space of M at a point $p \in M$. The cotangent bundle T^*M dual to the tangent bundle is defined similarly and we will say more on it shortly.

Definition 3. *A bundle* (P, π, M) *is a principal (fiber) G-bundle if*

1. *P is equipped with a right G-action which is free*
2. $P \xrightarrow{\pi} M \cong P \xrightarrow{\Pi} P/G$ *where* $\Pi : \epsilon \mapsto [\epsilon]$

these conditions tell us that the fibers of a principle bundle are isomorphic to G, $\Pi^{-1}([\epsilon]) \cong G$

A principal bundle is trivial if and only if there exists a smooth global section $s : M \to P$. There is also the notion of an associated bundle to a principal fiber bundle and although we will not rigorously define them here a word should be said about them. The associated bundle contains a different class of objects which a bundle can admit which transform in an appropriate way to those in

[1] There are some more subtlies to the definition. See chapter 9. of [**?**]

the principal bundle. For instance, if a manifold admits spinors then under a Lorentz transformation these spinors will transform according to their spin representation in the associated bundle while vectors (frames) will transform accordingly in the principal (frame) bundle. This basic concept is enough to define and understand spin bundles and the details of these claims can be found in the classic text by Lawson [17].

A spin bundle $S^{\pm}(M)$ is an associated bundle with structure group $G = Spin(n)$. They are the bundles associated to the (principal) frame bundle LM with structure group $SO(n)$, where $Spin(n)$ is the universal covering group of $SO(n)$. Sections of $K^{1/2}(M)$ are spinors which are in some spin representations of $Spin(n)$. The $\pm$ are used to denote spinors which decompose into irreducible representations defined by their chirality. When we consider two-dimensional sigma models we will denote these bundles by $S^{+} = K^{1/2}$ and $S^{-} = \overline{K}^{1/2}$ since spinors in the **2** of $SL(2,\mathbb{C})$ square to vector like quantity. We note that not all manifolds admit spin structures and therefore spinors give us topological information about the spacetime they live on. If the second Stiefel-Whitney class, which is a topological invariant, vanishes then the underlying (Riemannian) manfiold admits spinors [17].

The pull-back bundle, which we will use frequently, is defined as follows. Let $E \xrightarrow{\pi} M$ be a fiber bundle with fiber $F \cong \pi^{-1}(p)$ for all $p \in M$. For some map $\phi : \Sigma \to M$ between manifolds Σ and M we can construct the following total space

$$\phi^{*}E := \{(p, e) \in \Sigma \times E \mid \phi(p) = \pi(e)\} \tag{1}$$

over Σ. The pull-back bundle is therefore $\phi^{*}E \to \Sigma$ with the projection map $\pi^{*} : (p, e) \mapsto p$. We can similarly define a second projection map $\pi_2 : (p, e) \mapsto e$. This is summarized by the commutative diagram

$$\begin{array}{ccc} \phi^{*}E & \xrightarrow{\pi_2} & E \\ {\scriptstyle \pi^{*}}\downarrow & & \downarrow{\scriptstyle \pi} \\ \Sigma & \xrightarrow{\phi} & M \end{array}$$

Lastly we define the endomorphism bundle $End(E)$. Let $E \to M$ be a vector bundle, then the bundle $End(E) \to M$ (or $End(M)$) is the bundle whose sections are endomorphisms which act on sections in E. That is, if $T \in \Gamma(End(E))$ then $T : \Gamma(E) \to \Gamma(E)$. We can obtain local expressions through the isomorphism $End(E) \cong E \otimes E^{*}$ where E^{*} is the dual to E.

1.1.1 Connections on bundles & Holonomy

Over a vector bundle $E \to M$ a connection[2] ∇ on E is a linear map $\nabla : \Gamma(E) \to \Gamma(E \otimes T^*M)$ satisfying

$$\nabla(fe) = f\nabla s + s \otimes df \tag{2}$$

for $s \in \Gamma(E)$ and $f \in C^\infty(M)$. Connections allow us to determine how sections of the bundle vary as we go from fiber to fiber. This is known as parallel transport and we must first determine a local expression for the connection. Using the above information we may equivalently define a map $\nabla_X : \Gamma(E) \to \Gamma(E)$. Let U be a subset of the base manifold with $p \in U$. Then for $X = X^\mu \partial_\mu \in T_pU$, and e_j a local basis of E: $span\{e^i\} = F_p$ the connection takes the local expression

$$\begin{aligned}
\nabla(s)(X) &:= \nabla_X(s) \\
&= \nabla_{X^\mu \partial_\mu}(s) \\
&= X^\mu \nabla_\mu (s^i e_i), && \nabla_\mu := \nabla_{\partial_\mu} \\
&= X^\mu \left((\partial_\mu s^i) e_i + \Gamma^j_{\mu j} s^i e_j \right), && \nabla_\mu e_i := \Gamma^j_{\mu i} e_j \\
&= X^\mu (\partial_\mu s^i + \Gamma^i_{\mu j} s^j) e_j
\end{aligned}$$

where on the right side we have defined the expressions used on the left. The new term introducted Γ is a section the bundle $End(E|_U) \otimes T^*U$. There are remarks which need to be made about this. In gravitational theories the covariant derivative is denoted by ∇_μ while in gauge theories we instead use D_μ. The reason for this is each of these theories are defined on different types of bundles. In relativity Γ is simply the Christoffel symbol. This definition explicitly shows why they don't transform as a tensor on M: the first index is an index on M while the others come from the vector space fibers over M. Gauge theories are defined over principal fiber bundles which we found had fibers equal to the gauge group G. Therefore the endomorphism bundle over this bundle is valued in the Lie algebra $\mathfrak{g}$ and so $\Gamma = A$ where A is the familiar connection one-form. We summarize this information in the table below.

Net us now focus on parallel transport. Let $\gamma : [0,1] \to M$ be a smooth curve in M. By pulling back the bundle E with γ we construct the bundle $\gamma^*(E)$ over $[0,1]$ with the pullback of D a connection on $\gamma^*(E)$ which we refer to as the covariant derivative. We define the horizontal lift $\tilde{\gamma}$ of γ as the curve in the bundle satisfying $(\pi \circ \tilde{\gamma})(t) = \gamma(t)$ for all t. With this the covariant derivative is

$$D_{\tilde{\gamma}'(t)} v(t) = \frac{dv}{dt}(t) + A(\tilde{\gamma}'(t)) v(t). \tag{3}$$

[2]Strictly speaking a connection is a decomposition of the bundle into horizontal and vertical subspaces however these formalities will not be of use to us here. See [19] and [16].

	Gravity	Gauge-theory
Bundle	Vector bundle	Principal G-bundle
Fiber	V	G
Connection	∇_μ	D_μ
Potential	$\Gamma^i_{\mu j} \in End(E) \otimes \Gamma(T^*M)$	$(A_\mu)^i_j \in \mathfrak{g} \otimes \Gamma(T^*M)$

Table 1. Bundle structure of physical theories.

A vector $v(t)$ which satisfies $D_{\gamma'(t)}v(t) = \frac{dv}{dt}(t) + A(\tilde{\gamma}'(t))v(t) = 0$ for all t is said to be parallel transported along γ. A solution to the parallel transport equation is easily obtained by recursively replacing the solution $v(t) = u_0 - \int_0^t A(\tilde{\gamma}'(t))v(t_1)dt_1$ into itself. From this we obtain an explicit expression for how the vector $u(t)$ acts under parallel transport

$$v(t) = \mathcal{P}exp\left(-\int_0^t A(\gamma'(s))ds\right)v(0) \tag{4}$$

here $\mathcal{P}$ is the path-ordering operator which is exactly defined as the time-ordered product in quantum field theory. This motivates us to define the following

Definition 4. *For $\gamma : p \mapsto q$ we define the holonomy along γ on the bundle $E \to M$ as the map*

$$H(\gamma, D) : \Gamma(F_p) \to \Gamma(F_q) \tag{5}$$

where F_p and F_q are the fibers over p and q respectively. If γ is piecewise smooth defined by $\gamma = \gamma_n \circ \gamma_{n-1} \circ \cdots \gamma_i \circ \cdots \gamma_1$ and $\gamma_i : [t_i, t_{i+1}] \to M$ then

$$H(\gamma, A) = H(\gamma_n, A) \cdots H(\gamma_1, A) \tag{6}$$

Note that $H(\gamma, A) \in End(F)$ and if A is a Lie algebra valued one form

$$H(\gamma, A) = \mathcal{P}exp\left(-\int_0^t A(\gamma'(s))ds\right) \tag{7}$$

and so $H(\gamma, A) \in G$. In general the holonomy is an element of $GL(m, \mathbb{R})$ which acts on the vector u in the fiber. For a Riemannian manifold, if $g(H(\gamma, A)X, H(\gamma, A)Y) = g(X, Y)$ then the holonomy is a subgroup of $SO(1, m-1)$. Let γ and its lift be a loop. That is, $\gamma(0) = \gamma(1)$. Then under a gauge transformation at a point $p \in M$ the holonomy will transform in the adjoint $H \to g(p)H(\gamma, A)g(p)^{-1}$ where $g \in G$. In gauge theory we often characterize physical quantities by their invariance under gauge transformations since such a transformation moves us along the fiber of the principal bundle.

We define the *Wilson loop* (also called Wilson lines) $W(\gamma, A)$ as the trace of the holonomy

$$W(\gamma, A) := Tr\, \mathcal{P}exp\left(\oint_\gamma A\right). \tag{8}$$

which, by the cyclicity of the trace, is gauge invariant. These loops characterize a charged particle which moves around a closed loop and play an important role in understanding the topology of the manifold. In Chern-Simons theory these are precisely the physical observables which characterize knot invariants.

We can also consider parallel transport of spinors over spin bundles. The covariant derivatives on these bundles has a potential ω which is called the spin connection. We define non-coordinate basis of the tangent and co-tangent bundles by defining the vielbeins $e^\mu_\alpha \in GL(m, \mathbb{R})$ through $\hat{e}_\alpha := e^\mu_\alpha \partial_\mu$. If a bundle admits a twisted spin complex, i.e. if the total space is given by $S^\pm \otimes E$ where E is a vector bundle, then the covariant derivative on this bundle is of the form

$$D_\alpha \psi = e^\mu_\alpha (\partial_\mu + \omega_\mu + A_\mu)\psi. \tag{9}$$

The holonomy over a spin bundle $H_s(\gamma, D)$ will be an element of $Spin(m)$. The subgroup that the holonomy of such a manifold admits will play an important role in defining Calabi-Yau manifolds.

The cotangent bundle as a symplectic manifold

The target space of our topological string theories will turn out be endowed with a symplectic structure. For this reason we quickly recall some basic facts about symplectic manifolds.

Definition 5. *A 2-form ω on an even dimensional manifold M is called a symplectic form provided*

1. $d\omega = 0$
2. *ω is non-degenerate. That is, for $X \in T_pM$, $(i_X\omega)_j = X^i\omega_{ij} \neq 0$ meaning $det(\omega_{ij}) \neq 0$*

where i is simply the interior product. A manifold endowed with such a form is called a symplectic manifold and the pair (M, ω) a symplectic structure.

A Lagrangian submanifold $X \subset M$ is a submanifold of a symplectic manifold (M, ω) of dimension $dim(X) = 1/2 dim(M)$ such that the symplectic form on M vanishes on X, $\omega|_X = 0$.

The cotangent space T^*M (called the phase space in classical mechanics) is a symplectic manifold. Locally it is spanned by the coordinates (q^μ, p_μ) where q^μ are coordinates on M and p_μ are momentum coordinates on the fibers. It admits the following symplectic structure

$$\omega = dq^\mu \wedge dp_\mu \tag{10}$$

which can we written as an exact form $\omega = d\theta$ for $\theta = q^\mu dp_\mu$. Now since $p_\mu = 0$ on M then we find

$$\omega|_M = 0 \tag{11}$$

and since q^μ span exactly half of the coordinates we see that the base manifold M is Lagrangian.

1.2 Complex Manifolds:

We now switch our focus to the study of complex manifolds which are of paramount importance in our analysis of topological strings. A complex function $g : \mathbb{C} \to \mathbb{C}$ is said to be holomorphic if $\partial g/\partial \bar{z} = 0$. The Cauchy Riemann relation states that this is equivalent to the following

$$\frac{\partial g_1}{\partial x^\mu} = \frac{\partial g_2}{\partial y^\mu} \quad \frac{\partial g_2}{\partial x^\mu} = -\frac{\partial g_1}{\partial y^\mu}$$

where $g = g_1 + ig_2$ and $z^\mu = x^\mu + iy^\mu$. A function $f : \mathbb{C}^n \to \mathbb{C}^n$ is said to be holomorphic if it is holomorphic in each of its components. We define $\mathbb{C}^* := \mathbb{C}\setminus\{0\}$ for later use.

Definition 6. *Let M be a real 2m-dimensional manifold which admits a holomorphic atlas $\mathcal{A} = \{(\mathcal{U}_i, \phi_i)\}$ where the pair $(\mathcal{U}_i, \phi_i)$ are coordinate charts that cover M and $\phi_i(U_i) \to \mathbb{C}^n$. If the transition functions $\psi_{ij} = \phi_i \circ \phi_j^{-1} : \mathbb{C}^n \to \mathbb{C}^n$ are holomorphic then M is said to be a complex manifold.*

Complex Projective spaces

One of the main examples of such manifolds are complex projective spaces $\mathbb{CP}^m$. These spaces are defined as

$$\mathbb{CP}^m := (\mathbb{C}^{m+1}\setminus\{0\})/\sim \tag{12}$$

where $z \sim w$ if $z = \lambda w$ for some nonzero $\lambda \in \mathbb{C}$ and $z, w \in \mathbb{C}^{m+1}$. Intuitively speaking this is the space of complex lines through the origin of $\mathbb{C}^{m+1}$. We call the coordinates $[(z^0, \cdots, z^m)] := [z^0, \cdots, z^m]$ homogeneous coordinates of $\mathbb{CP}^m$. Choosing an open covering of these spaces by defining the open subsets $U_\alpha := \{(z^1, \cdot, z^\alpha, \cdots, z^m) | \, z^\alpha \neq 0\}$. Local coordinates on these charts are defined as $\xi^\mu_{(\alpha)} := z^\mu/z^\alpha$ which are referred to as inhomogeneous coordinates. On the intersection $U_\alpha \cap U_\beta$ the chart transition maps $\psi^\mu_{\alpha\beta} : \xi^\mu_{(\alpha)} \mapsto \xi^\mu_{(\beta)}/\xi^\alpha_{(\beta)}$ and is simply multiplication by $(\xi^\alpha_{(\beta)})^{-1}$ which is of course holomorphic.

Weighted Projective spaces

Also important is the weighted projective space $W\mathbb{P}^m$ which is similarly defined to $\mathbb{CP}^m$ except that we choose different weights in $\mathbb{C}^*$ to quotient each coordinate by. Let $q_i \in \mathbb{R}$ for $i = 0, \cdots, n$

and $gcd(q_0, \cdots, q_n) = 1$. Then the weighted projective space is

$$W\mathbb{P}^{(q_0,\cdots,q_n)} := (\mathbb{C}^{c+1}\{0\})/\sim \tag{13}$$

where $(z_0, \cdots, z_n) \sim (w_0, \cdots, w_n) \Leftrightarrow$ there exists $\lambda \in \mathbb{C}^*$ such that

$$(z_0, \cdots, z_n) = (\lambda^{q_0} w_0, \cdots, \lambda^{q_n} w_n) \tag{14}$$

which can also be shown to be a compact complex manifold.

For complex manifolds usual notion of a smooth structure such as $C^\infty(M)$ are replaced by holomorphic maps. With this in mind we can take many of the results from real analysis of manifolds to understand morphisms between complex manifolds as well as bundles over them. The simplest way to obtain these is to take a 2n real dimensional manfold and complexify it. Complexification amounts to making all objects defined in the real case to be complex valued. For example we take the tangent space T_pM of M at a point p and complexify it as

$$T_pM^{\mathbb{C}} := T_pM \otimes \mathbb{C} = \{V = X + iY | X, Y \in T_pM\}. \tag{15}$$

There are many possible choices of atlases on M and any two holomorphic atlases $\mathcal{A}$ and $\overline{\mathcal{A}}$ are equivalent if the composition $\phi_i \circ \overline{\phi}_j^{-1}$ is holomorphic for all i, j. Complex manifolds are thus characterized by equivalence classes of holomorphic atlases, each of which define a different complex structure. We can make the complex structure apparent by defining a globally defined section $J \in \Gamma(TM \otimes T^*M)$ over M [15]. Let V be a real vector space. Then a linear map $J : V \to V$ satisfying $J^2 = -1$ is called a complex structure on V. If we take $\mathbb{R}^2$ with standard Euclidean coordinates $(x_1, y_1, \cdots, x_n, y_n)$ then the map $J : (x_1, y_1, \cdots, x_n, y_n) \to (-y_1, x_1, \cdots, -y_n, x_n)$ is a complex structure and is simply multiplication by i. This means that J has eigenvalues $\pm i$ but since V is real the eigenspace of J is not spanned by elements in V. To obtain these eigenspaces we complexify V and extend our maps to $V^{\mathbb{C}}$. We then find that $V^{\mathbb{C}}$ has the following decomposition

$$V^{\mathbb{C}} = V^{(1,0)} \oplus V^{(0,1)}$$

$$J(v) = iv \implies v \in V^{(1,0)}, \quad J(w) = -iw \implies w \in V^{(0,1)}.$$

A more accurate notation would be to denote the composition by V^+ instead of $V^{1,0}$ however this would be notational overkill when we consider the field content of sigma models.

Definition 7. *An almost complex structure J on M is a smooth section $J \in \Gamma(End(TM))$ satisfying $J^j_i J^k_j = -\delta^k_i$.*

Upon complexification an almost complex structure will induce the decomposition $TM^{\mathbb{C}} = T^{(1,0)}M \oplus T^{(0,1)}M$ where $T^{(1,0)}M$ is the holomorphic tangent bundle over M and $T^{(0,1)}M$ the anti-

holomorphic tangent bundle. A similar decomposition is done for the cotangent bundle $T^*M^{\mathbb{C}} = T^*M^{(1,0)} \oplus T^*M^{(0,1)}$. Local frames of these bundles are fiberwise spanned by

$$\frac{\partial}{\partial z^\mu} := \frac{1}{2}\left(\frac{\partial}{\partial x^\mu} - i\frac{\partial}{\partial y^\mu}\right) \qquad \frac{\partial}{\partial \bar{z}^\mu} := \frac{1}{2}\left(\frac{\partial}{\partial x^\mu} + i\frac{\partial}{\partial y^\mu}\right) \tag{16}$$

$$dz^\mu := dx^\mu + idy^\mu \qquad d\bar{z}^\mu := dx^\mu - idy^\mu \tag{17}$$

Necessary & sufficient conditions

In order to determine whether a manifold can be considered complex we would like to have necessary and sufficient conditions. The Newlander Nirenberg theorem gives us exactly that [19]. Given vector fields $X, Y \in T^{1,0}M$ which satisfy $[X, Y] \in T^{1,0}M$ we say that the almost complex structure J is integrable.

Theorem 8. *For an almost complex manifold (M, J), if J is integrable then M is complex.*

furthermore we have the following theorem

Theorem 9. *An almost complex structure J is integrable if and only if the Nijenhuis tensor field $N : \Gamma(TM) \times \Gamma(TM) \to \Gamma(TM)$, which is defined as*

$$N(X, Y) := [X, Y] + J[JX, Y] + J[X, JY] - [JX, JY] \tag{18}$$

vanishes.

These two theorems allow us to easily determine when a $2m$ dimensional real manifold is complex.

1.3 Dolbeault Cohomology

Let M be a real n-dimensional manifold and T^*M it's cotangent bundle. The k-th exterior power of T^*M denoted $\Lambda^k T^*M$ is defined by the exterior product of sections $\Gamma(\Lambda^k T^*M)$ of T^*M. The exterior product

$$\wedge : \Gamma(\Lambda^k(T^*M)) \times \Gamma(\Lambda^l(T^*M)) \to \Gamma(\Lambda^{(k+l)}(T^*M)) \tag{19}$$

and exterior derivative $d_k : \Gamma(\Lambda^k(T^*M)) \to \Gamma(\Lambda^{(k+1)}(T^*M))$ on this space are defined as usual. Let $\alpha \in \Gamma(\Lambda^k(T^*M))$ and $\beta \in \Gamma(\Lambda^l(T^*M))$ then locally we have

$$(\alpha \wedge \beta)_{\mu_1 \cdots \mu_{k+l}} = \frac{(p+q)!}{p!q!}\alpha_{[\mu_1 \cdots \mu_k}\beta_{\mu_{k+1} \cdots \mu_{k+l}]} \tag{20}$$

$$(d_k\alpha)_{\mu_1 \cdots \mu_k} = (k+1)\partial_{[\mu_1}\alpha_{\mu_2 \cdots \mu_{k+1}]} \tag{21}$$

Define $\Lambda^{p,q}(T^*M) := \Lambda^p(T^*M^{(1,0)}) \otimes \Lambda^q(T^*M^{(0,1)})$, then a (smooth) section of this bundle is called a (p,q)-form which has p-holomorphic components and q antiholomorphic components. Denote $\Omega^{p,q}(M) := \Gamma(\Lambda^{p,q}M)$ and $\Omega^k_{\mathbb{C}}(M) := \Gamma(\Lambda^k((T^*M)^{\mathbb{C}})$ with[3] $\Omega^0_{\mathbb{C}}(M)$ the set of smooth complex functions on M. Then following decomposition is then obvious

$$\Lambda^k((T^*M)^{\mathbb{C}}) = \bigoplus_{p+q=k} \Lambda^{p,q}(T^*M) \qquad \Omega^k_{\mathbb{C}}(M) = \bigoplus_{p+q=k} \Omega^{p,q}(M) \tag{22}$$

The complexified cotangent bundle admits a $\mathbb{C}$-linear extenion of the exterior derivative $d : \Omega^k(M) \to \Omega^{k+1}(M)$ which decomposes into Dolbeault operators ∂ and $\overline{\partial}$, $d = \partial + \overline{\partial}$ which are defined as $\partial : \Omega^{p,q}(M) \to \Omega^{p+1,q}(M)$ and $\overline{\partial} : \Omega^{p,q}(M) \to \Omega^{p,q+1}(M)$. We write these forms as $\partial = dz^\mu \wedge \partial/\partial z^\mu$ and $\overline{\partial} = d\overline{z}^\mu \wedge \partial/\partial \overline{z}^\mu$. For an element $\alpha \in \Omega^{p,q}(M)$ we have the local expression

$$\begin{aligned} d\alpha =&(\partial + \overline{\partial})\alpha = (\partial + \overline{\partial})\frac{1}{p!q!}\alpha_{i_1\cdots i_p \bar{j}_1 \cdots \bar{j}_q} dz^{i_1} \wedge \cdots \wedge dz^{i_p} \wedge d\overline{z}^{j_1} \wedge \cdots \wedge \overline{z}^{j_q} \\ =&\frac{1}{p!q!}\left(\frac{\partial \alpha_{i_1\cdots i_p \bar{j}_1 \cdots \bar{j}_q}}{\partial z^k} dz^k \wedge dz^{i_1} \wedge \cdots \wedge dz^{i_p} \wedge d\overline{z}^{j_1} \wedge \cdots \wedge \overline{z}^{j_q}\right. \\ &\left. + \frac{\partial \alpha_{i_1\cdots i_p \bar{j}_1 \cdots \bar{j}_q}}{\partial \overline{z}^k} d\overline{z}^k \wedge dz^{i_1} \wedge \cdots \wedge dz^{i_p} \wedge d\overline{z}^{j_1} \wedge \cdots \wedge \overline{z}^{j_q}\right) \end{aligned} \tag{23}$$

From niloptency of d ($d^2 = 0$) we have $\partial\overline{\partial} + \overline{\partial}\partial = \partial^2 = \overline{\partial}^2 = 0$. The *k-th de Rham cohomology groups* $H^k_{dR}(M, \mathbb{R})$ of a manifold M is defined as

$$H^k_{dR}(M, \mathbb{R}) := \frac{Ker(d_k : \Omega^k(M) \to \Omega^{k+1}(M))}{Im(d_{k-1} : \Omega^{k-1} \to \Omega^k(M))}. \tag{24}$$

The de Rham cohomology ring $H^\bullet(M)$ is defined as the direct sum of these spaces

$$H^\bullet(M) := \bigoplus_{k=0}^{m} H^k_{dR}(M). \tag{25}$$

The dimension of k-th de Rham groups are referred to as Betti numbers $b^k := dim(H^k_{dR}(M, \mathbb{R}))$ and are topological invariants of M. A useful topological invariant defined from these Betti numbers is the Euler characteristic

$$\chi(M) := \sum_{k=0}^{m} (-1)^k b^k(M). \tag{26}$$

[3] We will often omit the $\mathbb{C}$ in $\Omega^k_{\mathbb{C}}$ as it should be apparent when we are considering the space Ω^k and not $\Omega^{p,q}$

Geometrically b^k counts the number of k-cycles which are not boundaries of M. The Dolbeault cohomology groups are similarly defined $H^{p,q}_{\overline{\partial}}(M) := Ker(\overline{\partial})/Im(\overline{\partial})$ with the sequence

$$\cdots \xrightarrow{\overline{\partial}} \Omega^{p,q}(M) \xrightarrow{\overline{\partial}} \Omega^{p,q+1}(M) \xrightarrow{\overline{\partial}} \cdots$$

is referred to as the Dolbeault complex. A similar complex could be made from ∂ but no new information is gained and we choose to work with $\overline{\partial}$ since the kernel of such operators are holomorphic. We define the Hodge numbers by $h^{p,q} := dim(H^{p,q}_{\overline{\partial}}(M))$ and are summarized by the Hodge diamond:

$$\begin{array}{ccccccc}
 & & & h^{m,m} & & & \\
 & & h^{m,m-1} & & h^{m-1,m} & & \\
 & \iddots & & \vdots & & \ddots & \\
h^{m,0} & \cdots & & \vdots & & \cdots & h^{0,m} \\
 & \ddots & & \vdots & & \iddots & \\
 & & h^{1,0} & & h^{0,1} & & \\
 & & & h^{0,0} & & &
\end{array}$$

From equation (22) $b^k = \sum_{j=0}^{k} h^{j,k-j}$, from hodge duality and complex conjugation $h^{p,q} = h^{q,p}$ and $h^{p,q} = h^{m-p,m-q}$ respectively.

We now refine the Dolbeault complex to include forms with different valued sections. We make use of such complexes in the study of twisted topological sigma models. Recall that sections of $\Lambda^k M$ were M valued k-forms. Let $E \to M$ be vector bundle over M. Then an E-valued $k$form ω is a section of $E \otimes \Lambda^k M$. With this we have the complex

$$0 \to \Gamma(E) \xrightarrow{\overline{\partial}} \Gamma(E) \otimes \Omega^{0,1}(M) \xrightarrow{\overline{\partial}} \cdots \xrightarrow{\overline{\partial}} \Gamma(E) \otimes \Omega^{0,m}(M) \xrightarrow{\overline{\partial}} 0$$

which defines the twisted Dolbeault cohomology groups

$$H^q_{\overline{\partial}}(E) := \frac{Ker(\overline{\partial} : \Gamma(E) \otimes \Omega^{0,q}(M) \to \Gamma(E) \otimes \Omega^{0,q+1}(M))}{Im(\overline{\partial} : \Gamma(E) \otimes \Omega^{0,q-1}(M) \to \Gamma(E) \otimes \Omega^{0,q}(M))} \tag{27}$$

1.4 Kahler Geometry

Let M be a complex manifold with an almost complex structure J then a Riemannian metric g on M which satisfies $g_p(J_pX, J_pY) = g_p(X, Y)$ for all $p \in M$ and $X, Y \in T_pM$ is called a Hermitian metric. In components this means $g_{\lambda\sigma}J^\lambda_\mu J^\sigma_\nu$ and so locally we find

$$g = g_{i\bar{j}} dz^i \otimes d\bar{z}^j + g_{\bar{i}j} d\bar{z}^i \otimes dz^j \tag{28}$$

since $g_{ij} = g_{\bar{i}\bar{j}} = 0$. An important property of complex manifolds is that every complex manifold admits a Hermitian metric so the addition of g on M does not constraint (M, J) any further. If g is Riemannian then $\hat{g}_p(X, Y) := 1/2(g_p(X, Y) + g_p(J_pX, J_pY)$ is Hermitian. The 2-form ω defined by

$$\omega(X, Y) := g(JX, Y) \tag{29}$$

is called the Kahler form of g. Locally this takes the form

$$\omega = ig_{i\bar{j}}dz^i \wedge d\bar{z}^j \tag{30}$$

In particular if we define $J_{i\bar{j}} := J_i^k g_{k\bar{j}}$ then the Kahler form takes the form $\omega = -J_{i\bar{j}}dz^i \wedge d\bar{z}^j$.

Definition 10. *Let (M, J) be a complex manifold which admits a Hermitian metric g and hence a Kahler form ω. M is called a Kahler manifold if the Kahler form is closed $d\omega = 0$. The metric g of a Kahler manifold M is called a Kahler metric.*

The class $[\omega] \in H^2_{dR}(M)$ is referred to as the Kahler class. The Ricci two-form

$$\mathcal{R} := iR_{i\bar{j}}dz^i \wedge d\bar{z}^j = -i\partial\bar{\partial} log\ \det(g_{i\bar{j}}) \tag{31}$$

is the curvature two-form of the holomorphic tangent bundle $F = -i\mathcal{R}$. It is an exact form $d\mathcal{R} = 0$. In coordinates this reads

$$(\partial + \bar{\partial})ig_{i\bar{j}}dz^i \wedge d\bar{z}^j = (i/2)(\partial_{[k}g_{i]\bar{j}}dz^k \wedge dz^i \wedge d\bar{z}^j + \partial_{[\bar{k}}g_{\hat{i}\bar{j}]}d\bar{z}^k \wedge dz^i \wedge d\bar{z}^j) = 0 \tag{32}$$

which gives the the following constraints on a Kahler metric

$$\partial_k g_{i\bar{j}} = \partial_j g_{k,\bar{j}}, \quad \partial_{\bar{k}} g_{i\bar{j}} = \partial_{\bar{j}} g_{i\bar{k}} \tag{33}$$

which are known as the Kahler conditions. This locally has the solution

$$g_{i\bar{j}} = \partial_i \partial_{\bar{j}} \mathcal{K} \tag{34}$$

where $\mathcal{K}$ is called the Kahler potential of g and is a local function that must be defined on each chart U of M. Note that $\mathcal{K}$ is unique up to the addition of holomorphic and antiholomorphic functions

$$\mathcal{K} \sim \mathcal{K} - f(z) - \overline{f}(\bar{z}). \tag{35}$$

The hodge star is the isomorphism $* : \Lambda^k T^*M^{\mathbb{C}} \to \Lambda^{2m-k} T^*_{\mathbb{C}} M$ which is defined by

$$\alpha \wedge *\beta = \alpha_{i_1 \cdots i_l} \beta_{j_1 \cdots j_m} g^{i_1 \bar{j}_1} \cdots g^{i_m \bar{j}_m} dV \tag{36}$$

where dV is the volume element on M. From this we can define the $L^2(M)$ inner-product of forms by where

$$\langle \alpha, \beta \rangle := \int_M \alpha \wedge *\beta = \int_M \alpha_{i_1 \cdots i_l} \beta_{j_1 \cdots j_m} g^{i_1 \bar{j}_1} \cdots g^{i_m \bar{j}_m} dV \tag{37}$$

The adjoint operators with respect to this inner product are given by

$$d^\dagger = - * d *, \quad \partial^\dagger = - * \partial *, \quad \overline{\partial}^\dagger = - * \overline{\partial} * . \tag{38}$$

As in the case of real variables we can define Laplacians

$$\Delta := dd^\dagger + d^\dagger d, \quad \Delta_\partial := \partial\partial^\dagger + \partial^\dagger\partial \quad \Delta_{\bar{\partial}} := \overline{\partial}\overline{\partial}^\dagger + \overline{\partial}^\dagger\overline{\partial} \tag{39}$$

which are related by $\Delta_\partial = \Delta_{\bar{\partial}} = \frac{1}{2}\Delta$ due to the Kahler structure of M. The theory of harmonic forms is constructed exactly as in the real case. From the Kahler condition the coefficients for the covariant derivatives is found to be non-zero for pure holomorphic and anti-holomorphic coordinates

$$\Gamma^k_{ij} = g^{k\bar{l}} \partial_i g_{j\bar{l}}, \quad \Gamma^{\bar{k}}_{\bar{i}\bar{j}} = g^{l\bar{k}} \overline{\partial}_{\bar{i}} g_{l\bar{j}}. \tag{40}$$

The non-zero components of Riemann tensor on a Kahler manifold are

$$\begin{aligned} R_{i\bar{j}k\bar{l}} &:= g_{i\bar{m}} R^{\bar{m}}_{\bar{j}k\bar{l}} = \partial_k \partial_{\bar{l}} g_{\bar{j}i} - g^{m\bar{n}} \partial_k g_{i\bar{n}} \partial_{\bar{l}} g_{\bar{j}m}, \quad R^i_{\bar{j}kl} = \partial_{\bar{k}} (g^{\bar{m}i} \partial_l g_{j\bar{m}}) \\ R_{i\bar{j}} &:= R^k_{ki\bar{j}} = -\partial_{\bar{j}} (g^{k\bar{l}} \partial_i g_{k\bar{l}}) = -\partial_{\bar{j}} \partial_i log(det(g_{i\bar{j}})) \end{aligned} . \tag{41}$$

For later use we note that the local action of the Laplacian Δ on a p-from α is

$$(\Delta\alpha)_{\mu_1 \cdots \mu_p} = -\nabla^\lambda \nabla_\lambda \alpha_{\mu_1 \cdots \mu_p} - pR_{\lambda[\mu_1} \alpha^\lambda_{\mu_2 \cdots \mu_p]} - \frac{1}{2} p(p-1) R_{\lambda\sigma[\mu_1\mu_2} \alpha^{\lambda\sigma}_{\mu_3 \cdots \mu_p]}. \tag{42}$$

1.4.1 Holomorphic line bundles

Given a vector bundle $E \to M$, if it's fibers have rank one ($F \cong \mathbb{C}$ then E is said to be a (holomorphic) line bundle. If L and $\tilde{L}$ are both line bundles then so too is $L \otimes \tilde{L}$ since $dim_{\mathbb{C}}(L \otimes \tilde{L}) = dim_{\mathbb{C}}(L) dim_{\mathbb{C}}(\tilde{L}) = 1$. Hence, any power $L^{\otimes r}$ is a line bundle. If $dim_{\mathbb{C}}(M) = m$ then the space

$$K := \Lambda^{m,0} M \tag{43}$$

is a holomorphic line bundle which we call the canonical (line) bundle. The reason this space is of complex dimension one is that its sections are $(m, 0)$-forms and is unique up to scaling. The

anticanoical line bundle is defined as $\overline{K} := \Lambda^{0,m}M$. The bundle

$$L^{-1} := \{(l, w) \in \mathbb{CP}^n \times \mathbb{C}^{n+1} | \ w \in l\} \tag{44}$$

is called the Tautological line bundle (on $\mathbb{CP}^m$). More concretely, for $([z_0, \cdots, z_n], (w_0, \cdots, w_n)) \in L^{-1}$ we have $(w_0, \cdots, w_n) = \lambda(z_0, \cdots, z_n)$ for some $\lambda \in \mathbb{C}^*$. This bundle is simply the assignment of the line through the origin of $\mathbb{C}^{m+1}$ as defined in $\mathbb{CP}^m$ and so each fiber of L^{-1} over p is simply the line in $\mathbb{CP}^m$ connecting p to the origin. In the literature it is common practice to denote this bundle by

$$L^{-1} := \mathcal{O}(-1) \tag{45}$$

where $\mathcal{O}(-1)$ is really the sheaf of holomorphic sections of L^{-1}. The dual to $(L^{-1})^* := L^1$ is called the Hyperplane bundle which is also denoted by its sheaf $L^1 := \mathcal{O}(1)$. Similarly for $L^k := L^{\otimes k}$ we have

$$\mathcal{O}(k) := \begin{cases} \mathcal{O}(1)^k & \text{if } k > 0 \\ \mathcal{O}(-1)^{-k} & \text{if } k < 0. \end{cases} \tag{46}$$

1.4.2 Characteristic Classes

The study of characteristic classes concerns polynomials which we define for some bundle. They classify the bundle and gives us topological invariants of the underlying manifold. See [19] for further details.

Let (P, π, M, G) be a principle bundle and $A_i \otimes \eta_i \in \mathfrak{g} \otimes \Omega^{p_i}(M)$ with $\mathfrak{g}$ the lie algebra of G. We define the map

$$\tilde{P} : (A_1\eta_1, \cdots, A_r\eta_r) \mapsto \eta_1 \wedge \cdots \wedge \eta_r \hat{P}(A_1, \cdots, A_r) \tag{47}$$

where $\hat{P} : G^{\otimes r} \to \mathbb{C}$ is a symmetric r-linear map. The restriction $\tilde{P}|_{\mathfrak{g}}$ is called G-invariant or adjoint (ad) invariant if $\hat{P}(Ad_gA_1, \cdots, Ad_gA_r) = \hat{P}(A_1, \cdots, A_r)$, $\forall g \in G$ and $A_i \in \mathfrak{g}$ where $Ad_g(A_i) := g^{-1}A_ig$.

$$P(A\eta) := \eta \wedge \cdots \wedge \eta \hat{P}(A, \cdots, A) \tag{48}$$

is called an (extended) invariant polynomial of degree r. Let $A = A_\mu dx^\mu \in \mathfrak{g} \otimes \Omega(M)$ be the connection on (P, M) and $F = dA + A \wedge A$ its curvature (locally this is given by $F = 1/2 F_{\mu\nu} dx^\mu \wedge dx^\nu$ where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]$). The Chern-Weil theorem [19] states for an invariant polynomial P, $dP(F) = 0$ so we can build a cohomology class of M $H^*(M)$. This cohomology class is called the characteristic class of M and is defined by the map $\chi : E \to \chi(E) \in H^*(M)$ which satisfies $\chi(f^*E) = f^*\chi(E)$ and $\chi(E) = \chi(E')$ if $E \cong E'$. A characteristic class of E is denoted by $\chi_E(P)$.

Let $E \to M$ be a complex vector bundle with typical fiber $\mathbb{C}^k$ and $G \subset GL(m, \mathbb{C})$. Then the

total Chern class $c(F)$ is an invariant polynomial defined by

$$c(F) := \tilde{det}\left(\mathcal{I} + \frac{iF}{2\pi}\right) \in \bigoplus_{i=0}^{m} \Omega^{2i}(M) \tag{49}$$

This decomposition is expressed as $c(F) = 1 + c_1(F) + c_2(F) + \cdots + c_m(F)$. The set of even degree forms $[c_j] \in H^{2j}(M)$ are called the jth Chern class (where $H^{2i}(M)$ is the homology group defined by the Chern-Weil theorm since c_j is exact). We find

$$\begin{aligned} c_0(F) &= 1 \\ c_1(F) &= tr\left(\frac{iF}{2\pi}\right) \\ &\vdots \\ c_m(F) &= \left(\frac{i}{2\pi}\right)^m \tilde{det}(F) \end{aligned} \tag{50}$$

Chern classes satisfy $c(f^*E) = f^*c(E)$, $c(E \oplus F) = c(E) \wedge c(F)$ which makes them a characteristic class of M. The total Chern character $ch(F)$ is a characteristic class defined by

$$ch(F) := tr\ exp\left(\frac{iF}{2\pi}\right) = \sum_{j=0} \frac{1}{j!} tr\left(\frac{iF}{2\pi}\right)^j \tag{51}$$

with $ch_j(F) := (1/j!)tr(iF/2\pi)^j$ being the jth Chern character. This characteristic class similarly satisfies $ch(f^*E) = f^*ch(E)$, $ch(E \otimes F) = ch(E) \wedge ch(F)$, and $ch(E \oplus F) = ch(E) + ch(F)$. The Todd class is defined by

$$td(F) := \prod_{j=1}^{m} \frac{x_j}{1 - e^{-x_j}} \tag{52}$$

where x_j is implicitly defined by the diagonalization of F: $g^{-1}Fg = diag(x_1, \cdots, x_m)$ for some $g \in G$. We find $td(E \otimes F) = td(E) \wedge td(F)$. Let $P_j(F) \in \chi_E(P)$ be an element of the characteristic class of some bundle. By Poincare's lemma it can be written locally as

$$P_j(F) = dQ_{2j-1}(A, F), \qquad Q_{2j-1}(A, F) \in \mathfrak{g} \otimes \Omega^{2j-1}(M). \tag{53}$$

The local $2j - 1$ form is the Chern-Simons form of the characteristic class of $P_j(F)$ and is given explicitly by the transgression of P_j. By invoking stokes theorem on a complex manifold with boundary we find $\int_M P_m(F) = \int_{\partial M} Q_{2m-1}$ and so the Chern-Simons forms describe the topology of ∂M. The second Chern-Simons form of the Chern character is given by

$$Q_3(A, F) = \frac{1}{2}\left(\frac{i}{2\pi}\right)^2 tr\left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A\right). \tag{54}$$

This term is the main focus of Chern-Simons theory.

2 Calabi-Yau Manifolds

In this section we will review the basics of Calabi-Yau manifolds and their moduli. The background geometries of the topological string will be Calabi-Yau and many of the properties of this space will unveil a great deal about the theory.

2.1 Calabi-Yau Moduli

There are a number of equivalent definitions of a Calabi-Yau manifold, each of which give us further insight into the geometry of the space.

Definition 11. *A real* $2m$ *dimensional compact Kahler manifold* (M, J, g) *which has*

1. *zero Ricci form* $\mathcal{R} = 0$*,*
2. *vanishing first Chern class* $c_1(TM) = 0$*,*
3. $SU(m)$ *holonomy,*
4. *a covariantly constant spinor,*
5. *trivial canonical bundle* K*,*
6. *covariantly constant spinors*

is called a Calabi-Yau manifold of complex dimension m which we denote CY_m*.*

The space CY_m is often referred to as a Calabi-Yau m-fold. In the next section we will use the physics of string compactification to show that all of these definitions are equivalent. Here we will only prove enough for us to better understand the classification of Calabi-Yau manifolds. let us suppose 1. is true and show that 5. follows from this assumption. The Ricci tensor of a Kahler manifold is locally expressed as $R_{i\bar{j}} = -\partial_{\bar{j}}\partial_i log(det(g))$ so for a Calabi-Yau manifold $R_{i\bar{j}} = 0$ we must have $det(g_{i\bar{j}}) = e^{f(z)}e^{\bar{f}(\bar{z})}$ for some holomorphic function $f(z)$ on M. In particular note that $e^{f(z)}dz^1 \wedge \cdots \wedge dz^m$ is by construction nowhere vanishing and since this is a top form on $(T^{1,0}M)^*$ it is unique up to the addition of constants. Therefore the bundle $K = \Lambda^m(T^{1,0}M)^*$ is trivial. The following theorem was proven by Yau after Calabi first conjectured it:

Theorem 12. *(Yau's theorem) Let M be a complex compact Kahler manifold with Kahler class* $[\omega]$ *and vanishing first Chern class* $c_1(TM) = 0$*. Then there exists a unique Ricci-flat metric g with associated Kahler form* $\tilde{\omega} \in [w]$*.*

We will use this theorm to motivate the discussion on classifying the moduli of Calabi-Yau manifolds.

Calabi-Yau hodge numbers

The Calabi-Yau conditions will decrease the number of independent hodge numbers. Recall that for a Kahler manifold $h^{p,q} = h^{m-p,m-q} = h^{q,p}$. As just shown, CY_3 admits a unique volume Ω 3-form which implies $h^{3,0} = 1$ so by complex conjugation $h^{0,3}$. The rest of the non-zero hodge numbers are obtained by consdiering harmonic fomrs [9] $h^{0,0} = 1 = h^{3,3}$, $h^{1,1}$, and $h^{2,1} = h^{1,2}$ with all others zero. The Hodge diamond for CY_3 is therefore

$$\begin{array}{ccccccc} & & & 1 & & & \\ & & 0 & & 0 & & \\ & 0 & & h^{1,1} & & 0 & \\ 1 & & h^{2,1} & & h^{2,1} & & 1 \\ & 0 & & h^{1,1} & & 0 & \\ & & 0 & & 0 & & \\ & & & 1 & & & \end{array}$$

and so the two numbers $h^{1,1}(M)$, $h^{2,1}(M)$ specify the cohomology classes of CY_3 and hence define topological properties of the manifold. In fact the Euler characteristic reduces to

$$\chi(CY_3) = \sum_{k=0}^{6} b^k = 2 + 2h^{1,1} - 2 - 2h^{2,1} \tag{55}$$

$$= 2(h^{1,1} - h^{2,1}) = \int_{CY_3} c_3(M) \tag{56}$$

As we shall now see, these values completely characterize the moduli spaces of CY_3 with $h^{1,1}$ classifying the Kahler structure and $h^{2,1}$ the complex structure of the Calabi-Yau threefold.

Before we look at the Calabi-Yau moduli spaces explicitly let us reason through what the moduli space should look like. Since a Kahler form ω is closed it is an element of $[\omega] \in H^{1,1}_{\bar{\partial}}(M)$. Given that a Kahler metric is Riemannian it must admit positive volumes on all submanifolds so we conclude that the Kahler form, as implicitly defined by g, must be closed under positive rescalings. Of course there are also submanifolds which admit zero volume. Hence we have the so called Kahler cone[4] $\mathcal{K} \subseteq H^{1,1}_{\bar{\partial}}(M)$. Calabi conjectured given a compact Kahler manifold with Kahler form ω and vanishing first Chern class $c_1 = 0$ then there exists a Ricci-flat Kahler metric on M $\tilde{g}$ whose form $\tilde{\omega}$ is of the same Kahler class as ω. Therefore each Kahler class admits a unique a Ricci-flat metric on M. Since $dim(\mathcal{K}) = h^{1,1}(M)$ we have an explicit expression for the number of possible Ricci-flat Kahler forms which M can have. The numbers $h^{1,1}(M)$ are called the Kahler moduli parameters. Elements of $\mathcal{K}$ can be used to deform the metric trivially. This characterization defined the Kahler moduli space $\mathcal{M}^{1,1}$ which has dimension $h^{1,1}(M)$. Locally $\mathcal{M}^{1,1}$ is the Kahler part of the tangent

[4] A vector subspace V over a field $\mathbb{F}$ is called a cone if $\lambda v \in V$ for $\lambda > 0$ and for all $v \in V$.

space of metric deformations. We will now formalize these claims by looking at deformations of the metric which preserve the structure of interest.

2.1.1 Calabi-Yau Moduli

The moduli spaces associated to Calabi-Yau threefolds are the Kahler and Complex moduli which are intimately related to the Hermitian metric which the manifold admits. Therefore we follow closely to the analysis of conformally equivalent metrics and look at deformations of the metric which preserve the Ricci-flat condition. Suppose we have the variation $g \mapsto g + \delta g$ then as in the study of conformally equivalent metrics[19] δg is orthogonal to the underlying vector field which defines the diffeomorphism v_μ for which $\int \sqrt{g}\delta g^{\mu\nu} 2\nabla_{(\mu} v_{\nu)} d^m\xi = 0$, hence we have the condition $\nabla^\mu \delta g_{\mu\nu} = 0$. Assuming this variation is also Ricci-flat $R_{ij}(g + \delta g) = 0$ then by linearizing this equation we obtain[4] the Lichnerowicz equation

$$\nabla^\lambda \nabla_\lambda \delta g_{ij} + 2R_i{}^k{}_j{}^m \delta g_{km} = 0. \tag{57}$$

Since CY_3 is Kahler the Riemann tensor splits into terms such as $R_i{}^{\bar{k}}{}_j{}^{\bar{m}}$, $R_i{}^{\bar{k}}{}_{\bar{j}}{}^m$, etc. the metric perturbations of mixed $\delta g_{i\bar{j}}$ and pure δg_{ij} type indicies satisfy the Lichnerowicz equation separately. Therefore we may analyze each of these cases separately.

If we consider mixed terms $\delta g_{i\bar{j}}$ then from the expression for a harmonic form, equation (42), the Lichnerowicz equation is equivalent to[5]

$$(\Delta_d g)_{i\bar{j}} = (dd^\dagger + d^\dagger d) g_{i\bar{j}} = 0.$$

This means that $\delta g_{i\bar{j}}$ is a harmonic complex (1, 1)-form. Let $span\{\eta^\alpha\} = H^{1,1}(CY_3)$ and $t^\alpha \in \mathbb{R}$, then by Yau's theorem

$$\delta g_{i\bar{j}} = \sum_{\alpha=1}^{h^{1,1}} t^\alpha \eta^\alpha_{i\bar{j}}. \tag{58}$$

Since we would like this metric deformation to be Hermitian and hence Kahler the parameters must be positive definite. This means that $\delta g_{i\bar{j}}$ is an element of the Kahler cone exactly as we reasoned above. The parameters t^α are the Kahler moduli and often refered to as Kahler parameters.

For pure indicies one can show that the the Lichnerowicz condition is equivalent to

$$\Delta_{\bar{\partial}}(g^{i\bar{k}} \delta g_{\bar{k}j}) = (\bar{\partial}\bar{\partial}^\dagger + \bar{\partial}^\dagger \bar{\partial})(g^{i\bar{k}} \delta g_{\bar{k}j}) = 0$$

where $(g^{i\bar{k}} \delta g_{\bar{k}j} d\bar{z}^{\bar{j}})$ is and element of $H^{0,1}_{\bar{\partial}}(M, T^{1,0}M)$. Its quite apparent that if we are to impose this metric deformation be Kahler then it would not be described by the same Kahler geometry of the

[5] The identifications made here are unique by the Hodge decomposition theorem.

original manifold as we cannot contract pure indicies with our mixed indexed metric. Therefore it must be Kahler with respect to some other complex structure. From this we can see pure index metric deformations classify the complex structure of manifold. To make better sense of this deformation we invoke an isomorphism between $H^{0,1}_{\bar{\partial}}(M, T^{1,0}M) \cong H^{2,1}_{\bar{\partial}}(M)$ which we now introduce. Let Ω be the unique holomorphic (3,0) form on the Calabi-Yau manifold. Then we can define

$$\Omega_{ijk} g^{k\bar{m}} \delta g_{\bar{m}\bar{l}} dz^i \wedge dz^j \wedge d\bar{z}^l \tag{59}$$

which is harmonic is we assume the Lichnerowicz condition holds. Let $span\{\alpha\} = H^{2,1}_{\bar{\partial}}(M)$ and $t^\alpha \in \mathbb{C}$ for $\alpha = 1, \cdots, h^{2,1}$, then this deformation can be expanded as

$$\Omega_{ijk} g^{k\bar{m}} \delta g_{\bar{m}\bar{l}} = \sum_{\alpha=1}^{h^{2,1}} t^\alpha \alpha^\alpha_{ij\bar{l}} \tag{60}$$

we call the parameters t^α complex structure moduli. There is a more sophisticated way to obtain this result which we now sketch for later use. We can consider infintesimal deformations of the complex structure $(J + \mu)^2 = -1$ to find $J\mu + \mu J = 0$. After linearizing the Nijenhuis tensor condition for a complex manifold one finds that the deformations μ are forms valued in the tangent space to $\mathcal{M}^{2,1}$. These are the well known Beltrami differentials which we express as

$$\mu = \mu^z_{\bar{z}}(z) \partial_z \otimes d\bar{z} \in \Gamma(H^1_{\bar{\partial}}(T^{1,0}M)) \tag{61}$$

$$\bar{\mu} = \mu^{\bar{z}}_z(\bar{z}) \partial_{\bar{z}} \otimes dz \in \Gamma(H^1_{\bar{\partial}}(T^{0,1}M)). \tag{62}$$

From this we have the following isomorphism $T\mathcal{M}^{2,1} \cong H^1(TM)$. These will be of use when we consider coupling twisted topological sigma models to gravity.

To summarize, we now seen that the Calabi-Yau moduli space, the moduli of Ricci-flat Kahler forms, splits up into harmonic representative of $H^{1,1}_{\bar{\partial}}$ and $H^{2,1}_{\bar{\partial}}(M)$. Such representations characterize the Kahler and complex moduli respectively characterized by $h^{1,1}$ and $h^{2,1}$ elements. Schematically this is written as $\mathcal{M}_{CY} = \mathcal{M}^{1,1} \times \mathcal{M}^{2,1}$. When we compactify string theories on Calabi-Yau manifolds we have an additional antisymmetric tensor field B which contributes exactly $h^{1,1}$ massless scalar fields to the compactified theory. In such cases we define the complexified Kahler moduli elements $\tilde{t}^\alpha_{\mathbb{C}}$ which is defined by variations of the complexified Kahler form $\mathcal{J} = B + iJ$. We will say more about this when we consider compactification of ten-dimensional theories on CY_3.

2.1.2 Special Geometry

Complex Moduli $\mathcal{M}_{\mathbb{C}}$

In order to describe the complex structure moduli we build a vector bundle over the moduli space. This bundle is known as the Hodge bundle which has a $H^3_{dR}(M) = H^{3,0}(M) \oplus H^{2,1} \oplus H^{1,2} \otimes H^{0,3}(M)$ fiber of dimension $2(h^{2,1} + 1)$. Locally this bundle is

$$Hod(M) \cong \mathcal{M}^{2,1} \times \left(H^{3,0}(M) \oplus H^{2,1} \oplus H^{1,2} \otimes H^{0,3}(M)\right). \tag{63}$$

An important (sub) line bundle is the vacuum line bundle which is simply the line bundle over $H^{3,0}(\mathcal{M}_{\mathbb{C}})$ which is one dimensional since the Calabi-Yau geometry admits a unique holomorphic three form Ω. This line bundle admits the metric

$$||\Omega||^2 = e^{-\mathcal{K}} := i \int_M \Omega \wedge \overline{\Omega} \tag{64}$$

which is not well define since a rescaling of $\Omega \mapsto e^{f(t)}\Omega$ define the same metric for some function f on the moduli space so this metric transorms as $g \mapsto g e^{f(t)+\overline{f}(\overline{t})}$. The expression $e^{-\mathcal{K}}$ was introduced to show that under such a rescaling the exponent transforms exactly as a Kahler potential $\mathcal{K} \mapsto \mathcal{K} - f - \overline{f}$ and so we define a Kahler metric on the moduli space by

$$g_{\alpha\bar{\beta}} := \partial_\alpha \partial_{\bar{\beta}} \mathcal{K} = -\partial_\alpha \partial_{\bar{\beta}} log\left(i \int_M \Omega \wedge \overline{\Omega}\right) \tag{65}$$

where α are holomorphic coordinates on $\mathcal{M}_{\mathbb{C}}$ and $\overline{\beta}$ anti-holomorphic coordinates. This is precisely the form of the Fubini-study metric of a complex projective space and as we shall see shortly the moduli space will turn out to admit projective inhomogeneous coordinates. Variations of the complex structure, and hence the tangent space to the moduli, were found to be $(2, 1)$ forms so we will write a variation of Ω with respect to the moduli coordinates t^α as

$$\partial_\alpha \Omega = k_\alpha \Omega + \chi_\alpha \tag{66}$$

where χ_α is the $(2, 1)$ form just mentioned. Considering $\partial_\alpha \Omega_{\overline{\mu}\overline{\nu}} = (\partial g / \partial t^\alpha)_{\overline{\mu}\overline{\nu}}$ then from our analysis of the complex moduli of Calabi-Yau we find

$$(\delta_\alpha g)_{\bar{i}\bar{j}} = -\frac{1}{||\Omega||^2} \overline{\Omega}_{\bar{i}}^{\,kl} (\chi_\alpha)_{kl\bar{j}} \quad \Leftrightarrow \quad (\chi_\alpha)_{ij\bar{k}} = -\frac{1}{2} \Omega_{ij}^{\ \ \bar{l}} \left(\frac{\partial g}{\partial t^\alpha}\right)_{\bar{k}\bar{l}} \tag{67}$$

which we can now use to determine the explicit form of the metric on this moduli

$$g_{\alpha\bar{\beta}} = -\frac{\int_M \chi_\alpha \wedge \overline{\chi}_{\bar{\beta}}}{\int_M \Omega \wedge \overline{\Omega}}. \tag{68}$$

This metric is called the Weil-Peterson metric. With this let us continue our analysis of the Hodge bundle. We wish to use the Hodge bundle as a tool to describe the complex moduli and we will find this is done by constructing local coordinates. We first define a symplectic basis of three cycles $\{A^I, B_J\}$ ($I, J \in \{0, \cdots, h^{2,1}\}$) of the homology class $H_3(M)$ with intersection numbers

$$A^I \cap B_J = -B_J \cap A^I = \delta^I_J, \quad A^I \cap A^J = B^I \cap B^J = 0. \tag{69}$$

Let $\{\alpha_I, \beta^J\}$ ($I, J \in \{0, \cdots, h^{1,2}\}$) be the cohomology basis of $H^3(M)$ which is Poincare dual to the three cycles

$$\begin{aligned} \int_{A^I} \alpha_J &= \int_M \alpha_I \wedge \beta^J = \delta^J_I \\ \int_{B_I} \beta^J &= \int_M \beta^J \wedge \alpha_I = -\delta^J_I \\ \int_M \alpha_I \wedge \alpha_J &= \int_M \beta^I \wedge \beta^J = 0. \end{aligned} \tag{70}$$

We can now introduce coordinates on the Hodge bundle space (X^I, F_I) which are called the periods of the unique holomorphic three form Ω

$$X^I := \int_{A^I} \Omega, \quad F_I := \int_{B_I} \Omega \tag{71}$$

for $I, J \in \{1, \cdots, h^{1,2}\}$. By the symplectic structure of the cycles chosen we can write

$$\Omega = X^I \alpha_I - F_I \beta^I. \tag{72}$$

To see why this is so take the integral the expression over the 3-cycle basis. The intention of introducing these coordinates was to describe the moduli of the complex of the theory so there should be $h^{2,1}$ and currently there are $2h^{2,1} + 2 = dim(H_3(M))$. To reduce these recall that Ω is unique up to rescalings of $\lambda \in \mathbb{C}^*$ so we should expect each of these coordinates to define a set of coordinates in $\mathbb{C}\mathbb{P}^{h^{1,2}}$. If we define charts $U_\mu := \{t^\alpha := X^\alpha / X^\mu | X^\mu \neq 0 \alpha = 1, \cdots h^{2,1}\}$ we now have a complex projective space with $h^{1,2}$ which describes the complex moduli. One can show that the second set of coordinates $F_I = F_I(X)$ are functions of X and by scaling must be of degree one in X. The form

of $F(X)$ can be determined by considering the following integral

$$\int_M \Omega \wedge \partial_I \Omega = \int_M (X^J \alpha_J - F_J \beta^J) \wedge (\alpha_I - \partial_I (F_J) \beta^J) = \int_M \Omega \wedge (k_I \Omega + \chi_I) = 0 \tag{73}$$

which can be solved by using the Riemann blinear relation

$$\int_M \Omega \wedge \partial_I \Omega = \sum_I \left(\int_{A^I} \Omega \int_{B_I} \partial_I \Omega - \int_{B_I} \Omega \int_{A^I} \partial_I \Omega \right) \tag{74}$$

$$= \sum_I \left(X^J \partial_I F_J - F_I \right) \tag{75}$$

giving us the constraint $F_I = X^J \frac{\partial F_J}{\partial X^I}$. Therefore we have

$$F_I = \frac{\partial F}{\partial X^I}, \qquad F := \frac{1}{2} X^I F_I \tag{76}$$

where the term F introduced is called the Calabi-Yau prepotential and is second order in X^I since $F(\lambda X) = \lambda^2 F(X)$. The metric of $\mathcal{M}_{\mathbb{C}}$ can be obtained directly from the prepotential by use of the Riemann blinear on $\int \Omega \wedge \overline{\Omega}$ for which

$$\begin{aligned} e^{-\mathcal{K}} &= i \int \Omega \wedge \overline{\Omega} = i \sum_{I=0}^{h^{2,1}} \left(\int_{A^I} \Omega \int_I \overline{\Omega} - \int_{B_I} \Omega \int_{A^I} \overline{\Omega} \right) \\ &= -i \left(X^I \frac{\partial \overline{F}}{\partial \overline{X}^I} - \overline{X}^I \frac{\partial F}{\partial X^I} \right) \\ &= -i \left[2(F - \overline{F}) - (t^\alpha - \bar{t}^\alpha) \left(\frac{\partial F}{\partial t^\alpha} + \frac{\partial \overline{F}}{\partial \bar{t}^\alpha} \right) \right] \end{aligned} \tag{77}$$

The second terms in the expression reflects the fact that the metric is invariant under deformations of the moduli. A geometry with the properties we have just developed is called special geometry and there is more geometric data we can build on it. For instance, we wrote $\partial_\alpha \Omega = k_\alpha \Omega + \chi_\alpha$ with out specifying what values k_α can take. It's exact form is obtained by considering

$$\begin{aligned} \frac{\partial}{\partial t^\alpha} e^{-\mathcal{K}} &= -e^{-\mathcal{K}} \frac{\partial \mathcal{K}}{\partial t^\alpha} = i e^{-\mathcal{K}} \int_M \frac{\partial \Omega}{\partial t^\alpha} \wedge \overline{\Omega} = i k_\alpha e^{-\mathcal{K}} \int_M \Omega \wedge \overline{\Omega} + i e^{-\mathcal{K}} \int_M \chi_\alpha \wedge \Omega \\ &= i k_\alpha e^{-\mathcal{K}} \int_M \Omega \wedge \overline{\Omega} \\ &= k_\alpha e^{\mathcal{K}} \end{aligned} \tag{78}$$

which shows that $k_\alpha = -\partial_\alpha \mathcal{K}$. The significance of this is once we replace this expression into our original anzatz we obtain the equation for a connection on the vacuum line bundle

$$\mathcal{D}_\alpha \Omega := \partial_\alpha \Omega + \partial_\alpha \mathcal{K} \Omega = \chi_\alpha \in H^{2,1}(M). \tag{79}$$

This allows us to define the Yukawa coupling

$$C_{ijk} := \int_M \Omega \wedge \frac{\partial^3 \Omega}{\partial X^i \partial X^j \partial X^k} \tag{80}$$

$$= \frac{\partial^3 F}{\partial X^i \partial X^j \partial X^k} \tag{81}$$

which will be important when we consider topoogical string theory. Geometrically this factor is given by

$$C_{ijk} \in \Gamma(\mathcal{L}^\otimes \otimes Sym^3(T^* \mathcal{M}_\}). \tag{82}$$

There is still much more to be said about the special geometry considered of the complex moduli $\mathcal{M}_\mathbb{C}$ but we will stop here. An important thing to note is that we have been considering the local structure of the theory and as we saw it is described by a complex projective space. This is the global structure of the theory which is general is an algebraic variety of degree five polynomials. These are discussed at the end of this section. Lastly the Kahler moduli can similarly be shown to give rise to a similar structure and prepotential [14] however we will not show this here.

2.1.3 Mirror Symmetry

There is an extremely interesting and deep duality between Calabi-Yau three-folds. For a Calabi-Yau three-fold M there exists a different Calabi-Yau three fold W such that

$$H^{p,q}(M) = H^{3-p,q}(W) \tag{83}$$

$$\therefore h^{1,1}(M) = h^{2,1}(W). \tag{84}$$

This is the statement of Mirror symmetry and it shows that the Kahler structure of one CY_3 is related to the comple structure of another. The pair (M, W) are called mirror manifolds. In both the physics and mathematical literature this conjecture has led to a large amount of new ideas and insights. We will say more on this when we consider topological sigma models and string theory.

2.2 Calabi-Yau geometries

Locally we can describe Calabi-Yau manifolds as a projective algebraic variety i.e. it is the solutions set to a polynomial in $\mathbb{CP}^k$ for some k. To be precise we look at bundles which have a base $\mathbb{CP}^m$. Let us now consider how this is done by deriving the Calabi-yau conditions for such spaces. The sections of the hyperline bundle $\mathcal{O}(1)$ are simply homogeneous coordinates of $\mathbb{CP}^m$. $T^{1,0}\mathbb{CP}^m = span\{s^i(z)\partial/\partial z^i\}$ where $s^i(z) \in \mathcal{O}(1)$. We can identify $\mathcal{O}(1)^{\oplus(m+1)} \cong T^{1,0}\mathbb{CP}^m$ given by the Euler sequence:

$$0 \to \mathbb{C} \to \mathcal{O}(1)^{\oplus(m+1)} \to T^{1,0}\mathbb{CP}^m \to 0 \tag{85}$$

which simply states $c(\mathbb{C}) = c(\mathcal{O}(1))/c(\mathbb{CP})$ and by the splitting principle,

$$c(T^{1,0}\mathbb{CP}) = c(\mathcal{O}(1)^{\oplus(m+1)}) = c(\mathcal{O}(1))^{(m+1)} = (1 + c_1(\mathcal{O}(1))^{(m+1)} \tag{86}$$

since $c(\mathbb{C}) = 1$. Therefore the Chern class of a general projective space is

$$c(\mathbb{CP}^m) = (1 + c_1(\mathcal{O}(1))^{(m+1)}. \tag{87}$$

The zeros of degree k polynomials in $\mathbb{CP}^m$ span a hypersurface $S = \{z \in \mathbb{CP}^m | p(z) = 0\}$ where $p \in \mathbb{C}[z_0, \cdots, z_m]$ is a polynomial of degree k. Therefore we can view such polynomials as sections of $\mathcal{O}(k)$ by the previous identification. Let us generalize this slightly. Consider $S = \otimes_l S_l$ where we consider l polynomials. On S we can define the normal bundle

$$\mathcal{N}_S := \{v \in T^{1,0}\mathbb{CP}^m \mid \langle v, w\rangle = 0,\ \forall w \in T^{1,0}S\} \tag{88}$$

to the surface S. Sections on S map points on S to zeros in $\mathcal{O}(k_l)^{\otimes l}$. Therefore a local trivialization of this bundle is looks like $\mathcal{O}(k_l)^{\otimes l}|_S$. Since we can write this normal nublde as a quotient $\mathcal{N}_S = \mathcal{O}(k_l)^{\otimes l}|_S = T^{1,0}\mathbb{CP}^m|_S/T^{1,0}S$ or equivalently the short exact seuqence

$$0 \to T^{1,0}S \to T^{1,0}\mathbb{CP}^m|_S \to \mathcal{O}(k_l)^{\otimes l}|_S \to 0 \tag{89}$$

for which we can conclude $c(S) = c(\mathbb{CP}^m)/c(\mathcal{O}(k)^{\otimes l})$. Since $c(\mathcal{O}(k_l)^{\otimes l}) = \prod_{i=1}^{l}(1 + k_i c_1(\mathcal{O}(1))$ we conclude the Chern class of this space is

$$c(S) = \frac{(1 + c_1(\mathcal{O}(1))^{(m+1)}}{\prod_{i=1}^{l}(1 + k_i c_1(\mathcal{O}(1))}. \tag{90}$$

which is called the (general) adjunction formula. Of course for this space to be Calabi-Yau we must have $c_1(S) = 0$

$$\begin{aligned}(1 + c_1(\mathscr{O}(1))^{(m+1)} &= \prod_{i=1}^{l}(1 + k_i c_1(\mathscr{O}(1)) \\ \implies (m+1) &= \sum_i k_i\end{aligned} \tag{91}$$

and in particular for a surface described by a single polynomial $l = 1$ on a CY_3 we have $m = 4$. This means we consider polynomials of degree 5 in $\mathbb{CP}^4$. We can therefore think of local Calabi-Yau manifolds which are described by the bundle $\mathcal{N}_S \to S$. When we introduce sigma models in the next section we will have embedding maps $\phi : \Sigma \to CY_3$ so we can take $S = \phi(\Sigma_g)$ and then pull-back this bundle to one over Σ_g. We will have more to say about this when we consider topological string theories.

Aside from the 'cones and fans' definition of toric geometries there is an equivalent way of describing such spaces. These are known as symplectic quotients and arise from the study of symplectic geometry. Let $(z_1, \cdots, z_k) \in \mathbb{C}^k$ and $\mu_a : \mathbb{C}^k \to \mathbb{C}$ for $a \in \{1, \cdots, k\}$ the moment map defined by

$$\mu_a(z_1, \cdots, z_k) := \sum_{i=1}^{k} Q_a^i |z_i|^2 = \Re(t^a) \tag{92}$$

where t^a are the Kahler moduli and $Q_a^i \in \mathbb{Z}$. If the target space is Calabi-Yau then following condition holds

$$\sum_{i=1}^{k} Q_a^i = 0. \tag{93}$$

A space which admits this structure is called a symplectic quotient and is written as

$$\mathcal{M} := \bigcap_{a=1}^{k-3} \mu^{-1}(\Re(t^a))/U(1)^{k-3} \tag{94}$$

This condition comes from the fact that toric Calabi-Yau manifolds are non-compact [14]. These spaces have a group action of $G = U(1)^{k-3}$ defined by

$$(z_1, \cdots, z_j, \cdots, z_k) \mapsto (e^{iQ_a^1\alpha_a} z_1, \cdots e^{iQ_a^j\alpha_a} z_j, \cdots, e^{iQ_a^k\alpha_a} z_k). \tag{95}$$

Let us now define two fibrations which will be of interest to us.

Local $\mathbb{C}\mathbb{P}^1$

The bundle $\mathscr{O}(-1) \oplus \mathscr{O}(-1) \to \mathbb{C}\mathbb{P}^1$ is referred to as local $\mathbb{C}\mathbb{P}^1$. We can represent this space as a weighted projective space

$$M = \left(\mathbb{C}^4 \backslash \{0\}\right) / \sim \tag{96}$$

with the equivalence

$$(x_1, x_2, z_1, z_2) \sim (\lambda^{-1} x_1, \lambda^{-1} x_2, \lambda z_1, \lambda z_2) \tag{97}$$

for $\lambda \in \mathbb{C}^*$. This bundle is a Calbi-Yau manifold as the Chern-class of the total space exactly cancels that of the base

$$c_1(\mathbb{C}\mathbb{P}^1) + c_1(\mathscr{O}(-1) \oplus (-1)) = 2 + (-2) = 0 \tag{98}$$

It is defined by the following toric weights $\mathbb{C}^* : [1, 1, -1, -1]$ and since these weights sum to zero it is CY. This space is defined by the surface

$$|z_1|^2 + |z_2|^2 - |z_3|^2 - |z_4|^2 = t \tag{99}$$

Local $\mathbb{C}\mathbb{P}^2$

The line bundle $\mathscr{O}(-3) \to \mathbb{C}\mathbb{P}^2$ is referred to as local $\mathbb{C}\mathbb{P}^2$. We can represent this space as a weighted projective space

$$M = \left(\mathbb{C}^4 \backslash \{0\}\right) / \sim \tag{100}$$

with the equivalence relation given by

$$(x_1, x_2, z_1, z_2) \sim (\lambda x_1, \lambda x_2, \lambda z_1, \lambda^{-3} z_2) \tag{101}$$

for $\lambda \in \mathbb{C}^*$. As in the previous case the Chern-classes of the fiber and base cancel

$$c_1(\mathbb{C}\mathbb{P}^2) + c_1(\mathscr{O}(-3)) = 3 + (-3) = 0 \tag{102}$$

The embedding into $\mathbb{C}\mathbb{P}^4$ is given by the solution to the polynomial

$$|z_1|^2 + |z_2|^2 + |z_3|^3 - 3|z_4|^4 = t \tag{103}$$

The two geometries presented here can related by looking at different limits of the conifold , which we now explain.

The conifold

The surface defined by the polynomial $ab - uv = 0$ in $\mathbb{C}^4$ is referred to as a confiold. This space gets its name from the fact that it defines a cone with the origin being the 'tip' of this cone. The origin turns out to be singular since for $p = ab - uv$, $dp = adb + bda - vdu - udv = 0$ is also zero at the origin. Because the conifold appears frequently in string theory we would like to make sense of the singularity which is often done by smoothing them out. This is often done by looking at slight deformations of the space by some parameters of the theory. To begin let us first consider a variation of the above equation $ab - uv = \mu^2$ where $\mu^2 \in \mathbb{R}$. By taking the following coordinate transformation

$$a = z_1 + iz_2, \quad b = z_1 - z_2, \quad u = z_3 + iz_4, \quad v = -z_3 + iz_4 \tag{104}$$

for $z_i \in \mathbb{C}$ this modified conifold equation becomes

$$z_1^2 + z_2^2 + z_3^2 + z_4^2 = \mu^2.$$

We can further decompose these coordinates into real coordinates by defining $z_i = x_i + iy_i$. Plugging these values in and collecting the real and imaginary parts we find the conifold conditions

$$(x_1^2 + x_2^2 + x_3^2 + x_4^2) - (y_1^2 + y_2^2 + y_3^2 + y_4^2) = \mu^2, \qquad x_1y_1 + x_2y_2 + x_3y_3 + x_4y_4 = 0. \tag{105}$$

To get a better intuition of this geometry we look at a slice away from the origin $|x|^2 + |y|^2 = r^2$ with $r \geq 0$. The first constraint tells us that both the coordinates x_i and y_i define three spheres via $|x|^2 = |y|^2 = 1/2r^2$. Therefore this manifold looks something like $S^3 \times S^3$ however we have not imposed the second condition. If we regard one set of coordinates in the second expression then we find that this is nothing more than the equation of a plane through the origin. Therefore we find that the y coordinates define a two sphere. Fibrations of this form are always trivial and so we conclude that the space we are dealing with is S^2 with radius $\sqrt{(r^2 - \mu^2)/2}$ fibered over S^3 with radius $\sqrt{(r^2 + \mu^2)/2}$. We now consider the differnt limits of theses spheres. In the case that $r = \mu$ we find that S^2 goes to zero radius. This space is called the deformed conifold and it turns out to be isomorphic to T^*S^3 the cotangent bundle to the sphere. This can be seen by defining coordinates $\tilde{\vec{x}} = \vec{x}/\sqrt{\mu^2 + |y|^2}$ so that the conifold conditions become $|\tilde{x}|^2 = 1$ and $\tilde{\vec{x}} \cdot \vec{y} = 0$ which are the equations which define the total space of T^*S^3. The case in which $\mu = 0$ is simply referred to as the singular conifold and we note that if $r = 0$ both spheres shrink to zero. Away from the singularity both of these spaces appear as $S^2 \times S^3$ and differ the closer we get to the singularity.

There is one other way in which we can vary the geometry of the singular conifold. This is done

by considering the subspace in $\mathbb{C}^4 \times \mathbb{CP}^1$ defined by

$$\begin{pmatrix} a & u \\ v & b \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = 0$$

for $(a, b, u, v) \in \mathbb{C}$ and homogeneous coordinates $(\xi_1, \xi_2) \in \mathbb{CP}^1$. Away from the singularity this space is exactly that of the singular conifold since this would require the determinant of the above matrix vanish. At the singularity $(a, b, u, v) = (0, 0, 0, 0)$ the homogeneous coordinates are arbitrarily so we have $\mathbb{CP}^1$ at the origin. This is the resolved conifold and it is isomorphic to $\mathcal{O}(-1) \oplus \mathcal{O}(-1) \to \mathbb{CP}^1$. We now have

$$\begin{aligned} \text{Deformed Conifold} &\longleftrightarrow T^*S^3 \\ \text{Resolved Conifold} &\longleftrightarrow \mathcal{O}(-1) \oplus \mathcal{O}(-1) \to \mathbb{CP}^1 \end{aligned}$$

Part II

Fundamentals II: Physics

3 String compactification

In this section we consider compactification of ten-dimensional string theories on Calabi-Yau threefolds which will physically motivate our understanding of topological strings. The different higher dimensional string theories are reviewed. We then give a brief account of the compactification of the Heterotic superstring to understand how the Calabi-Yau gemoetry is emergent from requiring supersymmetric invariance of the four-dimensional theory. The compatification of type IIA and IIB supergravity theories are also considered. These theories will be intamitaely related to the topological invariants we calculate from topological string theories. References for this section are [23], [9], [4], [13], [18].

3.1 Compactification basics

There are five different ten-dimensional string theories, two of which are open heterotic string theories with gauge gropus $SO(32)$ and $E_8 \times E_8$ which admits 16 supercharges, two closed string theories: type IIA and type IIB with 32 supercharges and the type I theory of branes which also has 16 supercharges. The appearance of ten-dimensional theories is attributed to the fact that the quantification of the supersymmetric theory is free of anomalies in $d = 10$. At low energies we can describe each of these superstrings as a ten-dimensional supergravity theories. By considering the low energy effective action of these theories we can use standard QFT formalism and consider perturbations to better understand how the theory acts at strong coupling. By introducing eleven-dimensional M-theory we can related each of these theories through a beautiful web of dualities. Strictly speaking M-theory is not a string theory as it does not contain any strings. Its low energy approximation is eleven-dimensional supergravity. The introduction of an eleven dimensional theory came from type IIA and heterotic $E_8 \times E_8$ appearing to be eleven dimensional at strong coupling. Although we can consider compactification of M-theory we will not do so here.

To make contact with four dimensional theories we follow a process which we call compactification of these higher dimensional string theories. The idea comes from Kaluza-Klein reduction where

we decompose the ambiant spacetime into a product of some 'internal' space M and external space $\mathbb{M}$ which decomposes the field content into modes on each of these spaces. For ten-dimensional theories this is schematically we written as

$$\mathbb{M}^{10} = \mathbb{M}^4 \times M \tag{106}$$

where $\mathbb{M}^4$is some 4-dimensional spacetime and M a compact six-dimensional internal manifold. If we consider only the zero modes of this reduction then the process is called dimensional reduction. Dimensional reduction tells us by considering a field A_M in $\mathbb{M}^10$ then we can decompose this field into massless degrees of freedom on the external space $\mathbb{M}^4$ while M carries the momenta and totally decouples as $A_M = (A_\mu, A_i)$ on $\mathbb{M}^4 \times M$. Generally the internal space will dimensionally be much smaller than the length scales observed in modern four dimensional theories since they are not observed.

The massless field content of this theory transform under representations of the Lorentz group $SO(1,9)$. If we consider compactifying the theory on M these representations decompose into representations of $SO(1,3) \times SO(6)$. To relate these theories to the standard model particle phenomenology would lead us to believe that an $\mathcal{N} = 1$ theory which has supersymmetry spontaneously broken at low energies. Since the heterotic string has this much supersymmetry is was the first theory used to try and obtain a consistent four-dimensional theory. It was later found that both type II theories admit intersting compacified theories when considering orientifolds and intersecting D-branes. In either case there was, and still is, a problem associated to the moduli of the compactified theory. The low energy effective actions obtained contains massless scalar fields which have no potential associated to them. Therefore we cannot make physical predictions about these theories. Becuase these massless scalars do not have a precise value for their vacuum expectation value (VEV) they are called the moduli fields and characterize the theory.

A solution to this problem is to introduce so called flux terms which give rise to a potential for the scalar moduli. The flux terms will come from fields already in the uncompactified theory. This process is known as moduli stabilization or flux quantization. By including flux terms in the compactified theories we find there is a 'back reaction' on the background geometry which in general is non-trivial. The flux terms will constrain the geometry of the allowed internal manifolds. In type IIA one finds that the allowed internal manifolds are complex and symplectic manifolds. Generalized geometry was introduced to describe the relationship between these two classes by considering B-field fluxes on the bundle $T \oplus T^*$. To simplify our anlaysis we assume flux terms are turned off so we will not go into the details of generalized geometry. An excellent review of type II theories with flux on generalized geometries is [12].

3.2 Compactification of the Heterotic String

For definiteness let us consider compactification of the heterotic string on a Calabi-Yau threefold. As mentioned this is an $\mathcal{N} = 1$ ten-dimensional superstring theory which has an $E_8 \times E_8$ or $SO(32)$ gauge group. The symmetries of the (super)Poincare group and gauge group of the ten-dimensional theory must be broken in this compactification. To understand how this occurs let us analyze the field content of the theory. The Heterotic string admits the following massless field content

$$\text{(gravity multiplet)} : \qquad \{g_{MN}, B^{(2)}_{MN}, \phi, \psi^{\alpha}_{M}, \chi_{\dot{\beta}}\} \tag{107}$$

$$\text{(gauge multiplet)} : \qquad \{A^{(1)}_{M}, \lambda_{\alpha}\} \tag{108}$$

where g_{MN} is the graviton, ϕ is the dilaton, A_M is an $SO(32)$ Yang-Mills gauge field, $B^{(2)}_{MN}$ is an anti-symmetric rank two tensor. The indicies α and $\dot{\beta}$ denote a spinor in the positive chiral representation **16** and negative chiral representation **16′** respectively.

The bosonic part of the low-energy effective action in ten-dimensions [4] is

$$S_B = \frac{1}{2\kappa^2} \int d^{10}x \sqrt{|g|} e^{-2\phi} \left(R + 4\partial_\mu \phi \partial^\mu \phi - \frac{1}{2}|H|^2 - \frac{\kappa^2}{30 g^2} Tr(|F|^2) \right) \tag{109}$$

where the field strength H is given by

$$H = dB + \alpha' \omega_L - \frac{\alpha'}{30} \omega_{YM}. \tag{110}$$

The second and third terms ω_L and ω_{YM} are Lorentz Chern-Simons and Yang-Mills Chern-Simons terms given by

$$\omega_L := Tr\left(\omega \wedge d\omega + \frac{2}{3} \omega \wedge \omega \wedge \omega \right) \tag{111}$$

$$\omega_{YM} := Tr\left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right) \tag{112}$$

here ω is the spin connection on $\mathbb{M}^{10}$. To simplify the analysis we consider a maximally symmetric space-time $\mathbb{M}^4$. This simply means $\mathbb{M}^4$ is homogeneous and isotropic so the curvature of the manifold is globally the same everywhere. From Einstien's equation we know the scalar curvature R will be proportional to the cosmological constant and we therefore have three possible choices for $\mathbb{M}^4$:

1. $(R = 0)$ $\mathbb{M}^4$ is simply Minkowski space-time,
2. $(R < 0)$ $\mathbb{M}^4 = AdS_4$ is an anti-de Sitter space-time,
3. $(R > 0)$ $\mathbb{M}^4 = dS$ is a de Sitter space-time.

For any such space-time the field strength H and F both vanish for pure indicies μ in $\mathbb{M}^4$ as well as mixed indicies. We will also assume that there is no background flux H-field in M and that the dilaton ϕ is constant. For the general case see [13].

3.2.1 Supersymmetry breaking

In going from a ten-dimensional theory to a four-dimensional one we will loose some of the supersymmetry the original theory posses. To understand this we recall some basics about supersymmetry breaking. Let Q be a supersymmetry generator which is a Majorana-Weyl spinor which transforms in the **16** representation of $SO(1,9)$. Then a vacuum state $|0\rangle$ (also called the Clifford vacuum) is said to be supersymmetric if $Q|0\rangle = 0$. A supersymmetric variation δ_ϵ is defined by

$$\delta_\epsilon \Psi = [\overline{\epsilon} Q, \Psi] \tag{113}$$

for some field Ψ of the theory and *local* infinitesimal fermionic parameters ϵ. The VEV will therefore vanish $\langle \delta_\epsilon \Phi \rangle = 0$. In a flux free background the VEV of fermionic fields will vanish so we have $\langle \delta_\epsilon \Phi_b \rangle \sim \langle \epsilon \Phi_f \rangle = 0$ where the subscript b is for a bosonic field and f for a fermionic one. We must therefore focus on the supersymmetric variations of fermionic fields in the theory to ensure that $\langle \delta_\epsilon \Phi_f \rangle = 0$. In what follows we will write $\delta_\epsilon \Phi_\epsilon = 0$ but we really mean the VEV of this quantity. Interestingly enough by imposing this condition we obtain a solution to the equation of motion for the bosonic fields of the theory. We will use this property when we consider the sigma models in the next section. We can now apply this to the Heterotic string to understand how the theory retains $\mathcal{N} = 1$ supersymmetry.

The supersymmetry transformations[6] for the fermionic field content of the ten-dimensional theory are [6]

$$\begin{aligned} \delta\psi_M &= \nabla_M \epsilon - \frac{1}{4} H_{MNP} \Gamma^{NP} \epsilon + (fermions) \\ \delta\lambda &= F_{MN} \Gamma^{MN} \epsilon + (fermions) \\ \delta\chi &= \not{\nabla} \phi \epsilon + \frac{1}{24} H_{MNP} \Gamma^{MNP} \epsilon + (fermions) \end{aligned} \tag{114}$$

and since we are assuming $H = F = 0$ and ϕ is constant then $\delta\lambda = \delta\chi = 0$ are satisfied. The remaining condition for unbroken supersymmetry is that ϵ satisfies the Killing spinor equation

$$\nabla_M \epsilon = 0. \tag{115}$$

The existence of covariantly constant spinors restricts the type of internal manifold we can consider since not all spaces admit spinors which satisfy this property. Since ϵ is also A Majorana-

[6] See the appendix for the conventions used here.

Weyl spinor it transforms under the **16** of $Spin(1,9)$. This means that the spin representation of $SO(1,3) \times SO(6)$ is given by $SL(2,\mathbb{C}) \times SU(4)$ since $SO(1,3) \cong SL(2,C)$ and $Spin(6) \cong SU(4)$. The representation **16** decomposes as

$$\mathbf{16} = (\mathbf{2},\mathbf{4}) \oplus (\overline{\mathbf{2}},\overline{\mathbf{4}}) \tag{116}$$

$$\epsilon = \epsilon_{4+} \otimes \epsilon_{6+} \oplus \epsilon_{4-} \otimes \epsilon_{6-} \tag{117}$$

where we have used $\pm$ to denote positive and negative Chirality of each representation. Since the $\mathbb{M}^{10}$ index decomposes as $M = (\mu, m)$ we may consider the Killing spinor equation for each type of spinor separately. For the external manifold $\mathbb{M}^4$ this reads $\nabla_\mu \epsilon_4 = 0$. From $[\nabla_\mu, \nabla_\nu]\epsilon_4 = \frac{1}{4} R_{\mu\nu\lambda\sigma}\Gamma^{\lambda\sigma}\epsilon_4 = 0$ and the restriction to space-times with maximal symmetry the curvature of this space is zero $R = 0$ making it Minkowski. In the compactified theory on $\mathbb{M}^4$ the spinor ϵ_4 becomes the infintesmal parameter of the supersymmetry transformations [4]. For the internal manifold M we have

$$\nabla_m \epsilon_6 = 0 \tag{118}$$

which is the first Calabi-Yau condition we impose on M. We can use the existence of a covariantly constant spinor to prove the equivalent properties we listed for a Calabi-Yau manifold.

3.3 Calabi-Yau conditions

Since ϵ_6 is covariantly constant then so too is its norm which we normalize to one everywhere $\epsilon_{6+}^\dagger \epsilon_{6+} = \epsilon_{6-}^\dagger \epsilon_{6-} = 1$. The internal gamma matricies γ_i are realted to the total space matricies Γ by $\Gamma_i = \gamma_5 \otimes \gamma_i$ where γ_5 is the chirality operator in $\mathbb{M}^4$. We will raise and lower indices with the einbein $e^i_{\ j}$. We now define the tensor[7]

$$J_j^{\ i} := i\epsilon_+^\dagger \gamma_j^{\ i} \epsilon_+ \tag{119}$$

which, after using the Fierz identities, satisfies

$$J_i^{\ k} J_k^{\ j} = -\delta_i^j \tag{120}$$

and so we see that M is an almost complex manifold with an almost complex structure J. To see that M is fully complex we note that since both the metric and the spinors are covariantly constant then $\nabla_i J_k^l = 0$ and hence the Nijenhuis tensor vanishes. Therefore by the Newlander-Nirenberg theorem

[7] We will drop the 6 label as was are now only considering the internal manifold.

M is complex and we can define complex coordinates on M such that

$$J = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \tag{121}$$

and so M admits a Hermitian metric and sarisfies $g_{ij} = J_i^k J_j^l g_{kl}$ by our analysis of Kahler manifolds. Solving this expression for J we obtain an expression for the closed two-form $\omega_{i\bar{j}} := J_{i\bar{j}} = ig_{i\bar{j}}$ so that the Kahler form is given by

$$\omega = ig_{i\bar{j}} dz^i \wedge d\bar{z}^j. \tag{122}$$

From this we learn that M is a Kahler manifold. To see that this Kahler manifold is Ricci-flat we consider the commutator of $\nabla_i \epsilon_\pm = \partial_i \epsilon_\pm + \omega_{ijk}\gamma^{jk}\epsilon_\pm$ to obtain the integrability condition

$$[\nabla_i, \nabla_j]\epsilon_\pm = \frac{1}{4} R_{ijkl}\Gamma^{kl}\epsilon_\pm = 0. \tag{123}$$

Multiplying this expression by γ^m and using the identity $\gamma^i\gamma^{kl} = \gamma^{ijk} + g^{ij}\gamma^k - g^{ik}\gamma^j$ we find

$$R_{ij} = 0. \tag{124}$$

Locally we can write this as $R_{i\bar{j}} = -\partial_i \partial_{\bar{j}} \log\det(g)$ from which we define the Ricci-form

$$\mathcal{R} = iR_{i\bar{j}} dz^i \wedge d\bar{z}^j \tag{125}$$

which is trivially closed. It therefore defines an element in the first Chern class $c_1(TM)$. However since M is Ricci-flat this class must vanish

$$c_1(TM) = 0 \tag{126}$$

To see that the conditions so far imply an $SU(3)$ holonomy let us consider parallel transport of the spinor $\epsilon_\pm$ around a loop in the $SU(4)$ spin bundle $S^\pm$. Let us consider the positive chirality spinor $\epsilon_+ \in \Gamma(S^+)$ in the fundamental representation **4** of $SU(4)$. Since this spinor is covariantly constant after parallel transport it returns to itself. Therefore we can fix one of the four components in **4** by an $SU(4)$ transformation

$$\epsilon_+ = (0, 0, 0, \epsilon_+)^T \in \mathbf{4} \tag{127}$$

which has a little group equal to $SU(3) \subset S(4)$. This implies the existence of a metric which has $SU(3)$ holonomy on M and we have the decomposition $\mathbf{4} = \mathbf{3} \oplus \mathbf{1}$. More concretly on a Kahler manifold [19] this decomposition in the holonomy is of the form $U(3) = SU(3) \times U(1)$ which

preserves the Kahler form on M. The $U(1)$ part of the decomposition comes from the determinant of $U(3)$ which geometrically is equivalent to parallel transporting $(3,0)$ forms which transform as determinant densities. This was of course the claim made when we first considered Calabi-Yau manifolds where we concluded the canonical line bundle $K = \Lambda^3 T^* M$ was trivial. The globally defined ($(3,0)$ form) section of this bundle is given by

$$\Omega_{ijk} = \epsilon_+^T \Gamma_{ijk} \epsilon_+. \tag{128}$$

By assuming no flux $H = 0$ and $\phi = const.$ we have now proven every equivalence mentioned in the definition of a Calabi-Yau manifold by considering the conditions imposed on the geometry of a compactified superstring theory. In general the addition of flux terms will give rise to much more complicated geometries and a much more in depth analysis is required. This analysis will not be considered here and we will add a B-field terms at the end of our analysis of the twisted sigma models.

This same analysis can be done for all the other string theories which will give rise to different four-dimensional theories with varying amounts of supersymmetry. We now gives some basic facts about the type IIA/B theories compactified on CY_3. This will be breif as we will expand on this topic in a later section. See [13] and [23] for a complete analysis.

3.3.1 Type II supergravity

We can compactify non-chiral type IIA $\mathcal{N} = 2$ superstring theory on a Calabi-Yau threefold to obtain a d=4, $\mathcal{N} = 2$ super-gravity theory. There are $2h^{1,1} + 4(h^{2,1} + 1)$ massless scalar fields that occur in the theory which paramterize the moduli space $\mathcal{M}^{1,1} \times \mathcal{M}^{2,1}$. The massless field content of the theory before compactification is

$$\left\{ g_{MN},\ C_M^{(1)},\ \phi, C_{MNP}^{(3)},\ B_{MN},\ \psi_M^\alpha,\ \overline{\psi}_M^{\dot{\alpha}},\ \lambda_\alpha,\ \lambda_{\dot{\alpha}} \right\} \tag{129}$$

where $g_{\mu\nu}$ is the graviton, $C^{(n)}$'s are n-forms, B_{MN} is the 2-fom B-field, ψ_M^α and $\overline{\psi}_M^{\dot{\alpha}}$ are gravitinos of opposite chirality and λ_α are dilatinos. This field content is easily obtained by dimensional reduction of the M-theory fields $\{g_{MN}, C_{MNK}, \phi_M^\alpha\}$. Although we did not give the effective action or the four-dimensional field content obtained for the heterotic string we will give this field content for both type II theories. The reason for this is we can explicitly see how to relate the moduli of the internal Calabi-Yau to the moduli of the compactified four dimensional theory. The four-dimensional theory breaks up into the following multiplets:

Each multiplet is characterized by their highest helicity state. For instance the gravity multiplet contains the gravition $g_{\mu\nu}$ which has a helicity of (2). By acting with the supersymmetric operators we see that ψ^μ is of helicity $3/2$ and C_μ the lowest helicity state of 1 which is a vector. The vector

Multiplet	Multiplicity	Field content
Gravity	1	$g_{\mu\nu}$, ψ_μ, $\overline{\psi}_\mu$, C_μ
Vector	$h^{1,1}$	$C_{\mu i\bar{j}}$, $g_{i\bar{j}}$, $B_{i\bar{j}}$, + (fermions)
Hypermultiplet	$h^{2,1}$	$C_{ij\bar{k}}$, g_{ij}, + (fermions)
Universal hypermultiplet	1	C_{ijk}, Φ, $B_{\mu\nu}$, + (fermions)

Table 2. Field content for type IIA supergravity compactified on a Calabi-Yau threefold.

multiplet is characterized by a highest helicity of 1 and so on.

The ten-dimensional field content of chiral type IIB supergravity is

$$\left\{g_{MN},\ C^{(0)},\ \phi, C^{(2)}_{MN},\ B_{MN},\ C^{(4)}_{MNKL},\ \psi^{\alpha}_{M},\ \psi^{\alpha}_{M},\ \lambda_{\alpha},\ \lambda_{\dot{\alpha}}\right\} \tag{130}$$

where we now have two different types of dilatinos ϕ and $C^{(0)}$, ψ^{α}_{M} and $\overline{\psi}_{M}$ are gravitinos of the same chirality. $C^{(4)}$ is a self-dual 4-form. The four dimensional theory decomposes into similar multiplets:

Multiplet	Multiplicity	Field content
Gravity	1	$g_{\mu\nu}$, ψ_μ, $\overline{\psi}_\mu$, C
Vector	$h^{2,1}$	$C_{\mu i\bar{j}\bar{k}}$, g_{ij}, + (fermions)
Hypermultiplet	$h^{1,1}$	$C_{\mu\nu i\bar{j}}$, $g_{i\bar{j}}$,$B_{i\bar{j}}$, $C_{i\bar{j}}$, + (fermions)
Universal hypermultiplet	1	ϕ, $C^{(0)}$, $C_{\mu\nu}$, $B_{\mu\nu}$, + (fermions)

Table 3. Field content for type IIB supergravity compactified on a Calabi-Yau threefold.

Where the total number of scalar fields is $2h^{2,1} + 4(h^{1,1} + 1)$. At the level of supergravity theory we see that mirror symmetry between type IIA and IIB is manifest since the theories have switched Kahler and complex moduli.

4 Conformal Field Theory

The purpose of this section is to give a conceptual overview of the subject without going into the details. Choosing to omit such detials allows us to state many of the important traits that CFT's posses. In particular there is an intamiate relationship between the gemoetry of the theory and the CFT it admits. Our analysis will not contain much rigor and the reader interested in the details omitted here are referred to [10]

In short conformal field theory is the field theory built from representations of an 'extended' Poincaré group. The two extra coordinate transformations added are defined as Dilations and SuperConformal transformations (SCT) and are trasnformations which preserve the geometry of the space. To understand the relationship between CFT and geometry we state the following definition:

Definition 13. *A smooth map $f : M \to N$ between Riemannian manifolds (M, g) and (N, g') is a Conformal Transformation (CT) if*

$$f^* g'_{f(p)} = e^{\sigma(p)} g_p \tag{131}$$

for $p \in M$ and some function $\sigma : M \to \mathbb{R}$ which is determined by f.

This transformation will add extra symmetries to the theory. We will restrict ourselves to the case in which g is Lorentzian. Let $M = \mathbb{R}^{p,q}$, that is $dim(M) = d = p + q$ and $\eta_{\mu\nu} = diag(-1, -1, \cdots, -1, +1, \cdots, +1)$ be the diagonalized (p,q) metric on M, and consider a CT. Then under this transformation the metric transforms as

$$\eta'_{\mu\nu}(p') = e^{\sigma(p)} \eta_{\mu\nu}(p) = \frac{\partial x'^{\alpha}}{\partial x^{\mu}} \frac{\partial x'^{\beta}}{\partial x^{\nu}} \eta_{\alpha\beta}(p) \tag{132}$$

for $x^{\mu} := \varphi^{\mu}(p)$. This expression states how we can deform the local geometry of the space. Under a general CT the angles between vectors are preserved.

Definition 14. *The Conformal Group $Conf(M)$ is the connected component of the set of CT's containing the identity.*

Although they will not (explicitly) be part of our analysis we state the following theorem to show the added symmetries obtained from a CT.

Theorem 15. *Conf(M) is generated by the following coordinate transformations*

$$x^{\mu} \mapsto x^{\mu} + c^{\mu}, \qquad c^{\mu} \in \mathbb{R}^d \quad \textit{(Translations)} \tag{133}$$

$$x^{\mu} \mapsto \Lambda^{\mu}_{\nu} x^{\nu} \qquad \Lambda^{\mu}_{\nu} \in SO(p, q) \quad \textit{(Rotations)} \tag{134}$$

$$x^{\mu} \mapsto \lambda x^{\mu} \qquad \lambda \in \mathbb{C} \quad \textit{(Dilations)} \tag{135}$$

$$x^{\mu} \mapsto \frac{x^{\mu} - b^{\mu} x^2}{1 - 2b \cdot x + b^2 x^2} \qquad b^{\mu}(p) \in \mathbb{R} \quad \textit{(SCT)} \tag{136}$$

The explicit form of b^μ will not be important to us.

The idea behind deriving such terms is simple: we consider the case in which the CT f is an infinitesimal variation $x^\mu \mapsto x^\mu + \epsilon^\mu(x)$ and plug it back into the transformation above. Constraints are then found on the possible types of infinitesimals one can consider.

4.1 Primary Fields and Operator Product Expansion

Let $\Phi(x)$ be a a field in our theory. Then under CT's this field transforms actively and under a dilation we find $\Phi(x) \mapsto \Phi(x') = \lambda^{-\triangle}\Phi(\lambda^{-1}x)$ where $\triangle_a$ is called the scaling dimension of the field. A quick look at any familiar field theory action shows that the scaling dimension is the mass-scale dimension we associate fields and their quantized counterparts via dimensional analysis. Let s be the spin eigenvalue associated to the action of the rotational generators in the eigenstate $|\Phi(x)\rangle$[8], then the holomorphic and antiholomorphic *Conformal dimension* of the field $\Phi(x)$ are defined as $h := \frac{1}{2}(\triangle + s)$, $\overline{h} := \frac{1}{2}(\triangle - s)$ respectively and are often denoted by the pair $(h, \overline{h})$. These will play an important role in our understanding of $\mathcal{N} = (2, 2)$ superconformal theories.

We now define complex coordinates $z := x^1 + ix^2$ and $\overline{z} := x^1 - ix^2$ and analytically continue our field onto the complex plane. Let $w : z \mapsto w(z)$ and $\overline{w} : z \mapsto \overline{w}(z)$ be holomorphic and anitholomorphic maps. A field which transforms as

$$\Phi'(w, \overline{w}) = \left(\frac{dw}{dz}\right)^{-h} \left(\frac{d\overline{w}}{d\overline{z}}\right)^{-\overline{h}} \Phi(z, \overline{z}) \tag{137}$$

is called a *primary field of conformal weight* $(h, \overline{h})$. We will find most of the fields in our theory are primary fields. The most important is the energy momentum tensor

$$T_{\mu\nu} = \frac{1}{\sqrt{g}} \frac{\delta S}{\delta g^{\mu\nu}} \tag{138}$$

where S is the action of the theory and g the metric of the spacetime. We find it to be symmetric[9] and (in the present case) traceless $T^\mu_\mu = 0$. It is also of course conserved $\partial_\mu T^{\mu\nu} = 0$. Once expanded in complex coordinates we find $\partial_{\overline{z}} T_{zz} = 0$, $\partial_z T_{\overline{z}\overline{z}} = 0$ and the only non-zero components are given by T_{zz} and $T_{\overline{z}\overline{z}}$. The vanishing (anti-)holomorphic derivatives of T show that it splits into a holomorphic and anti-holomorphic part so we define

$$T(z) := T_{zz} \quad \overline{T}(\overline{z}) := T_{\overline{z}\overline{z}}. \tag{139}$$

Using a series of relations known as the Ward identities we obtain expressions for the so called Operator Product Expansions (OPE) which encode the all the information of the primary fields of

[8] We will define this state more precisely in a later section

[9] It is possible to have an extra antisymmetric 'B' term but we will not consider it here.

our theory. Obtaining these results are straight forward and we refer the reader again to [10] to see how these are obtained. The basic idea is after we've quantizated our theory we calculate a vacuum expectation value of a *radial*[10] ordered product of fields which are varied by a symmetry of the theory. After taking some derivatives and contour integrals one finds an expression for the OPE for a field $\Phi(w, \overline{w})$ of conformal weight $(h, \overline{h})$ with $N = 0$ supersymmetry is given by

$$T(z)\Phi(w, \overline{w}) = \cdots + \frac{h\Phi(w, \overline{w})}{(z-w)^2} + \frac{\partial\Phi(w, \overline{w})}{(z-w)} + \cdots \tag{140}$$

$$\overline{T}(\overline{z})\Phi(w, \overline{w}) = \cdots + \frac{\overline{h}\Phi(w, \overline{w})}{(\overline{z}-\overline{w})^2} + \frac{\partial\Phi(w, \overline{w})}{(\overline{z}-\overline{w})} + \cdots . \tag{141}$$

Where the first set of dots represent Laurent expansion of the negative integers past -2 and the second the Laurent expansion of regular polynomials. We see that the (anti)holomorphic energy momentum tensor yields a term which is directly proportional to the field multiplied by its (anti)holomorphic dimension. OPE's take two operators defined at nearby points and approximation their short distance behavior by a series of operators at a single one of these points. For primary fields we find the lowest order term in the Laurent expansion is given by $(z - w)^{-2}$. In the present text we will denote the OPE of primary fields as

$$T(z)\Phi(w, \overline{w}) \sim \frac{h\Phi(w, \overline{w})}{(z-w)^2} + \frac{\partial\Phi(w, \overline{w})}{(z-w)} \tag{142}$$

which simply means equivalence up to the addition of a regular complex function which will not be relevant. It should be noted that when we write and OPE what we really mean is the vacuum expectation value of the radially ordered product of the fields in the expansion. Further, the above fields are the quantized operators on the Hilbert space of states of our theory and we will suppress the operator hat notation throughout.

There's a natural way to identify the conformal dimension of a given tensor field. The holomorphic energy momentum tensor $T(z)$ has conformal weight $(2, 0)$ while the antiholomorphic counterpart has a conformal weight of $(0, 2)$. A tensor which has p-holomophic and q-antiholomorphic indicies will have a conformal weight (p, q). As defined, the value of the conformal dimensions need not be an integer and in such cases we find the underlying field to be a section of some spin-bundle product. We will have more to say about this when we look at the field content of $\mathcal{N} = (2, 2)$ supersymmetric sigma models.

[10] We will explain this shortly.

4.2 Conformal Compactification of $\mathbb{R}^{2,0}$

When looking at CFT's in 2 dimensions (CFT_2) we find that our quantum theories yield projective unitary representations of $Conf(\mathbb{R}^{2,0})$. When studying the causal structure of black holes we often extend four dimensional Minkowski space $\mathbb{M}^4$ by adding a point at infinity which allows us to compactify the spacetime. As the same suggests, we do the same for our CFT by compactifying $\mathbb{M}^2$. For coordinates (σ^0, σ^1) on $\mathbb{M}^2$ we wick rotate into $\mathbb{R}^2 \cong \mathbb{C}$ with coordinates $u := \sigma^0 + i\sigma^1$ and $\overline{u} := \sigma^0 - i\sigma^1$. We are really considering the space $\bar{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ which is the Riemann two sphere. The addition of this point at infinity will lead to extremely profound results as known from the study of black holes. When adding infinity to $\mathbb{M}^2$ one should think of the infinite past and future of a future (past) directed world line of a photon as being the same infinity. This identification leads to map $\mathbb{R}^2 \to \mathbb{R} \times S^1$ where S^1 is a sphere of radius 1 with coordinate $\sigma^1 \in [0, 2\pi)$ which is periodically identified and $\sigma^0 \in (-\infty, \infty)$ is the coordinate along the cylinder. Now, the interesting configuration comes when we now map the cylinder back into the complex plane using the coordinates

$$z := e^{\sigma^0 + i\sigma^1}, \quad \overline{z} := e^{\sigma^0 - i\sigma^1}. \tag{143}$$

Let $\alpha, \theta \in \mathbb{R}$ with $\theta \in [0, 2\pi)$. Then we find following points of interest

$$(\sigma^0 = \infty, \sigma^1 = \theta) \quad \mapsto \quad (z = \infty, \overline{z} = \infty) \tag{144}$$

$$(\sigma^0 = -\infty, \sigma^1 = \theta) \quad \mapsto \quad (z = 0, \overline{z} = 0) \tag{145}$$

$$(\sigma^0 = \alpha, \sigma^1 = \theta) \quad \mapsto \quad (z = e^{\alpha} e^{i\theta}, \overline{z} = e^{\alpha} e^{-i\theta}) \tag{146}$$

The infinite future gets mapped to the 'usual' infinite that we think of when doing real analysis on $\mathbb{R}^2$ while the infinite past gets mapped to the origin. Furthermore, constant slices along the cylinder get mapped to circles of constant radius on the new compactified complex plane. We think of time as a a radial coordinate on the plane while space is our angular coordinate. When we build irreducible representations of the conformal group the compactified spacetime will be advantageous as defining in states amounts to finding the value of those states at the origin. Likewise evolution of our operators will be determined by it's initial value at the origin. Under a time translation $\sigma^0 \mapsto \sigma^0 + a$ we obtain a dilation $z \mapsto e^a z$ while time reversal $\sigma^0 \mapsto -\sigma^0$ yields $z \mapsto \frac{1}{\overline{z}}$.

4.3 The Conformal Algebra

At the level of fields we can derive an expression for the generators of the superconformal transformations as a local algebra. Under the coordinate variation $z \mapsto z' = z + \epsilon(z)$ where $\epsilon(z)$ is an infinitesimal meromorphic function (it has isolated singularities outside of the domain of interest). A primary field then undergoes the following variation

$$\Phi(z,\bar{z}) \mapsto \Phi'(z',\bar{z}') = \Phi(z,\bar{z}) + \epsilon(z)\partial_z\Phi(z,\bar{z}) + \bar{\epsilon}(\bar{z})\partial_{\bar{z}}\Phi(z,\bar{z})$$

since ϵ is assumed to be meromorphic we can expand it as $\epsilon(z) = \sum_{n\in\mathbb{Z}} \epsilon_n z^{n+1}$ which yields

$$\delta\Phi = \sum_{n\in\mathbb{Z}} \left(\epsilon_n z^{n+1}\partial_z\Phi(z,\bar{z}) + \bar{\epsilon}_n\bar{z}^{n+1}\partial_{\bar{z}}\Phi(z,\bar{z})\right) := \sum_{n\in\mathbb{Z}} \left(\epsilon_n\ell_n\Phi(z,\bar{z}) + \bar{\epsilon}_n\bar{\ell}_n\Phi(z,\bar{z})\right) \tag{147}$$

where we have introduced the generators of the infinite dimensional Witt Algebra $\ell_n := z^{n+1}\partial_z$ and $\bar{\ell}_n := \bar{z}^{n+1}\partial_{\bar{z}}$. A simple application of derivatives shows that

$$[\ell_n,\ell_m] = (n-m)\ell_{n+m}, \quad [\bar{\ell}_n,\bar{\ell}_m] = (n-m)\bar{\ell}_{n+m} \quad [\ell_n,\bar{\ell}_m] = 0 \tag{148}$$

The subset of operators $\{\ell_{-1},\ell_0,\ell_1\}\cup\{c.c.\}$ generate the global $Conf(M)$ transformations mentioned previously. Moreover $\{\ell_{-1},\ell_0,\ell_1\}$ generate an $SL(2,\mathbb{C})$ subgroup which turns out to be deeply connected to the conformal structure of theory. to see this we note that the map $f(z) := z + \epsilon(z)$ is of course a conformal transformation given that $ds^2 = dz\otimes d\bar{z} \to \frac{\partial f}{\partial z}\frac{\partial\bar{f}}{\partial\bar{z}}dz\otimes d\bar{z}$ with the conformal factor $\Lambda(z) := |\frac{\partial f}{\partial z}|^2$. This means that f can only have simple poles and it turns out the most general form of f is given by $f = \frac{az+b}{cz+b}$ and for the inverse to exist we require $ad-bc \neq 0$. In the case $ad-bc = 1$ we see this is nothing more than the action of an element of $SL(2,\mathbb{C})$ which is well known to generate the connected component of the projective group $PSL(2,\mathbb{C})$. The group of conformal transformations on $\mathbb{C}\cup\{\infty\}$ is called the special conformal group and we find a one-to-one correspondence between the conformal group generated by maps f and $SL(2,\mathbb{C})$. We conclude

$$Conf(\mathbb{C}\cup\{\infty\}) \cong SL(2,\mathbb{C}).$$

Upon quantization the Witt algebra is extended to include terms which are associated with anomalies that arise. This extended algebra is called the Virasorro Algebra and is given by

$$[L_m,L_n] = (m-n)L_{m+n} + \frac{c}{12}m(m^2-1)\delta_{m+n,0} \tag{149}$$

$$[\bar{L}_m,\bar{L}_n] = (m-n)\bar{L}_{m+n} + \frac{c}{12}m(m^2-1)\delta_{m+n,0} \tag{150}$$

$$[L_m,\bar{L}_n] = 0 \tag{151}$$

where c is called the central extension of the Witt algebra and it's value characterizes the anomollies of the theory. To understand where this comes from one expands the quantized holomorphic and aniholomorphic energy momentum tensor as

$$T(z) = \sum_{n\in\mathbb{Z}} z^{-n-2}L_n \implies L_n = \frac{1}{2\pi i}\oint dz T(z)z^{n+1} \tag{152}$$

and then by using the OPE of T

$$T(z)T(w) \sim \frac{(c/2)}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial_w T(w)}{z-w} \tag{153}$$

we obtain the Virasorro commutation relations above. Notice that the power of the expansion has a factor of -2 which corresponds to the conformal dimension of T. This is a general feature. Let ϕ be a holomorphic primary field of conformal dimension h then we have the following expansion

$$\phi(z) = \sum_{n\in\mathbb{Z}} z^{-n-h}\phi_n. \tag{154}$$

4.4 Hilbert Space of CFT's

One of the greatest utilities of the CFT formalism is the Operator-State correspondence which allows us to look at operators defined in a neighborhood of the origin. As previously mentioned the CFT operators are radially quantized so that the origin $z = 0$ of the conformally compactified complex plane is thought of as the infinite past. We will now see that the Hilbert space of intereset is built from highest representations of the conformal group generated by the Virasorro algebra. The vacuum state $|0\rangle$ as used in this correspondence is refered to as the conformal vacuum. Let $\Phi(z,\bar{z})$ be a quantized primary field of the theory. Then the operator state correspondence simply states

$$|\Phi\rangle := \lim_{z,\bar{z}\to 0} \Phi(z,\bar{z})|0\rangle \tag{155}$$

which we now use to analyze the underlying Hilbert space of the theory. Our Hilbert space will be built from highest representations of the conformal group which we The vacuum state of interest must be $SL(2,\mathbb{C})$ invariant to ensure that they are globally conformally invariant. That is, they are invariant under the sub-algebra generated by $\{L_0, L_{-1}, L_1\}$. By the operator state correspondence this amounts to ensuring that $T(z)$ and $\overline{T}(\bar{z})$ are regular functions at the origin

$$T(z)|0\rangle = \sum_{n\in\mathbb{Z}} z^{-n-2}L_n|0\rangle \implies L_n|0\rangle = 0, \forall n \geq -1$$

and likewise for $\overline{T}$. This constraint ensures the vacuum is invariant under the desired subgroup. Using the expression for L_n above and the OPE of a primary field we find[11]

$$[L_n, \Phi(w,\overline{w})] = \oint \frac{dz}{2\pi i} T(z)\Phi(w,\overline{w}) = \frac{1}{2\pi i}\oint dz\, z^{n+1}\left(\frac{h\Phi(w,\overline{w})}{(z-w)^2} + \frac{\partial_w \Phi(w,\overline{w})}{z-w} + (reg)\right) \tag{156}$$

$$= h(n+1)w^n\Phi(w,\overline{w}) + w^{n+1}\partial_w\Phi(w,\overline{w}) \tag{157}$$

in particular for $n = 0$ this gives $h\Phi(w,\overline{w})$ which shows that the primary field is an eigenstate of L_0 ($\overline{L}_0$) with eigenvalue h ($\overline{h}$). By the operator state correspondence we label these states as $|h,\overline{h}\rangle := \Phi(0,0)|0\rangle$ for which $L_0|h,\overline{h}\rangle = h|h,\overline{h}\rangle$. It is possible to decompose these states into $|h,\overline{h}\rangle \approx |h\rangle \otimes |\overline{h}\rangle$ These are the highest weight states of the Hilbert space which we conclude decomposes as

$$\mathcal{H} = \bigoplus_{h,\overline{h}} N_{h,\overline{h}} Vir(h) \otimes \overline{Vir}(\overline{h})$$

where $N_{h,\overline{h}} \in \mathbb{Z}$ keeps track of degeneracies and $Vir(h)$ is the Verma module which is composed of holomorphic representations of the Virasoro albegra generated by acting on the highest weight state $\Phi_{-h}|0\rangle = |\Phi\rangle$

$$Vir(h) = \left\{ |\Phi\rangle, L_{-n_1}L_{-n_2}\cdots L_{-n_k}|\Phi\rangle \;\middle|\; \sum_{i=1}^{k} n_i = n, n_i \geq 0 \forall i \right\}$$

the elements of this group are called the descendants of the highest weight state.

4.5 D-branes & Classical Strings

D_p branes are $(p+1)$ dimensional hypersurfacse that open strings can end on. Branes can be found in both the bosonic string and superstring formalisms. The dualities relating the different string theories has a simple interpretation in terms of branes, the CFT/AdS correspondence is a duality between D-branes in open and closed string theories. D-branes are almost even more fundamental to string theory than strings in that they are an extremely powerful 'computational' tool. Roughly speaking, we can think of D-branes as arising from the boundary conditions that are imposed on the worldsheet. Without loss of generality consider the case in which the real axis is a boundary and consider a conformal trasnformation $z \mapsto z + \epsilon_m z^m$. If we wish to keep the theory conformal them we must impose $\epsilon_m \in \mathbb{R}$ which restricts both the holomorphic and antiholomorphic decomposition to the condition $\epsilon_m = \overline{\epsilon}_m$ so we loose half of the symmetries. Along the real axis we must further

[11] The reader unfamiliar with CFT may be confused as to why there is only a single integral on the right hand side of the first equality since we are considering a commutator. The reason for this is we in fact do consider the difference of two contour integrals however we choose these two contours and their orientation in such a way that their difference is a closed contour around the point w. See [10] chp. 6.

impose the condition

$$T(z) = \overline{T}(\overline{z}), \quad \text{on} \qquad \mathfrak{R}(z). \tag{158}$$

In this way it is now possible to simply work with $T(z)$ on $\mathfrak{I}(z) \geq 0$ and define $\overline{T}(\overline{z})$ on $\mathfrak{I}(z) \leq 0$ by the method of images.

Let us now consider the two dimensional sigma model for a bosonic field ϕ which admits the action

$$S = \frac{1}{2}\int d^2\xi \sqrt{-\eta}\eta^{\mu\nu}\partial_\mu\phi\partial_\nu\phi = \int d^2z\partial\phi\overline{\partial}\phi \tag{159}$$

where the first expression is given for any two-dimensional coordinate system and the latter for the complex plane. The equations of motion for the scalar field (in complex coordinates) are simply

$$\partial\overline{\partial}\phi = 0 \tag{160}$$

while the energy momentum tensor splits as

$$T_{\mu\nu} = \frac{1}{\sqrt{-\eta}}\frac{\delta S}{\delta h^{\mu\nu}} = \partial_\mu\phi\partial_\nu\phi - \frac{1}{2}\eta_{\mu\nu}\eta^{\lambda\sigma}\partial_\lambda\phi\partial_\sigma\phi \tag{161}$$

$$T(z) = \frac{1}{2}\partial\phi\partial\phi, \qquad \overline{T}(\overline{z}) = \frac{1}{2}\overline{\partial}\phi\overline{\partial}\phi. \tag{162}$$

Along the real axis we therefore must have the following

$$\partial\phi\partial\phi = \overline{\partial}\phi\overline{\partial}\phi \implies (\partial\phi)^2 - (\overline{\partial}\phi)^2 = 0 \tag{163}$$

$$(\partial\phi - \overline{\partial}\phi)(\partial\phi - \overline{\partial}\phi) = 0 \tag{164}$$

and so there are two conditions that we obtain from considering boundary conditions.

- $(\partial\phi) = (\overline{\partial}\phi)$: These boundary conditions are called Neumann boundary conditions and to better understand what this condition means we can rewrite this expression in terms of real coordinates $z = x + iy$. In doing so we find

$$n^y\partial_y\phi\,\big|_{y=0} = 0 \quad \text{(Neumann)} \tag{165}$$

  where n^y is the unit normal to the x axis ($\hat{y}$). The reason for writing the expression in this way is that it allows us to easily consider arbitrary boundary conditions. For Neumann boundary conditions there is no momentum flowing out of the ends of the string so we can think of these conditions as allowing the string to end anywhere in spacetime.

- $(\partial\phi) = -(\overline{\partial}\phi)$: These are Dirchlet boundary conditions. Which are written as

$$\partial_x\phi|_{y=0} \implies \phi = c, \quad c \in \mathbb{R} \quad \text{(Dirichlet)} \tag{166}$$

These boundary conditions state that along the boundary the fields must be constant. This amounts to keeping the ends of string fixed in spacetime. D-branes arise from considering Dirchlet boundary conditions as they specify a surface in which the string can end on.

More generally we can consider the case in which we have $\phi : \Sigma \to \mathbb{R}^N$ so that we have several scalar fields ϕ^I, $I = \{1, \cdots, N\}$. We can consider having mixed boundary conditions

$$\text{(Neumann)} \qquad n^\mu \partial_\mu \phi\,|_{\partial\Sigma} = 0, \quad \mu = 1, \cdots, k \tag{167}$$

$$\text{(Dirichlet)} \qquad \phi^I = c^I, \quad I = k+1, \cdots N \tag{168}$$

the boundary of the worldsheet gets mapped to a $(N-k)$-dimensional submanifold U_p which we call an $(N-k)-$brane[12]. Common terminology is to say that a D_p brane wraps U^p if the boundary of a Riemann surface is mapped to U_p with imposed Neaumann boundary conditions for directions normal to U_p.

The same analysis can be done for the free fermion showing that the worldsheet supersymmetry gives us restrictions on boundary conditions. If we consider the full sigma model action these boundary conditions only preserve half of the overall supersymmetries.

4.6 The N=(2,2) Superconformal algebra

When ten dimensional string theories are compactified on Calabi-Yau threefolds they admit a two dimensional CFT which is extended to include supersymmetric partners. This superconformal algebra will underpin the nonlinear models considered in this thesis. As such we explain aspects of this theory following [**?**]. $\mathcal{N} = 2$ Superconformal algebra (SCA) is generated by four sets of generators

$$T(z) = \sum_{n\in\mathbb{Z}} L_n z^{-n-2}, \quad G^\pm(z) = \sum_{n\in\mathbb{Z}} G_{n+a} z^{-(n\pm a)-3/2}, \quad J(z) = \sum_{n\in\mathbb{Z}} J_n z^{-n-1} \tag{169}$$

and their conjugates, where $0 \le a \le 1$ and corresponds to the boundary conditions of the fermionic fields. Under a phase transformation a can be explicitly defined by $G^\pm(e^{2\pi i} z) = e^{\mp 2\pi i a} G^\pm(z)$. Anti-periodic boundary condition $a \in \mathbb{Z}$ are called Ramond boundary conditions and Periodic boundary condition s $a \in \frac{1}{2}\mathbb{Z}$ are called Naveu-Schwarz (NS) boundary conditions. The operators T, $\overline{T}$ are the holomorphic and antiholomorphic energy momentum tensors with $\hat{c} = c/3$ and G, $\overline{G}$ their respective superpartners. J and $\overline{J}$ are the generators of the $U(1)$ symmetry of the theory. When considering representations of this algebra the weights of the $U(1)$ operator J are often written as Q and correspond to the Lorentz charge of the state being acted on. These operators split in the following way

[12] In the case of nonlinear sigma models it is natural to think of branes as a vector bundle $E \to U_p$ with a connection ∇_E.

Left Moving: T, G^+, G^-, J
Right Moving: $\overline{T}, \overline{G^+}, \overline{G^-}, \overline{J}$

this is due to the fact that the spin states split into positive and negative chirality. Since all the fields split into holomorphic and antiholomorphic parts in the liturature it is common to describe fields by the pair[13] (h, Q). The OPE's of these operators are given by

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial_w T(w)}{(z-w)} \tag{170}$$

$$T(z)G^{\pm}(w) \sim +\frac{(3/2)G^{\pm}(w)}{(z-w)^2} + \frac{\partial_w G^{\pm}(w)}{(z-w)} \tag{171}$$

$$T(z)J(w) \sim +\frac{J(w)}{(z-w)^2} + \frac{\partial_w J(z)}{(z-w)} \tag{172}$$

$$G^+(z)G^-(w) \sim +\frac{(2c/3)}{(z-w)^3} + \frac{2T(w) + \partial_w J(w)}{(z-w)} \tag{173}$$

$$J(w)G^{\pm}(w) \sim \pm\frac{G^{\pm}}{(z-w)} \tag{174}$$

$$J(z)J(w) \sim \frac{(c/3)}{(z-w)^2} \tag{175}$$

and from them we could see that the theory contains anomolies which arise from the $U(1)$ current J. As we shall see these anomolies come from the so called R-symmetries of the topological sigma models. If we expand each field using their operator expansion and take the approriate commutators we obtain the $N = 2$ superconformal algebra

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}m(m^2-1)\delta_{m+n,0} \tag{176}$$

$$[J_m, J_n] = \frac{c}{3}m\delta_{m+n,0} \tag{177}$$

$$[L_n, J_m] = -mJ_{m+n} \tag{178}$$

$$[L_m, G^{\pm}_{n\pm a}] = \left(\frac{n}{2} - (m \pm a)\right)G^{\pm}_{m-n\pm a} \tag{179}$$

$$[J_m, G^{\pm}_{n\pm a}] = \pm G^{\pm}_{m-n\pm a} \tag{180}$$

$$[G^{\pm}_{n+a}, \bar{G}^{\pm}_{m-a}] = 2L_{m+n} + \frac{c}{12} + (n-m+2a)J_{n+m} + \frac{c}{3}\left((n+a)^2 - \frac{1}{4}\right)\delta_{m+n,0} \tag{181}$$

[13] It should be clear from context when we write out terms such as (2,0) we are not talking about the conformal weight of a general primary field.

Part III

Topological String Theory

In this section we present the basics of topological field theories. We will see there are two main types topological theories which are characterized by different properties. The first is Chern-Simons theory which can be used calculate topological invariants of three manifolds and knots. The second are cohomological field theories which span a greater class of theories. We then proceed to define two dimensional sigma models and introduce the concept of topological twisting, making them cohomological. These theories will be called the A and B model and we will focus on the A-model. In the last section we couple these twisted theories to gravity to obtain a formulation for topological string theory

5 Topological Field Theories

We begin by stating the basics of his Atiyah's definition of Topological field theories which we use to motivate the topological field theories we construct in this thesis.

Definition 16. *A Topological Quantum Field Theory (TQFT) associates to each m-dimensional manifold M with boundaries Σ_i a Hilbert space $\mathcal{H}(\Sigma_i) := \mathcal{H}_{\Sigma_i}$ with $\mathcal{H}(M) \in \mathcal{H}(\partial M)$ and to each 'cobordism' $\partial\tilde{M} = \Sigma_i \sqcup \Sigma_j$ a morphism $\Phi(\tilde{M}) : \mathcal{H}_{\Sigma_i} \to \mathcal{H}_{\Sigma_j}$. Each Hilbert space is subject to the following axioms*

1. $$H(\Sigma_i^*) \cong \mathcal{H}(\Sigma_i)^* \tag{182}$$

   where Σ_i^* is Σ_i defined with the opposite orientation and $\mathcal{H}(\Sigma_i)^*$ is the dual space of $\mathcal{H}(\Sigma_i)$. Likewise we have

   $$\Phi(M^*) = \Phi(M)^* : \mathcal{H}^*_{\Sigma_i} \to \mathcal{H}^*_{\Sigma_j}. \tag{183}$$

2. *$\mathcal{H}(\Sigma_i \sqcup \Sigma_j) = \mathcal{H}_{\Sigma_i} \otimes \mathcal{H}_{\Sigma_j}$ and $\mathcal{H}(\Sigma_i^* \sqcup \Sigma_j) = \mathcal{H}^*_{\Sigma_i} \otimes \mathcal{H}_{\Sigma_j} \cong Hom(\mathcal{H}_{\Sigma_i}, \mathcal{H}_{\Sigma_j})$.*

3. *If $\partial M_1 = \Sigma_1 \sqcup \Sigma_2$ with $\Phi(M) : \mathcal{H}_{\Sigma_1} \to \mathcal{H}_{\Sigma_2}$ and $\partial M_2 = \Sigma_2^* \sqcup \Sigma_3$ with $\Phi(M_2) : \mathcal{H}^*_{\Sigma_2} \to \mathcal{H}_{\Sigma_3}$ for*

$M = M_1 \cup_{\Sigma_2} M_2$ *then*

$$\Phi(M) = \Phi(M_1) \otimes \Phi(M_2) : \mathcal{H}_{\Sigma_1} \otimes \mathcal{H}_{\Sigma_2} \otimes \mathcal{H}^*_{\Sigma_2} \otimes \mathcal{H}_{\Sigma_3} \to \mathcal{H}_{\Sigma_1} \otimes \mathcal{H}_{\Sigma_3} \tag{184}$$

4. $\mathcal{H}(\emptyset) := \mathbb{C}$ *for which* $\Phi(\emptyset) : \mathbb{C} \to \mathbb{C}$.

The morphism Φ presented here is often written as Z but we refrain from using this common notation as we reserve this symbol for the partition function. There are two types of major TQFTs. These are called topological theories of Schwarz type and Witten type. In either case these theories are naturally paired with surgery theory. This is the study of cutting and gluing together topological surfaces in some homeomorphic manner. The view point allows us better understand the classification of the theory and will play an important role in determining topological invariants of the underlying manifold. Here we will only present ideas which are sufficient for use in this thesis. For an excellent review of these ideas see [8].

The idea is now to consider some manifold M and think of 'evolution' from the left to right hand side of the object. For a closed topological surface the morphism Φ is given by the partition function of the theory $\Phi = Z : \mathbb{C} \to \mathbb{C}$ by property 4. This is written as

$$Z[\phi] = \int \mathcal{D}\phi e^{-S[\phi]}. \tag{185}$$

We can make better sense of this identification by splitting the manifold in half $M = M_1 \sqcup M_2$ with each end having opposite orientations. M_1 now has a closed end on the left and an open suface Σ where it was cut on the right. From properties 3. we find the morphism on M_1 is a map $\Phi_1 : \mathbb{C} \to \mathcal{H}_\Sigma$ which we identify as the 'ket' vector

$$\Phi_1(M_1) = |M_1\rangle \in \mathcal{H}_\Sigma. \tag{186}$$

As one might have imagined M_2 gives the 'bra' vector by property 1. and 3.:

$$\Phi_2 = \langle M_2| \in \mathcal{H}^*_\Sigma. \tag{187}$$

We can now use property 4. to glue these two surfaces back together to obtain

$$\Phi(M) = \Phi_1\Phi_2 = \langle M_1||M_2\rangle = Z. \tag{188}$$

Amazing. Now, if we consider $M = \Sigma \times [0, 1]$ the cylinder with the time evolution parameter $t \in [0, 1]$ then the morphism is a map $\phi_M : \mathcal{H}_\Sigma \to \mathcal{H}_\Sigma$. This is the ever so familiar Hamiltonian time

evolution operator in the Heisenberg picture

$$\Phi = e^{tH} \tag{189}$$

Note that if we glue the cylinder together we obtain a closed surface. In this case the map is simply the trace of the evolution operator

$$\Phi = Tr(e^{-tH}). \tag{190}$$

In Chern-Simons theory this splitting will play an important role in understanding the invariants we can obtain from the gauge theory. When we introduce the topological cohomology theories we will not give an an abstract decomposition of the manifold via surgery theory and instead withhold this discussion for the case of the topological sigma models.

5.1 Chern-Simons Theory

Chern-Simons theory is unarguably the most well known Schwarz type topological field theory. In the mid to late 80's Chern-Simons theory was a heavily studyied subject as many suprising relationships were unveiled through it. To name a few, in [31] Witten constructed a topological formulation of lower dimensional gravity in terms of Chern-Simons theory. It can be found can be used to determine topological invariants and in particular knot invariants which also dates back to Witten in [29]. There are many other uses of the Chern-Simons formulation and in the present we will focus on giving a brief outline of how we can obtain topological data from it. In a later section we will formulate a non-commutative version of the Chern-Simons action contructed by Witten to understand dualities between topological string theories. All the information presented here can be found in [24], [29], [3].

Let M be a manifold with $dim_{\mathbb{R}}(X) = 3$ and $E \to M$ be a principal G-bundle. Let $A \in \Omega^1(M, \mathfrak{g})$ be a connection on E. Let $\mathcal{G} := \{g : M \to G\}$ be the gauge group of E and $\mathcal{A}/\mathcal{G}$ the set of gauge equivalent connections on E. Then the Chern-Simons action is defined as the integral of the second Chern form of the Chern-character over X [3]

$$\begin{aligned} S_{CS}[A] &:= \frac{k}{4\pi}\int_M Tr\left(A \wedge dA + \frac{2}{3}A \wedge A \wedge A\right) \\ &= \frac{k}{8\pi}\int_M \varepsilon^{ijk} tr\left(A_i(\partial_j A_k - \partial_k A_j) + \frac{2}{3}A_i f_{jk}{}^l A_l\right) \end{aligned} \tag{191}$$

where $f_{ij}{}^k$ are the structure constants of G and k is called the level of the theory. The reason for considering such a term is apparent from its origin as Chern-polynomials are used to determine topological information of X. This action is invariant under gauge transformations of the form

$A \mapsto A + D\lambda$ which means that the action is independent of the local trivialization of the bundle. Given a symmetry of the theory we find the variation

$$\delta S_{CS} = \frac{k}{4\pi}\int_{\partial M} tr(\delta A \wedge A) + \frac{k}{2\pi}\int_{M} tr(\delta A \wedge F)$$

meaning we require the boundary conditions $\int_{\partial M} tr(\delta A \wedge A) = 0$. Requiring $\delta S_{CS} = 0$ we conclude $F = dA + A \wedge A = 0$ which are the equations of motion of A. Hence the classical solutions of Chern-Simons is characterized by the space of flat connections on E. If G is compact and simple then the homology group $\pi_3(G) = \mathbb{Z}$ which implies that $k \in \mathbb{Z}$. This comes from the fact that under a gauge transformation $A \mapsto g^{-1}Ag - g^{-1}dg$ the action contains a term purely in terms of g over the boundary

$$\int_{\partial M} tr(g^{-1}dg \wedge g^{-1}dg \wedge g^{-1}dg) \tag{192}$$

which depends on the mapping class homomorphism $\mu : \pi_3(M) \to \mathcal{G}$. The term above gives the degree of this mapping class which is an integer and a topological invariant. This is a well known result from the study of instantons and is associated to the 'winding number' of the theory. Physically the degree of the mapping class is simply the number of times a charged particle loops around a magnetic source. The path traced out by the particle determines a curve in spacetime and so the study of knots and Chern-Simons (from a physical perspective) it apparent: we use knots to describe the path a charged particle traces out around a magnetic source and the Chern coupling k determines its winding number. It is important to note that physical theories do not have to be given by the Chern-Simons action. In most cases they either contain or give rise to Chern like terms. With this in mind we now formalize these claims by quantizing the theory.

5.1.1 Quantization & knots

In the beautiful paper [29] Witten showed how to use quantized Chern-Simons gauge theory to determine topological information of three manifolds. Furthermore, using surgery theory he showed that the quantized theory was exactly solvable. This was done by considering knots embedded in S^3 and using the properties of topological field theories which we previously defined. The invariants obtained were related the well known Jones polynomials and their generalized HOMFLY polynomials. Our physical interpretation motivates the study of embedding knots into the manifold. Knots tell us about the three manifold they are embedded in since two knots are, roughly speaking, equivalent if their compliment in M are homeomorphic. In this section we will not give a detailed analysis as this would cause too much of a digression. Instead we state some of the general ideas so that we may use them when we consider topological string theories. See [3] for a physically motivated introduction to knot theory.

A knot $\mathcal{K}$ is a submanifold of $\mathbb{R}^3$ of codimension 2 diffeomorphic to S^1. Let $\gamma : [0, 1] \to \mathbb{R}^3$ be

a loop in $\mathbb{R}$ such that $Im(\gamma) = \mathcal{K}$. A link $L = \bigsqcup_i S^1_i$ is a disjoint union of knots. We will interchably use γ_i for a knot $\mathcal{K}_i$ and γ for a link L for $\gamma = \sqcup_i \gamma_i$. The circle, S^1 is called the unknot. As one would expect two knots are equivalent if they are homotopic, that is, if you can deform one into the other without cutting the knot. In knot theory such a homotopy is called an ambient isotopy. A common way to classify knots is to project them onto a two-dimensional subspace of $\mathbb{R}^3$. Once the projection is made there are a set of 'rules' one must follow to manipulate the knot, these are called Reidemeister moves. To obtain knot invariants, values which are unique up to ambient isotopies, one must place a framing on the knot (or link). A framing is a vector field on $\mathcal{K}$ which is no where tangent to $\mathcal{K}$. The importance of framing in the quantization of Chern-Simons is that the partition function is dependent of the framing of a knot and the tanget space of the three manifold. The partition function of the theory is given by

$$Z_0[A] = \int \mathcal{D}[A] \exp\left[-\frac{k}{4\pi}\int_M Tr\left(A \wedge dA + \frac{2}{3}A \wedge A \wedge A\right)\right] \tag{193}$$

Topological invariants of a link γ are obtained from the correlator of Wilson loops [29]

$$\langle W(\gamma, A)\rangle = \frac{1}{Z_0}\int_{\mathcal{A}/\mathcal{G}} \mathcal{D}\mathcal{A} \exp\left[-\frac{k}{4\pi}\int_M Tr\left(A \wedge dA + \frac{2}{3}A \wedge A \wedge A\right)\right] \prod_{i=1}^{N} W_{\mathcal{R}_i}(\gamma_i, A) \tag{194}$$

where $\gamma = \sqcup_i \gamma_i$ and each Wilson loop over γ_i ($\mathcal{K}_i$) in some link $L = \gamma$ in the appropriate representation $\mathcal{R}_i$ of $U(N)$:

$$W_{\mathcal{R}_i}(\gamma_i, A) = Tr_{\mathcal{R}_i}\mathcal{P}\exp\left(\oint_{\mathcal{K}_i} A\right). \tag{195}$$

The Wilson loop of a general link, which we used above, is expanded as

$$W(\gamma, A) := W_{\mathcal{R}_i \cdots \mathcal{R}_N}(\gamma, A) = \prod_{i=1}^{N} W_{\mathcal{R}_i}(\gamma_i, A). \tag{196}$$

From its construction it is easy to see that the observables of the theory, Wilson loops, are metric independent. If the base manifold admits a metric g then

$$\frac{\partial}{\partial g^{\mu\nu}}\langle W(\gamma, A)\rangle = 0 \tag{197}$$

this is the other property which makes the theory topological. Let us now consider the case in where the gauge group is $SU(N)$ and all representations are in $\mathcal{R}_i = \underline{\mathbf{N}}$, the fundamental representation of $SU(N)$. From this we obtain invariants of the HOMFLY polynomials of a link γ. These polynomials

are characterized by two variables (q, λ): $P(q, \lambda)(\gamma)$. This equivalence is [18]

$$\langle W_{\underline{\mathbf{N}}\cdots\underline{\mathbf{N}}}(\gamma, A)\rangle = \lambda^{lk(L)}\left(\frac{\lambda^{1/2} - \lambda^{-1/2}}{q^{1/2} - q^{-1/2}}\right) P(q, \lambda)(\gamma) \tag{198}$$

where $lk(L)$ is the linking number defined by counting crossing terms of a framed link projected onto a 2d surface (see [3]) and

$$q = \exp\left(\frac{2\pi i}{k + N}\right), \quad \lambda = q^N. \tag{199}$$

In the case $G = SU(2)$ this reduces to invariants of the Jones polynomial. We can obtain a generating function for link invariants. Let us define the holonomy terms

$$U := \mathcal{P}exp\left(\oint_{\mathcal{K}} A\right) \in U(N), \quad V := \mathcal{P}exp\left(\oint_{\mathcal{K}} \overline{A}\right) \in U(M) \tag{200}$$

where U is the holonomy over a link and V is a probing source term. The generating function is [21]

$$Z(U, V) := \exp\left(\sum_{n=0}^{\infty} Tr(U^n)Tr(V^n)\right) \tag{201}$$

and for the unknot this is given by [18]

$$Z(V) = \exp\left(\sum_{n=1}^{\infty} \frac{\lambda^{n/2} - \lambda^{-n/2}}{n(q^{n/2} - q^{-n/2})} Tr(V^n)\right). \tag{202}$$

When we consider the open string field theory and the topological string we will find a relationship between these knot invariants and the A-model topological string theory.

Let us now mention how to relate Chern-Simons, surgery theory and the general definition of topological field theories relate to obtain topological information about three manifolds. The Wess-Zumino-Witten (WZW) model is a twodimensional conformal field theory which is defined over an affine lie algebra[14]. Along with showing the above relationship between knot invarants and Chern-Simons theory in [29] Witten found that the surgery associated to Chern-Simons theory on Riemannian surfaces is related to the WZW model. After canonically quantizing Chern-Simons theory the hilbert space obtained from the boundaries of surfaces is precisely the conformal blocks of the WZW. These objects are the basic 'building blocks' for the correlation functions of the theory. Because the WZW model had been well studied before this discovery the results of the WZW model

[14]These algebras are related to 'normal' lie algebras by a tensoring of representations of the so called loop group. See [10] for the details of this and the WZW model.

could be used to better understand the quantization of Chern-Simons theories. In our analysis we make no distinction between the two and refer the reader to [29].

Let S be a three-manifold whose boundary $\partial S = \Sigma$ is Riemann surface. Such a space is called a handle body. We will consider what is known as Heegaard splitting: We will cut S at Σ so that it is the connected sum of two handle bodies which we can glue together at Σ to obtain S. For simplicity suppose S split into two handle bodies

$$S = M_2 \bigsqcup_{\Sigma} M_1. \tag{203}$$

Recall that to each boundary of a manifold S we associate to it a Hilbert space $\mathcal{H}_\Sigma$ and a map $\Phi_\Sigma : \mathcal{H}_\Sigma \to \mathcal{H}_\Sigma$. Using the results presented in the previous section we conclude that this splitting is simply the statement

$$Z(S) = \langle \Psi_{M_2} | \Phi_\Sigma | \Psi_{M_1} \rangle. \tag{204}$$

where $\langle \Psi_{M_2}| \in \mathcal{H}_\Sigma^*$ and $|\Psi_{M_1}\rangle \in \mathcal{H}_\Sigma$. If we consider the disjoint union of two closed handle bodies B_1 and B_2 we can cut two of its ends and glue these two ends together while simultaneously gluing the two ends to make S^3. That is $B_1 \sqcup B_2 \to B_1 \# B_2 \sqcup S^3$ where $\#$ denotes the direct product which glues the two cut handle bodies together. Let $B := B_1 \# B_2$ then by axiom 3. we find

$$Z(B)Z(S^3) = Z(B_1)Z(B_2). \tag{205}$$

If we now consider embedding links in S^3 and evaluate the VEV of Wilson loops over these knots we can consider certain monodromy actions [29]. From these considerations we obtain the expression knot invariant which we presented earlier. It is also possible to use the topology of the space to obtain information about the quantum theory. In particular through representation theory one can obtain the expression

$$Z(S^3) = \sqrt{\frac{2}{k+2}} sin\left(\frac{\pi}{k+2}\right). \tag{206}$$

For a Riemann surface of genus g and gauge group $G = SU(2)$ this expression can be used to obtain the Verlinde formula [24]

$$dim(\mathcal{H}_\Sigma) = \left(\frac{k+2}{2}\right)^{g-1} \sum_{j=0}^{k} \left(sin\left(\frac{(j+1)\pi}{k+2}\right)\right)^{2-2g}. \tag{207}$$

How does the dimension of the Hilbert space obtained from Heegaard splitting related to the topology of the handlebody? Recall that the moduli space of Chern-Simons theory was the space of

gauge equivalent flat connections over the base three manifold. Therefore in computing the dimension of this space we can understand how many states the theory will have. To see let us consider a specific example. this consider the gluing of two tori as $S^1 \times S^2 \cong S^3$. However, we can also think of this as S^2 periodically identified. In doing so the morphism reduces to the trace of the Hamiltonian evolution operator

$$\Phi(S^1 \times S^2) = Tr\left(e^{-itH}\right) = dim(\mathcal{H}) = 1 \tag{208}$$

using the fact that $H = 0$ for Chern-Simons. Now, the moduli space to S^3 is simply a point since every bundle over it is trivial. This fact is exactly reflected in the results just obtained.

5.2 Cohomological Field Theories

In the present we consider the cohomological approach to topological field theories which are also refered to as "Witten" type TQFT's. The Faddeev-Popov gauge fixing ghost procedure of Yang-Mills gauge theories resolves the issue of infinities that occur from integrating over gauge equivalent fields. In this formalism one introduces so called (anit)ghost fields which add a new fermionic BRST symmetry with corresponding operator Q_{BRST}. The physical operators of this theory are described by cohomological equivalence classes of the BRST symmetry generator. In string theory the same procedure gives rise to ghost fields with corresponding BRST symmetry. As in the case of gauge theory the physical string operators are given by 'exact' terms written as $T = \{Q_{BRST}, b\}$ where T is the quantized energy momentum tensor and b a ghost operator. We call theories with this property cohomological and the present section is intended to generalize them. We follow [26] in our presentation.

Recall that Noether's theorem states that a classical system which has a global symmetry gives rise to a conserved current and with it an associated conserved charge Q. After quantzation this conserved charge becomes an operator which acts on the physical Hilbert space of the theory. Let $\mathcal{O}_i$ be an operator of the theory and $|\Psi\rangle$ a state in our Hilbert space. Then the state is said to be symmetric under the action of Q if $Q|\Psi\rangle = 0$. Let ϵ be an infintesimal parameter of the symmetry so that $\delta_\epsilon \mathcal{O} = i\epsilon[Q, \mathcal{O}]$ if atleast one is bosonic and $\delta_\epsilon \mathcal{O} = i\epsilon\{Q, \mathcal{O}\}$ if both are fermionic. Without loss of generality we will denote this as $i\epsilon\{Q, \mathcal{O}\}$ until we specify Q explicitly. With this, we state the following axiomatic properties of a Toplogical Quantum Field theory:

Definition 17. *A (topological) Cohomological Quantum Field Theory is a collection* $(M, \mathcal{H}(M), \{\cdot, \cdot\}, Q)$ *where* Q

(i) Q *is nilpotent.* $Q^2 = 0$

(ii) The vacuum state of theory is symmetric. $Q|0\rangle = 0$

(iii) The physical observables of the theory are defined by operators $\mathcal{O} : \mathcal{H} \to \mathcal{H}$ which satisfy

$$\{Q, \mathcal{O}\} = 0 \tag{209}$$

(iv) The Lagrangian of the theory is Q-exact, meaning $\mathcal{L} = \{Q, G\}$ for some other (non physical) operator G

An immediate consequence of (*iv*) is that the quantized energy momentum tensor $T_{\alpha\beta} = \frac{1}{\sqrt{-h}} \frac{\delta S}{\delta h^{\alpha\beta}}$ is also Q exact. This is a desirable property as we know the importance of T when we study CFT's. Note that vacuum expectation values such as $\langle \mathcal{O} \rangle$ are invariant under such variations

$$\langle \mathcal{O} \rangle \to \langle \mathcal{O} + \delta_\epsilon \mathcal{O} \rangle = \langle \mathcal{O} \rangle + i\epsilon \langle Q\mathcal{O} + \mathcal{O}Q \rangle = \langle \mathcal{O} \rangle$$

where the second term vanishes from (*ii*). In fact any correlator which contains any exact terms vanishes. Let $\mathcal{O}_i$ be a physical observable be The cohomological properties are defined as follows: an operator A is said to be *exact* if there exists a (nonphysical) operator V such that $A = \{Q, V\}$. A is said to be *closed* if $\{Q, A\} = 0$. We then have a cohomology ring defined by Q

$$H_Q := \frac{Ker(Q)}{Im(Q)} \tag{210}$$

with the equivalence relation given by $\mathcal{O}_i \sim \Lambda_i \implies \mathcal{O}_i - \Lambda_i = \{Q, A\}$ which classifies the redundancies of physical observables of the theory. The action of the theory should be invariant under this symmetry so by (iv) the lagrangian is Q-exact $\mathcal{L} = \{Q, V\}$ and so

$$\delta_\epsilon S = \{Q, \delta_\epsilon \int_M V\} = \{Q, i\epsilon\{Q, \int_M V\}\} = 0.$$

We now make the claim of metric invariance explicit. Let $\mathcal{O}$ be physical observables which commute with Q, then

$$\begin{aligned} \langle \mathcal{O}_{i_1} \mathcal{O}_{i_2} \cdots \mathcal{O}_{i_k} \{Q, V\} \cdots \mathcal{O}_{i_n} \rangle &= \langle (\mathcal{O}_{i_1} \mathcal{O}_{i_2} \cdots \mathcal{O}_{i_k} QV \cdots \mathcal{O}_{i_n}) + (\mathcal{O}_{i_1} \mathcal{O}_{i_2} \cdots \mathcal{O}_{i_k} VQ \cdots \mathcal{O}_{i_n}) \rangle \\ &= 0. \end{aligned} \tag{211}$$

Therefore we find that our correlators do not depend on the choice of metric[15]

[15] We are ignoring the term coming from the derivative of Z_0 which yields the same result. We are also Writing the path integral measure $D\phi$ with respect to a single field ϕ simply for simplicity of notation

$$\begin{aligned}\frac{\partial}{\partial h^{\alpha\beta}}\langle \mathcal{O}_{i_1}\mathcal{O}_{i_2}\cdots\mathcal{O}_{i_n}\rangle &= \frac{1}{Z_0}\frac{\partial}{\partial h^{\alpha\beta}}\left(\int \mathcal{D}\phi \mathcal{O}_{i_1}\mathcal{O}_{i_2}\cdots\mathcal{O}_{i_n}e^{-S(\phi)}\right)\\ &= \frac{1}{Z_0}\int_M \mathcal{D}\phi\langle \mathcal{O}_{i_1}\mathcal{O}_{i_2}\cdots\mathcal{O}_{i_n}\frac{\partial S}{\partial h^{\alpha\beta}}e^{-S(\phi)}\rangle\\ &= \langle \mathcal{O}_{i_1}\mathcal{O}_{i_2}\cdots\mathcal{O}_{i_n}T_{\alpha\beta}\rangle\\ &= \langle \mathcal{O}_{i_1}\mathcal{O}_{i_2}\cdots\mathcal{O}_{i_n}\{Q, G_{\alpha\beta}\}\rangle\\ &= 0\end{aligned} \tag{212}$$

hence showing that the correlators of a topological field theory are independent of choice of metric. Furthermore, if we do not suppress the factor of $\hbar$ in the partition function so that the integration measure reads $\int \mathcal{D}\phi e^{-S/\hbar}$ we also find that it is indepent $\hbar$ since $d/d\hbar$ applied to the correlators brings down an exact term which annihilates the vacuum states. This is quite a remarkable feature as we can take the limit $\hbar \to 0$ so that all the quantities of interest are determined exactly in the semiclassical limit. Stated otherwise we have achieved a theory which admits localization which we define below.

5.2.1 Descent equations

Since the Hamiltonian is given by $H = \int d^{m-1}xT_{00}$ is a physical observable of our theory and is exact by (*iv*). By this same property the momentum operator P_α is also exact $P_\alpha = \{Q, G_\alpha\}$. The exactness of the momentum operator allows us to uncover a chain of physical operators related by the exterior derivative. Suppose we have a local operator $\mathcal{O}^{(0)}$ in our theory that is Q closed which transforms as a scalar under coordinate transformations. By taking the exterior derivative of this operator we find

$$\begin{aligned}d\mathcal{O}^{(0)} &= \frac{d}{dx^\alpha}\mathcal{O}^{(0)}dx^\alpha = i[P_\alpha, \mathcal{O}^{(0)}]dx^\alpha = i[\{Q, G_\alpha\}, \mathcal{O}^{(0)}]dx^\alpha\\ &= \{Q, \mathcal{O}^{(1)}\}\end{aligned}$$

where $\mathcal{O}^{(1)} := \mathcal{O}^{(1)}_\alpha dx^\alpha = i\{G_\alpha, \mathcal{O}^{(0)}\}dx^\alpha$ is an exact operator valued 1-form. If we continue applying the exterior derivative to $\mathcal{O}$ we obtain operator valued p-forms $\mathcal{O}^{(p)}$ recursively defined by

$$\begin{aligned} 0 &= [Q, \mathcal{O}^{(0)}] \\ d\mathcal{O}^{(0)} &= \{Q, \mathcal{O}^{(1)}\} \\ &\vdots \\ d\mathcal{O}^{(n-1)} &= \{Q, \mathcal{O}^{(n)}\} \\ d\mathcal{O}^{(n)} &= 0. \end{aligned} \tag{213}$$

This collection of equations are referred to as descent equations/relations and $\mathcal{O}^{(i)}$ as descendant fields. Let $\beta_p \in H_p(M)$ be a p-cycle. If we integrate $d\mathcal{O}^{(p-1)}$ over β_p we find, by Stokes theorem, a set of physical operators

$$0 = \left\{ Q, \int_{\beta_p} \mathcal{O}^{(p)} \right\}.$$

In particular, the top form operator $\int_M \mathcal{O}^{(m)}$ can be added to the action of the theory since the action itself is Q exact. This additional term is referred to as a deformation term and as we shall see they play an important role in understanding the deformations of the twisted sigma models. We will find for a set of complex modui parameters t^α much more general action of a cohomological theory can be written as

$$S = \int_\Sigma \mathcal{L} + \sum_{i=1}^m t^i \int_\Sigma \mathcal{O}_i^{(n)} + \sum_{i=1}^m \bar{t}^i \int_\Sigma \overline{\mathcal{O}}_i^{(n)} \tag{214}$$

6 Topological Sigma Models

A sigma model is collection (ϕ, Σ, M) where $\phi : \Sigma \to M$ is a map between a manifold Σ called the world sheet and a different manifold M called the target space. This model is of particular interest in string theory as we would like to embed the world sheet of a string into some portion of spacetime so the map ϕ is naturally thought of as the worldsheet embedding. The Polyakov string action is one of the simplest cases of a sigma model which contains a bosonic fields. We begin this section by introducing localization. After doing so we will writing down the general two dimensional supersymmetric sigma model on Σ and then specialize to the case of $\mathcal{N} = (2, 2)$ supersymmetry. The latter is obtained by considering the action of the super Kahler potential associated to a chiral superfield (defined below). As we shall see, such theories have a target space which admits a Kahler geometry. This is one of many beautiful relationships found between supersymmetric theories and geometry. It is also possible to obtain different target spaces geometries by considering different underlying superfields. The main reference for this is [14] which we follow closely along with [30] [26] [7].

6.1 Localization and Supersymmetry

One of the most appealing properties of supersymmetric theories is that they often allow for localization: the infinite dimensional partition function becomes finite with contributions coming from the critical points of this supersymmetric action. Aside from being a computational tool it turns out that there is an intimate relationship between supersymmetric theories and geometry of the space which was first observed by Witten in [28]. This property will allow the theory to posses properties similar to those defined in cohomological field theories. We motive this idea with a simple example and refer the reader to [14] where this is based on. Consider the simplest 0 dimensional quantum field theory whose action and partition function[16] are given by

$$S(\phi, \psi_1, \psi_2) := \frac{1}{2}(\partial h(\phi))^2 - \partial^2 h(\phi)\psi_1\psi_2 \qquad \Longrightarrow \qquad Z_0 = \int d\phi d\psi_1 d\psi_2 e^{-[\frac{1}{2}(\partial h(\phi))^2 - \partial^2 h(\phi)\psi_1\psi_2]}$$

where $\phi(x) : p \mapsto \mathbb{R}$ is the embedding map from a single point into $\mathbb{R}$, $h = h(\phi)$ is some function of this embedding and ψ_1, ψ_2 real valued fermionic Grassmann variables. This system is invariant under the following (super)symmetry transformations

$$\begin{aligned} \delta\phi &= \epsilon(\psi_1 + \psi_2) \\ \delta\psi_1 &= \epsilon\partial h \\ \delta\psi_2 &= -\epsilon\partial h \end{aligned}$$

[16] In this zero dimensional example the integration measure reduces $D \to d$ to a normal integral.

that is $\delta S = 0$. It is also easy to see that the partition function measure $\delta(D\phi D\psi_1 D\psi_2)$ is invariant under this symmetry. The partition function reduces to an integral over the bosonic measure since $\int d\psi_1 d\psi_2 \psi_1\psi_2 = 1$ and zero otherwise. Expanding h about its critical points ($h'(\phi_c) = 0$) as

$$h(\phi) = h(\phi_c) + \frac{\partial^2 h(\phi_c)}{2}(\phi - \phi_c)^2 + \cdots \tag{215}$$

the partition function evaluated to Thus

$$Z_0 = \int d\phi(\partial^2 h) e^{-\frac{1}{2}(\partial h)^2} = \sum_{\phi_c} \frac{\partial^2 h(\phi_c)}{|\partial^2 h(\phi_c)|} \tag{216}$$

We see that the theory localizes on critical points of h which we can interpret geometrically. The above expression is in a sense a simple example of the supersymmetric index since its contributions come from critial points with positive slope (+1) as well as those with negative slope (-1). The index theorem does just this. The Atiyah-Singer index theorems allow us to relate the number of sections in the kernel of the operators of the theory to the topology of the underlying manifold. These sections are referred to as zero modes since they are eigenstates with zero eigenvalue. The use of such ideas[17] will be a defining property of how we deal with localization of our twisted models.

Given this we now switch focus back to our toy model and show that this theory is also topological. The symmetry of the theory is not nilpotent meaning the supersymmetry does not close off shell as written. We must add an auxiliary field F so that the action is given by $S = \frac{1}{2}F^2 + i\partial h F - \psi_1\psi_2\partial^2 h$ with the offshell supersymmetric variations now

$$\begin{aligned} \delta\phi &= \epsilon(\psi_1 + \psi_2), & \delta F &= 0 \\ \delta\psi_1 &= i\epsilon F, & \delta\psi_2 &= -\epsilon i F. \end{aligned}$$

If Q is the generator of this system then the action can be written as

$$S = \frac{1}{4} Q[(\psi_1 - \psi_2)(2\partial h + iH)]$$

which shows that it is Q exact. From our previous definitions we conclude that this theory is topological. We now have a better idea of how localization, a general property of supersymmetry, can be related to cohomological theories.

Let us make a few remarks on the mathematics associated to this model. The localization observed here can be cast into the language of Morse theory. Let $h : M \to \mathbb{R}$ be some function on

[17] In this thesis we do not make explicit use of index theorems and the reader interested should see [19] for an introduction.

the manifold with critical points $_i$. The Morse index $\mu(p_i)$ at a critial point is defined as the number of negative eigenvalues of the Hessian matrix[18] $\partial^2 h(\phi)/\partial x^\mu \partial x^\nu$. Let N_q denote the number of critial points which have Morse index q. It turns out we can obtain topological information about the manifold through

$$\chi(M) = \sum_{q=0}^{m} (-1)^q M_q \tag{217}$$

where χ is the Euler characteristic. From all this we see that our toy model is simply calculating the sum of Morse indicies.

Now there is one more thing to be said about topological invariance of this model. Before taking the final integral above we could consider a change of variables $h \mapsto th$ then the partition function reads

$$Z_0 = \int \phi e^{-t^2(\partial h)^2} t\partial^2 h \tag{218}$$

differentiating with respect to t the integral vanishes as it becomes a surface term. This means the theory does not depend on the scaling parameter t. This is a general property known as deformation invariance. Physical theories naturally come with a scaling parameter: $1/\hbar$ and so if we take $t^2 = 1/\hbar$ we can take the semi-classical limit $\hbar \to 0$. This limit contributes only at critical points of the classical action, hence the theory is exactly solvable classically. This is one of the central ideas in the construction of twisted topological sigma models and the reader should keep the results of this section in mind when we construct these theories.

6.2 Supersymmetric non-linear sigma-models

Here we will show how to explicitly derive the sigma model from the superspace formalism as this is often omitted in most texts. After doing so we begin to analyze properties of the theory including the CFT associated to $\mathcal{N} = (2,2)$ non-linear sigma model as they each give us greater insight to the theory. We begin by assuming our theory lives on a super manifold $\Sigma^{(2|2)}$ where the notation denotes the fact that we have superspace coordinates $(z^\mu, \theta^\pm, \overline{\theta}^\pm)$. The bosonic part of this superspace is a Riemannian surface of genus g. The supersymmetry generators of the theory transform in the spinor representations of the Lorentz group and the number of such generators is denoted by $\mathcal{N}$. In the case that representation is reducible the spinors will split into spinors that transform under the irreducible representations of this group. In two dimensions the Lorentz group is $U(1)$ so a general spinor will split into two components that transform oppositely. The notation $\mathcal{N} = (2,2)$ simply means we have two generators that transform with positive $U(1)$ and two (conjugate) spinors that transform oppositely. The Poincare generators of the theory are given by the Hamiltonian H, the moemtum operator P and the Lorentz rotational generator M. The supersymmetry generators and covariant

[18] One should compare this to the analysis of spontaneous symmetry breaking.

derivatives are given by

$$Q_\pm = \frac{\partial}{\partial\theta^\pm} + i\overline{\theta}^\pm\partial_\pm, \qquad \overline{Q}_\pm = -\frac{\partial}{\partial\overline{\theta}^\pm} - i\theta^\pm\partial_\pm \tag{219}$$

$$D_\pm = \frac{\partial}{\partial\theta^\pm} - i\overline{\theta}^\pm\partial_\pm, \qquad \overline{D}_\pm = -\frac{\partial}{\partial\overline{\theta}^\pm} + i\theta^\pm\partial_\pm \tag{220}$$

and the only non-zero brackets between these operators are

$$\{Q_\pm, \overline{Q}_\pm\} = -2i\partial_\pm = P \pm H, \quad \{D_\pm, \overline{D}_\pm\} = 2i\partial_\pm = -P \mp H. \tag{221}$$

For the reader unfamiliar with these operators they arise from 'parellel transport' along the bundles over super-Riemannian manifolds in going from bosonic to Grassmann valued coordinates and vice-versa. For later use we write the expressions for the Hamiltonian and momentum operators in terms of the supersymmetry generators

$$\{\overline{Q}_+ + Q_-, Q_+ - \overline{Q}_+\} = 2H \qquad \{\overline{Q}_+ + Q_-, Q_+ + \overline{Q}_+\} = 2P \tag{222}$$

$$\{\overline{Q}_+ + \overline{Q}_-, Q_+ - Q_-\} = 2H \qquad \{\overline{Q}_+ + \overline{Q}_-, Q_+ + Q_+\} = 2P. \tag{223}$$

These are not the only symmetries which the theory possesses. Since the spinors spit into irreducible representations of opposite chirality we have symmetries that acts only these spinors. This symmetry is known as *R*-symmetry and will be fundamental to our analysis as it turns out it gives rise to anomalies. We will return to this matter shortly.

6.2.1 The $N = 2$ non-linear Sigma Model

The purpose of this section is mainly to introduce notation. Let x^I be local coordinates on the target space M. We then use the familiar notation $\phi^I := (x \circ \phi)^I$. In complex coordinates the $N = 2$ supersymmetric sigma model Lagrangian is given by

$$S = \int_\Sigma d^2x \left(\frac{1}{2} g_{IJ}\partial_z\phi^I\partial_{\bar{z}}\phi^J + \frac{i}{2} g_{IJ}\psi^I_- D_z\psi^J_- + \frac{i}{2} g_{IJ}\psi^I_+ D_{\bar{z}}\psi^J_+ + \frac{1}{4} R_{IJKL}\psi^I_+\psi^J_+\psi^K_-\psi^L_- \right) \tag{224}$$

where $g_{IJ} := \frac{\partial^2 K}{\partial\phi^I\partial\phi^J}$ is the Kahler metric of the target space which we will later define, R_{IJKL} the Riemann tensor of M and D_z is the Douboult complex ∂ of the bundle $K^{1/2} \otimes \phi^*(TM)$ which we obtain by pulling back the Levi-Civita connection on TM

$$D_{\bar{z}}\psi^I_+ = \frac{\partial\psi^I_+}{\partial\overline{z}} + \Gamma^I_{JK}\frac{\partial\phi^J}{\partial\overline{z}}\psi^K_+ \tag{225}$$

assuming a local trivialization of $K^{1/2}$ so that no spin connection appears. The fermi fields of the theory are sections

$$\psi_+^I \in \Gamma(K^{1/2} \otimes \phi^*(TM)) \tag{226}$$

$$\psi_-^I \in \Gamma(\overline{K}^{1/2} \otimes \phi^*(TM)) \tag{227}$$

where $K^{1/2}$ and $\overline{K}^{1/2}$ are the square roots of the canonical and anti-canonical line bundles of Σ. In our analysis the scripts + and − for spinors will refer the the coordinate on $K^{1/2}$ and $\overline{K}^{1/2}$ respectively. Since $dim_{\mathbb{C}}(\Sigma) = 1$ then the canonical and anitcanonical line bundles are $K = \Omega^{1,0}$ and $\overline{K} = \Omega^{0,1}$ which are simply bundles of forms of type $(1, 0)$ and $(0, 1)$ respectively. We will also use the notation $\pm$ to denote complex Riemann coordinates as well. The reader should be warned that many texts, including [14], use this notation to denote light-cone coordinates if the explicit complex nature of the theory is not stated. All Riemann surfaces considered here will be complex. The reason for using the $\pm$ notation is that it allows us to generalize equations so in the appropriate sense we will switch between $\pm$ and the z and $\bar{z}$ notation.

6.2.2 $\mathcal{N} = (2, 2)$ supersymmetric nonlinear sigma model

Let us now construct the field content of the $N = (2, 2)$ sigma model. A (left-handed)chiral superfield satisfies the condition $\overline{D}_{\dot{\alpha}}\Phi = 0$ and can be expanded in superspace coordinates $(z^\mu, \theta^\pm, \overline{\theta}^\pm)$ with $y^+ := z - i\theta^+\overline{\theta}^+$ and $y^- := \bar{z} - i\theta^-\overline{\theta}^-$ as [14]

$$\begin{aligned}\Phi^i(y, \theta) =& \phi^i(y^\pm) + \theta^\alpha \psi^i_\alpha(y) + \theta^+\theta^- F^i(y^\pm) \\ =& \phi^i(x) + \theta^+\psi^i_+(x) + \theta^-\psi^i_-(x) + \theta^+\theta^- F^i - i\theta^+\overline{\theta}^+\partial_+\phi^i - i\theta^-\overline{\theta}^-\partial_-\phi^i \\ & - i\theta^+\theta^-\overline{\theta}^-\partial_-\psi^i_+ - i\theta^-\theta^+\theta^-\overline{\theta}^+\partial_+\psi^i_- - \theta^-\theta^+\overline{\theta}^+\partial_+\partial_-\phi^i\end{aligned} \tag{228}$$

Where ϕ^i is a complex scalar field and ψ^i it's fermionic super partner. The field F is an auxilary field added to close the supersymmetry off shell. Define $\overline{\phi}^i := \phi^{\bar{i}}$ and likewise for all other fields. We can now take the action of a super Kahler potential $K(\Phi, \overline{\Phi})$ and integrate it over all of the superspace

$$S = \int_{\Sigma^{(2,2)}} d^4\theta d^2 z K(\Phi^i, \bar{\Phi}^i) \tag{229}$$

More generally we could also consider adding a superpotential $W(\phi)$ and its conjugate to the action but we will not do so here. See [14] for this more general case. It is well known that under the spacetime integral taking a fermionic integral is equivalent to taking a fermionic derivative. As such, the above integral picks out the $\theta^2\overline{\theta}^2$ component of $K(\Phi, \overline{\Phi})$ which is often referred to as the D-term of the superfield expansion. Before deriving this expression we first define the following

quantities

$$\Delta^i(x) := \Phi^i - \phi^i(x) \qquad K_i := \left.\frac{\partial K(\Phi,\overline{\Phi})}{\partial \phi^i}\right|_{\Phi=\phi}$$
$$\Delta^{\bar{i}}(x) := \Phi^{\bar{i}} - \phi^{\bar{i}}(x) \qquad K_{\bar{i}} := \left.\frac{\partial K(\Phi,\overline{\Phi})}{\partial \phi^{\bar{i}}}\right|_{\Phi=\phi}$$

with higher derivatives of $K(\Phi,\overline{\Phi})$ similarly defined. Because the supersymmetric algebra is $\mathbb{Z}_2$ graded superspace parameters which are higher order than two vanish: $\theta^n = 0$ for $n > 2$. Therefore the expansion of $K(\Phi,\overline{\Phi})$ will terminate and so the terms $\Delta^i\Delta^j\Delta^k = 0$ and its complex conjugate will vanish. If for the moment we ignore the i index and expand $K(\Phi,\overline{\Phi})$ around $(\phi,\overline{\phi})$ we find

$$\begin{aligned} K(\Phi,\overline{\Phi}) =& K(\phi,\overline{\phi}) + K_i\Delta^i + K_{\bar{i}}\Delta^{\bar{i}} + \frac{1}{2}K_{ij}\Delta^i\Delta^j + \frac{1}{2}K_{\bar{i}\bar{j}}\Delta^{\bar{i}}\Delta^{\bar{j}} + K_{i\bar{j}}\Delta^i\Delta^{\bar{j}} \\ &+ \frac{1}{2}K_{ij\bar{k}}\Delta^i\Delta^j\Delta^{\bar{k}} + \frac{1}{2}K_{\bar{i}\bar{j}k}\Delta^{\bar{i}}\Delta^{\bar{j}}\Delta^k + \frac{1}{4}K_{ij,\bar{k}\bar{l}}\Delta^i\Delta^j\Delta^{\bar{k}}\Delta^{\bar{l}} \end{aligned}$$

note that we can get rid of the $K(\phi,\overline{\phi})$ term since is has no D-term, it is simply a scalar. The second through fifth terms contribute

$$-(K_i\partial_+\partial_-\phi^i + K_{\bar{i}}\partial_+\partial_-\phi^{\bar{i}} + \partial_+\phi^i\partial_-\phi^j K_{ij} + \partial_+\phi^{\bar{i}}\partial_-\phi^{\bar{j}}K_{\bar{i}\bar{j}})$$

and it is easy to see this is equal to $-\partial_+\partial_- K$ which is a total derivative that will vanishes at the boundary of Σ assuming the fields have compact support. Therefore these terms do not contribute to the action. The remaining terms are

$$\begin{aligned} \int d^4\theta K(\Phi,\overline{\Phi}) =& K_{i\bar{j}}\left(\partial_+\phi^i\partial_-\phi^{\bar{j}} + \partial_-\phi^i\partial_+\phi^{\bar{j}} - 2i\phi^i\partial_-\psi_+^{\bar{j}} - 2i\psi_-^i\partial_+\psi_-^{\bar{j}} + F^iF^{\bar{j}}\right) + \\ &+ \frac{1}{2}K_{ij\bar{k}}\left(-2\psi_+^i\psi_-^j F^{\bar{k}} - i\psi_+^i\partial_-\phi^j\psi_+^{\bar{k}} - i\psi_-^i\partial_+\phi^j\psi_-^{\bar{k}}\right) \\ &+ \frac{1}{2}K_{\bar{i}\bar{j}k}\left(-2\psi_-^{\bar{i}}\psi_+^{\bar{j}}F^k + i\psi_-^{\bar{i}}\partial_+\phi^{\bar{j}}\psi_-^k + i\psi_+^{\bar{i}}\partial_-\phi^{\bar{j}}\psi_+^k\right) \\ &+ \frac{1}{4}K_{ij\bar{k}\bar{l}}\psi_+^i\psi_-^j\psi_+^{\bar{k}}\psi_-^{\bar{l}} \end{aligned} \tag{230}$$

which can be given off shell by solving the equations of motion for the auxiliary field F and its conjugate. These are given by $F^l = K^{l\bar{k}}\partial_i(K_{j\bar{k}})\psi_+^i\psi_-^j$ and we can now identify the factor in front of the fermionic fields as $\Gamma^l_{ij} = K^{l\bar{k}}\partial_i(K_{j\bar{k}})$. The Kahler geometry of the theory is beginning to become apparent. In this spirit we also define $R_{i\bar{j}k\bar{l}} := \partial_k\partial_{\bar{l}}K_{\bar{j}i} - K^{m\bar{n}}\partial_k K_{i\bar{n}}\partial_{\bar{l}}K_{\bar{j}m}$. By replacing F by its

equations of motion, making use of $K_{l\bar{k}}\Gamma^{l}_{ij} = \partial_i(K_{j\bar{k}})$ and collecting terms we obtain

$$S = 2t\int_{\Sigma}\left(g_{i\bar{j}}\partial_z\phi^i\partial_{\bar{z}}\phi^{\bar{j}} + g_{\bar{i}j}\partial_z\phi^{\bar{i}}\partial_{\bar{z}}\phi^j + ig_{\bar{i}j}\psi^{\bar{i}}_{-}D_z\psi^j_{-} + ig_{\bar{i}j}\psi^{\bar{i}}_{+}D_{\bar{z}}\psi^j_{+} + R_{i\bar{j}k\bar{l}}\psi^i_{+}\psi^{\bar{j}}_{+}\psi^k_{-}\psi^{\bar{l}}_{-}\right) \tag{231}$$

where we have defined the Kahler metric

$$g_{i\bar{j}} := \frac{\partial^2 K}{\partial\phi^i\partial^{\bar{j}}} = K_{i\bar{j}} \tag{232}$$

and we have reverted to the notation $+ \to z$, $- \to \bar{z}$ for our derivatives and have added a coupling constant t.

We obtain the supersymmetric variation of the field content by applying an infinitesimal supersymmetric variation of the superfield Φ with respect to the operators previously defined. The explicit calculation is quite long but nontheless straight forward and refer the reader to [28]. The resulting supersymmery is generated by

$$\delta\phi^i = i\epsilon_-\psi^i_+ + i\epsilon_+\psi^i_-, \qquad \delta\phi^{\bar{i}} = i\epsilon_-\psi^{\bar{i}}_+ + i\epsilon_+\psi^{\bar{i}}_- \tag{233}$$

$$\delta\psi^i_+ = -\bar{\epsilon}_-\partial_z\phi^i - i\epsilon_+\psi^j_-\Gamma^i_{jk}\psi^m_+, \qquad \delta\psi^{\bar{i}}_+ = -\epsilon_-\partial_z\phi^{\bar{i}} - i\bar{\epsilon}_+\psi^{\bar{j}}_-\Gamma^{\bar{i}}_{\bar{j}\bar{k}}\psi^{\bar{k}}_+ \tag{234}$$

$$\delta\psi^i_- = -\bar{\epsilon}_+\partial_{\bar{z}}\phi^i - i\epsilon_-\psi^j_+\Gamma^i_{jk}\psi^m_-, \qquad \delta\psi^{\bar{i}}_- = -\epsilon_+\partial_{\bar{z}}\phi^{\bar{i}} - i\bar{\epsilon}_-\psi^{\bar{j}}_+\Gamma^{\bar{i}}_{\bar{j}\bar{k}}\psi^{\bar{k}}_- \tag{235}$$

By the Noether prescription we obtain the corresponding conserved (super)currents $G^{\alpha}_{\pm}$ and $\overline{G}^{\alpha}_{\pm}$ defined by the action

$$\delta\int_{\Sigma}\mathcal{L} = \int_{\epsilon}d^2z\left(\partial_\mu\epsilon_+G^\mu_- - \partial_\mu\epsilon_-G^\mu_+ + \partial_\mu\bar{\epsilon}_-\overline{G}^\mu_+ - \partial_\mu\bar{\epsilon}_+\overline{G}^\mu_-\right) \tag{236}$$

from this we can write down the classical expressions of the energy-momentum tensor and the supercurrents

$$G^0_{\pm} = G^{\pm} = g_{i\bar{j}}\partial_{\pm}\phi^{\bar{j}}\psi^i_{\pm} \tag{237}$$

$$\overline{G}^0_{\pm} = \overline{G}^{\pm} = g_{\bar{i}j}\psi^{\bar{i}}_{\pm}\partial_{\pm}\phi^j_{\pm} \tag{238}$$

$$T(z) = -g_{i\bar{j}}\partial_+\phi^i\partial_+\phi^{\bar{j}} + \frac{1}{2}g_{i\bar{j}}\psi^i_+\partial_+\psi^{\bar{j}}_+ + \frac{1}{2}g_{i\bar{j}}\psi^{\bar{j}}_+\partial_+\psi^i_+ \tag{239}$$

with corresponding supercharges

$$Q_+ = \oint G^-, \qquad \overline{Q}_+ = \oint G^+ \tag{240}$$

$$Q_- = \oint \overline{G^-}, \qquad \overline{Q}_- = \oint \overline{G}^+ \tag{241}$$

We now list the spinor field content in terms of sections over the appropriate bundles. This way of thinking will be useful when we introduce the concept of topological twisting thin the next section.

$$\psi_+^i \in \Gamma(\Sigma, K^{1/2} \otimes \phi^*(T^{1,0}M)) \qquad \psi_-^i \in \Gamma(\Sigma, \overline{K}^{1/2} \otimes \phi^*(T^{1,0}M)) \tag{242}$$

$$\psi_+^{\bar{i}} \in \Gamma(\Sigma, K^{1/2} \otimes \phi^*(T^{0,1}M)) \qquad \psi_-^{\bar{i}} \in \Gamma(\Sigma, \overline{K}^{1/2} \otimes \phi^*(T^{0,1}M)) \tag{243}$$

$$\epsilon_+, \overline{\epsilon}_+ \in \Gamma(\Sigma, K^{1/2}), \qquad \epsilon_-, \overline{\epsilon}_+ \in \Gamma(\Sigma, \overline{K}^{1/2}) \tag{244}$$

6.2.3 R-symmetry & the Axial Anomaly

There are two R-symmmetries that the theory possessess, the axial and vector symmetry. These are both $U(1)$ rotational symmetries defined on a superfield Φ. Let F_V and F_A be the generators of the vector and axial symmetry respectively with associated 'R' charges q_V and q_A. Then we define $R_{V/A}$-symmetry rotations as

$$e^{i\alpha F_V} : \Phi(z^\mu, \theta^\pm, \overline{\theta}^\pm) \mapsto e^{i\alpha q_V}\Phi(z^\mu, e^{-i\alpha}\theta^\pm, e^{i\alpha}\overline{\theta}^\pm) \tag{245}$$

$$e^{i\alpha F_A} : \Phi(z^\mu, \theta^\pm, \overline{\theta}^\pm) \mapsto e^{i\alpha q_A}\Phi(z^\mu, e^{\mp i\beta}\theta^\pm, e^{\pm i\beta}\overline{\theta}^\pm) \tag{246}$$

which has the following nonzero commutation relations

$$[F_V, Q_\pm] = Q_\pm, \qquad [F_A, Q_\pm] = \pm Q_\pm \tag{247}$$

$$[F_V, \overline{Q}_\pm] = -\overline{Q}_\pm, \qquad [F_A, \overline{Q}_\pm] = \mp\overline{Q}_\pm. \tag{248}$$

Although the nonlinear sigma model admits R-symmetries classically in order for them to become symmetries of the quantum theory the target space must admit certain properties. This is due to the localization properties previously discussed. To determine whether the symmetries are anomalous we will focus on the fermionic part of the $\mathcal{N} = (2,2)$ action and suppress the Lorentz index. Expanding the fields ψ_-,$\overline{\psi}_-$ as eigenstates of $D_{\bar{z}}D_z$ and the fields ψ_+,$\overline{\psi}_+$ as eigenstaes of $D_zD_{\bar{z}}$ formally as

$$\psi_- = \sum_\alpha \sum_{n=0}^{\infty} c_{n\alpha}\psi_-^{(n,\alpha)} = \sum_\alpha c_{0\alpha}\psi_-^{(0,\alpha)} + \sum_{n=1}^{\infty} c_n\psi_-^n \tag{249}$$

$$\psi_+ = \sum_\alpha \sum_{n=0}^{\infty} b_{n\alpha}\psi_-^{(n,\alpha)} = \sum_\alpha b_{0\alpha}\psi_-^{(0,\alpha)} + \sum_{n=1}^{\infty} b_n\psi_-^n \tag{250}$$

where $\psi_-^{(0,\alpha)}$ corresponds to states with zero eigenvalue $\lambda^{(0)} = 0$ and α counts the degeneracy of such states. If $\psi_- \in ker(D_{\bar{z}}D_z)$ then this field is also an element of $ker(D_z)$ since

$$0 = \int d^2z \overline{\psi}_+ D_z D_{\bar{z}} \psi_- = \int d^2z \, |D_z \psi_-|^2$$

implies $D_z\psi_- = 0$. The path integral measure will now contain the term

$$\prod_a dc_{0\alpha} db_{0\alpha} \prod_{n=1}^{\infty} dc_n d\overline{c}_n db_n d\overline{b}_n$$

but the zero modes will no longer be part of the action so the path integral simply vanishes. In order to remedy this we insert fermionic terms into our correlators to 'absorb' the zero modes. If we insert an equal number of each type of fermions then the correlator will have zero q_V and q_A charge so care must be taken we considering inserting different types of zero modes. Let

$$\ell_+ := dim(ker(D_{\bar{z}})) \tag{251}$$

count the number of ψ_+ zero modes and similarly for

$$\ell_- := dim(ker(D_z)). \tag{252}$$

Note since $(D_{\bar{z}}\psi_+)^* = D_z\overline{\psi}_-$ the number of zero modes of ψ_+ is equal to the number of $\overline{\psi}_-$ zero modes and like wise for ψ_- and $\overline{\psi}_+$. Since we are considering Kahler geometry there are equal number of ψ_- and $\overline{\psi}_+$ and so the correlator has neutral R_V charge. A general correlator will now carry an extra weight which takes the form $(\psi_+)^{\ell_+}(\overline{\psi}_+)^{\ell_-}(\psi_-)^{\ell_-}(\overline{\psi}_-)^{\ell_+}$. Under the R_A charge these fields in this weight transform in the same way

$$\langle (\psi_+)^{\ell_+}(\overline{\psi}_+)^{\ell_-}(\psi_-)^{\ell_-}(\overline{\psi}_-)^{\ell_+} \rangle \overset{R_A}{\mapsto} \langle (e^{-i\beta}\psi_+)^{\ell_+}(e^{i\beta}\overline{\psi}_+)^{\ell_-}(e^{i\beta}\psi_-)^{\ell_-}(e^{-i\beta}\overline{\psi}_-)^{\ell_+} \rangle \tag{253}$$

$$= e^{-i(\ell_+ - \ell_-)\beta} \langle (\psi_+)^{\ell_+}(\overline{\psi}_+)^{\ell_-}(\psi_-)^{\ell_-}(\overline{\psi}_-)^{\ell_+} \rangle \tag{254}$$

so in order for the path integral to be invariant we must have $\ell_+ - \ell_- = 0$. This condition[19] is satisfied when the target space admits a Calabi-Yau geometry since

$$\ell_+ - \ell_- = \int_\Sigma \phi^* c_1(T^{1,0}M). \tag{255}$$

If the target space is not Calbi-Yau then the theory will be anomalous. We therefore conclude upon quantization the R_V is symmetry is preserved for any Kahler target space and R_A is preserved if the target space is Calabi-Yau.

[19] This is derived in the appendix along with the conditions corresponding of the A and B model geometries.

6.3 Topological Twists

As previously discussed, we would like to preserve supersymmetry when we compactify our theories on some internal Calabi-Yau manifold. At the classical level when we define global supersymmetric symmetries our infinitesimal parameters of the symmetry $\epsilon_\pm$ are constant. The reason for this is that we are usually dealing with worldsheets that are globally flat, i.e. they are trivial bundles. Because we are taking Σ to be any Riemannian surface with arbitrary metric then we cannot simply do so and must require that these infinitesimals be covariantly constant $\nabla\epsilon = 0$ throughout the entire worldsheet. The sigma models as they are currently defined do not admit a global supersymmetry transformation since we cannot be sure that our supersymmetries commute with the nontrivial covariant derivative of the curved bundle $[\delta_\epsilon, \nabla] \neq 0$ where $\delta_\epsilon \Phi = (\epsilon^+ Q_+ + \overline{\epsilon}^- \overline{Q}_-)$. If we are to turn our sigma model into a cohomological field theory we must modify the bundles in such a way that these parameters takes values in a trivial bundle leaving $\nabla\epsilon_\pm = 0$. From our previous analysis of string compactification one may jump to the conclusion that we must choose the manifold to be Calabi-Yau but this would require fixing a background metric. This is not an appealing option as we would like to couple the theory to gravity which requires us to integrate over *all* possible (conformally equivalent) metrics. Hence, we cannot single restrict the choice of metric in this way. In [28] Witten invented a twisting of the underlying bundles which allows us to consider the theory over non-trivial bundles with which admit global supersymmetry. The theories obtained this way are called the A and B model twists of the underlying sigma model. Witten showed that by twisting $K^{1/2} \otimes \phi^*(TM)$ in a way that we will describe below none of the fields are sections over $K^{1/2}$ and $\overline{K}^{1/2}$ but instead are sections over K and $\overline{K}$.

There are two natural ways to obtain the twisting of these bundles At the level of fields we define the positive (+) and negative (-) twists (following the notation in [7] and ignoring spacetime indicies) as

$$(\psi_+, \overline{\psi}_+) \in (\Gamma(K^{1/2} \otimes \phi^*(TM)), \Gamma(K^{1/2} \otimes \phi^*(TM))) \mapsto \begin{cases} (\Gamma(\mathbb{C} \otimes \phi^*(TM)), \Gamma(K \otimes \phi^*(TM))) & (+)\text{twist} \\ (\Gamma(K \otimes \phi^*(TM)), \Gamma(\mathbb{C} \otimes \phi^*(TM))) & (-)\text{twist} \end{cases} \tag{256}$$

$$(\psi_-, \overline{\psi}_-) \in (\Gamma(\overline{K}^{1/2} \otimes \phi^*(TM)), \Gamma(\overline{K}^{1/2} \otimes \phi^*(TM))) \mapsto \begin{cases} (\Gamma(\mathbb{C} \otimes \phi^*(TM)), \Gamma(K \otimes \phi^*(TM))) & (+)\text{twist} \\ (\Gamma(K \otimes \phi^*(TM)), \Gamma(\mathbb{C} \otimes \phi^*(TM))) & (-)\text{twist} \end{cases} \tag{257}$$

where, depending on the twisting done, half of our spinors have become scalar fields. It should also be emphasized that these fields are still Grassmann odd and so they are still in this sense fermionic. The twists corresponding to the A and B model are defined as $(+,-)$ twists on (ψ_+, ψ_-) for the A-model and $(-,-)$ twists on (ψ_+, ψ_-) for the B-model. This should be compared with the fact that

type IIA string theory has opposite chirality spinors and type IIB has spinors of the same chiriality.

Changing the spin of the field content means that the corresponding operators must admit a change in its conformal weight. Since the Lorentz generator determines the spin of the field the above twists amount to modify the Lorenz algebra by one of the internal $U(1)_R$ R-symmetries. To see why this makes sense first recall that we are dealing with Euclidean sigma models so the Lorentz group $SO(1,1)$ is now the Euclidean group $SO(2)_E \cong U(1)_E$. Under this group we have $z \mapsto e^{i\theta}z$ which is a symmetry of the theory if the fermionic terms in the action transform oppositely to $e^{i\theta}D_z$ and $e^{-i\theta}D_{\bar{z}}$, that is $\psi_\pm\overline{\psi}_\pm \mapsto e^{\pm i\theta}\psi_\pm\overline{\psi}_\pm$. Since the fields ψ are no longer valued in $K^{1/2}$ they must admit a $SO(2)$ symmetry which we naturally inherit from the R-symmetries previously defined. Therefore we conclude

$$\tilde{M}_E := M_E - F_V \quad (A-twist) \tag{258}$$

$$\tilde{M}_E := M_E - F_A \quad (B-twist) \quad . \tag{259}$$

where F_V and F_A are simply the generators of the vector and axial symmetry respectively. In summary the twisting procedure amounts to replacing $SO(2)_E$ by its diagonal embedding in $SO(2)_E \times U(1)_R$ with the generator of this subgroup being $\tilde{M}_E$. Although the choice of generator F in our definitions seems rather arbitrary, a thorough analysis will show that the moduli space of the theories restrict us to such choices. Table 1. lists the $U(1)$ charges of the generators $Q_\pm$ and $\overline{Q}_\pm$ before and after twisting.

	Before twisting				A-twist		B-twist	
	$U(1)_V$	$U(1)_A$	$U(1)_E$	L	$U(1)'_E$	L	$U(1)'_E$	L
Q_+	1	1	-1	$K^{1/2}$	-2	K	-2	K
$\overline{Q}_+$	-1	-1	-1	$K^{1/2}$	0	$\mathbb{C}$	0	$\mathbb{C}$
Q_-	1	-1	1	$\overline{K}^{1/2}$	0	$\mathbb{C}$	2	K
$\overline{Q}_-$	-1	1	1	$\overline{K}^{1/2}$	2	$\overline{K}$	0	$\mathbb{C}$

Supercharges before and after applying a topological twist. Here L is the line bundle which it acts on. Recreated from [14].

At the level of the SCA twisting becomes

$$T' = T + \frac{1}{2}\partial J, \qquad \overline{T}' = \overline{T} + \frac{1}{2}\overline{\partial} J, \qquad J' = J, \qquad G^{\pm\prime} = G^\pm \quad (A-model)$$

$$T' = T - \frac{1}{2}\partial J, \qquad \overline{T}' = \overline{T} + \frac{1}{2}\overline{\partial} J, \qquad J' = -J, \qquad G^{\pm\prime} = G^\pm \quad (B-model)$$

whose modes take satisfies the relation $L'_n = L_n - \frac{m}{2}J_n$. We now take L'_m to generate the twisted

Virasoro algebra

$$[L'_n, L'_m] = (n-m)L'_{n+m}. \tag{260}$$

The action of the theory now picks up terms with are proportional to the spin connection

$$S \mapsto S + \int_\Sigma \left(\pm J(\frac{i}{2}\overline{\omega}) \pm \overline{J}(\frac{i}{2}\omega) \right) \tag{261}$$

where the sign is determined by the twist made. This means that the covariant derivatives in the twisted models will contain a spin connection which we cannot get rid of by trivializing the bundle as in the untwisted case.

As mentioned the twists which are performed are not unique. Once can twist the Lorentz algebras by considering the addition of the R-symmetry generators instead of subtraction as we did. This will change the conformal dimensions of each operator and lead to different (but equivalnt) assignments of the topological BRST operators. The choice made here allows us to continue using the same notation as the untwisted theories.

6.3.1 tt* equations

We can use the machinery of the CFT associated to our sigma model to better understand the correlation functions of the theory. This is in fact a general feature of topological field theories which we pointed out but in the present we shall see how we can relate our discussion of special geometry and the vacuum Hodge bundle to conformal fields. The idea is to build a geometry on the Hilbert space of our conformal field theory $\mathcal{H}$ and interpret this as surgery theory on the Riemannian surface Σ_g of genus g. What we mean by this is that given a Σ_g it is possible to break apart the surface into more 'fundamental' components that we define through our topological field theory morphisms Φ. To see pictorial representation see [8]. It turns out for topological field theories which harbor a CFT the surgery considered can be restricted to studying the $g = 0$ surface with marked points. Our analysis follows [14].

We begin by defining the topological metric bilinear $\eta : \mathcal{H} \otimes \mathcal{H} \to \mathbb{C}$ such that

$$\eta(\phi_i, \phi_j) := \eta_{ij} = \langle \phi_i \phi_j \rangle_0 \tag{262}$$

where the 0 in the correlator is with respect to the vacuum state $|0\rangle$ of the theory. It is customary to drop this as it is generally understood which vacuum we are talking about. In the case that we need to specify we will in stead use a g to denote the genus. This can geometrically be though of as a sphere with two operator insertions i.e. two marked points. Note that this metric is invariant under the representation chosen for a physical operator by our previous exposition By the operator-state correspondence we write $|\phi_i\rangle = \lim_{z,\bar{z}\to 0} \phi_i|0\rangle := |i\rangle$. With this it is possible to write a 'completeness

relation' for the theory which factorizes the n point functions and the Riemann surface. is given by

$$\mathcal{I} = |\phi_i\rangle\eta^{ij}\langle\phi_j| \tag{263}$$

where η^{ij} is the inverse defined through $\eta^{-1} : \mathbb{C} \to \mathcal{H}\otimes\mathcal{H}$. The factorization of an n-point function is of the form

$$\langle O_{i_1}\cdots O_{i_n}\rangle_g = \sum_{j,k}\langle O_{i_1}\cdots O_{i_s}O_j\rangle\eta^{jk}\langle O_k O_{i_s+1}\cdots O_{i_n}\rangle \tag{264}$$

We could also define the bilinear $[\cdot,\cdot] : \mathcal{H}\otimes\mathcal{H} \to \mathcal{H}$ also known as the 'pair of pants' map defined as one would expect

$$[\phi_i,\phi_j] := C_{ij}{}^k\phi_k \tag{265}$$

however it is not antipysmmetric and instead symmetric in its lower indicies $C_{ij}{}^k = C_{ji}{}^k$. It is common to write this expression as $\phi_i,\phi_j = C_{ij}{}^k\phi_k$ in the same way we write OPE's knowing that we really mean vacuum expectation value of a radial order product of fields. In this way one may also think of this as the topological analog of an OPE. We can define a Frobenius algebra by defining the fundamental

$$c_{ijk} := c_{ij}{}^l\eta_{lk} = \langle O_i O_j O_k\rangle. \tag{266}$$

The reason this correlator is fundamental is due to the fact that it can no longer be factorized. It corresponds to a sphere with three marked points which are fixed by the $SL(2,\mathbb{C})$ subgroup defined through the Witt algebra of our conformal theory. At this point it is clear there should be some connection between c_{ijk} defined here and the Yukawa coupling of special geomtry on complex moduli and we will now see how to defining this coupling term in present formalism. Strictly speaking the current discussion, in the context of twisted sigma models, has been for primary fields on weight $h = 1$. It is possible to also define an 'anti-topological' metric

$$\tilde{\eta}_{i\bar{j}} = \langle i||\bar{j}\rangle \tag{267}$$

which admits the same properties and in particular we note that they describe the same vacuum states of physical operators. This means that the two metric should be related by some 'matrix' like element $|i\rangle = M_i^{\bar{j}}|\bar{j}\rangle$ and by CPT invariance ($|i\rangle^* = |\bar{i}\rangle$) this matrix should be Hermitian $MM^* = \mathcal{I}$. The ground states $Q|i\rangle = 0$ of the theory define a vector bundle called the Vacuum bundle. For some basis e_i, the connection of this bundle is given by

$$A^i_{jk} = \eta^{il}\langle e_l|\partial_j|e_k\rangle \tag{268}$$

and the covariant derivative D_j is defined with respect to this connection

$$(D_i)^k_j = \delta^k_l \partial_i - (A_i)^k_l \tag{269}$$

The tt^* equations read

$$[D_i, \overline{D}_j] = \partial_i A_{\bar{j}} - \partial_{\bar{j}} A_i = -[C_i, \overline{C}_j] \tag{270}$$

$$[D_i, C_j] = [D_j, C_i] \tag{271}$$

and their complex conjugates. All other expressions are zero. These equations describe the topology of the CFT.

7 A-model

Since we have $(-,+)$ twitsts on ψ_+, ψ_- our fermionic field content is given by

$$\psi^i_+ := \psi^i_z \in \Gamma(K \otimes \phi^*(T^{1,0}M)) \qquad \psi^{\bar{i}}_+ := \chi^{\bar{i}} \in \Gamma(\phi^*(T^{0,1}M)) \tag{272}$$

$$\psi^i_- := \chi^i \in \Gamma(\phi^*(T^{1,0}M)) \qquad \psi^{\bar{i}}_- := \psi^{\bar{i}}_{\bar{z}} \in \Gamma(\overline{K} \otimes \phi^*(T^{0,1}M)) \tag{273}$$

The fermionic parameters of the symmetry also transform accordingly. After integrating by parts so that the covariant derivatives act on the scalar fields χ the A-model action reads

$$S_A = 2t \int_\Sigma \left(g_{i\bar{j}} \partial_z \phi^i \partial_{\bar{z}} \phi^{\bar{j}} + g_{i\bar{j}} \partial_{\bar{z}} \phi^i \partial_z \phi^{\bar{j}} + i g_{\bar{i}j} \psi^{\bar{i}}_{\bar{z}} D_z \chi^j + i g_{i\bar{j}} \psi^i_z D_{\bar{z}} \chi^{\bar{j}} + \frac{1}{2} R_{i\bar{j}k\bar{l}} \psi^i_z \psi^{\bar{j}}_{\bar{z}} \chi^k \chi^{\bar{l}} \right) \tag{274}$$

where the covariant derivative is now given by

$$D_z \chi^i = \partial_z \chi^i - \frac{i}{2} \omega_z \chi^i + \Gamma^i{}_{jk} \partial_z \phi^j \chi^k \tag{275}$$

as mentioned in the previous section. The new OPE's of the conformal theory are given by

$$T(z)T(w) \sim \frac{2T(w)}{(z-w)^2} + \frac{\partial_w T(w)}{(z-w)} \tag{276}$$

$$T(z)G^\pm(w) \sim \frac{(3 \mp 1)G^\pm(w)}{2(z-w)^2} + \frac{\partial_w G^\pm(w)}{(z-w)} \tag{277}$$

$$T(z)J(w) \sim +\frac{J(w)}{(z-w)^2} + \frac{\partial_w J(z)}{(z-w)} - \frac{c}{3(z-w)^3} \tag{278}$$

$$G^+(z)G^-(w) \sim +\frac{(2c/3)}{(z-w)^3} + \frac{2T(w) + \partial_w J(w)}{(z-w)} \tag{279}$$

To see that this theory is topological we must define a BRST Q_A operator for which we can build a cohomology from. Recall that one of the requirements of a cohomological field theory was that our physical operators are Q_A exact. From the equations for Hamiltonian and momentum generators above we see they are exact in $\overline{Q}_+ + Q_-$. As seen in table 1. this combination also transforms as a scalar so it can be globally defined! We have found the global supersymmetry that we can extend to arbitrary curved worldsheets. We call

$$Q_A := \overline{Q}_+ + Q_- \tag{280}$$

the A-model BRST operator. By setting $\epsilon := \epsilon_- = \overline{\epsilon}_+$ and $\epsilon_+ = \overline{\epsilon}_- = 0$ then equation (**??**) satisfies the conditions for a globally defined supersymmetry. Furthermore the supersymmetric generator of this symmetry satisfies the desired nilpotency condition $Q_A^2 = 0$. The A-model twisted supersymmetry transformations are

$$\delta\phi^i = i\epsilon\chi_+^i, \qquad \delta\phi^{\bar{i}} = i\epsilon\chi^{\bar{i}} \tag{281}$$

$$\delta\psi_z^{\bar{i}} = -\epsilon\partial_z\phi^{\bar{i}} - i\epsilon\chi^{\bar{j}}\Gamma^{\bar{i}}_{\bar{j}\bar{k}}\psi_z^{\overline{m}}, \qquad \delta\psi_{\bar{z}}^i = -\epsilon\partial_{\bar{z}}\phi^i - i\epsilon\psi_z^j\Gamma^i_{jk}\chi^m \tag{282}$$

$$\delta\chi^i = 0, \qquad \delta\chi^{\bar{i}} = 0 \tag{283}$$

There is one last condition that we must check before we can truly declare this theory to be topological we must show that the action (hence the energy momentum tensor) is Q_A exact. It turns out [26] that the actions is almost exact in that there is an extra factor

$$S_A = it\int_\Sigma \{Q_A, V\} + t\int_\Sigma \phi^*(\omega) \tag{284}$$

where $V = g_{i\bar{j}}(\psi_z^i\partial_{\bar{z}}\phi^{\bar{j}} + \partial_z\phi^i\psi_{\bar{z}}^{\bar{j}})$ and ω is the Kahler form $\omega = ig_{i\bar{j}}dz^i \wedge d\overline{z}^j$. The energy momentum 'density G in $T = \{Q_A, G\}$ is

$$G_{\mu\nu} = itg_{IJ}\left(\psi_\mu^I\partial_\nu\phi^I + \psi_\nu^I\partial_\mu\phi^I - h_{\mu\nu}h^{\lambda\sigma}\psi_\lambda^I\partial_\sigma\phi^J\right) \tag{285}$$

where we have generalized the expression for later use and h is the worldsheet metric. The second integral in the expression for the action takes the form $\int_{\phi(\Sigma)}\omega$ and since the Kahler form is closed it can be uniquely characterized by its homology class $H_2(M)$. If we consider cycles $[\beta] \in H_2(M)$ then the A-model partition function takes the form

$$Z_0 = \sum_{\beta\in H_2(M,\mathbb{Z})} e^{-t(\beta,\omega)}\int_{\phi(\Sigma)\subset\beta} D\phi D\chi D\psi e^{-it\int_\Sigma\{Q_A,V\}} \tag{286}$$

where (β,ω) denotes the natural pairing $(\beta,\omega) = \int_\beta\omega$.

The partition function is obviously independent of the complex structure and dependent on the Kahler structure by the pairing just mentioned. In this sense the theory is not fully topological but instead depends on half of the moduli of Ricci-flat metrics which were considered for Calabi-Yau manifolds. Indeed as we shall see, the twisted A-model on some Calabi-Yau manifold can be related to a twisted B-model theory on a different Calabi-Yau manifold through mirror symmetry. Before we can make this claim explicit we must understand what topological information we can obtain from these models. To do this we must determine what the allowed observables of the theory are since we are restricted by the axial anomaly.

7.0.1 Localization & Gromov-Witten invariants

Aside from desiring our observables be invariant under both the vector and axial R-symmetries we also want them to be Q_A closed since we would like any exact operator to commute and annihilate the vacuum state. From equation the supersymmetry transformations we see that $\{Q_A, \chi^{\bar{i}}\} = 0$ and so we can expand local operators of the theory as products of such fields. A general local operator will take the form

$$O(z) := O_{i_1 \cdots i_p \bar{j}_1 \cdots \bar{j}_q}(\phi(z)) \chi^{i_1} \cdots \chi^{i_p} \chi^{\bar{j}_1} \cdots \chi^{\bar{j}_q} \tag{287}$$

the fields ψ are not included as they contain a z spacetime index which can only be contracted with a metric and since we are considering metric independent topological theories such terms cannot be added. Further, since we would like to couple our theories to gravity the correlators should not contain any metric terms. By the Grassmann nature of the fermionic fields the function O is antisymmetric

$$\{Q_A, O(z)\} = -\frac{\partial O_{i_1 \cdots i_p \bar{j}_1 \cdots \bar{j}_q}(\phi(z))}{\partial \phi^{\bar{k}}} \chi^{\bar{k}} \chi^{i_1} \cdots \chi^{i_p} \chi^{\bar{j}_1} \cdots \chi^{\bar{j}_q} - \frac{\partial O_{i_1 \cdots i_p \bar{j}_1 \cdots \bar{j}_q}(\phi(z))}{\partial \phi^{k}} \chi^{k} \chi^{i_1} \cdots \chi^{i_p} \chi^{\bar{j}_1} \cdots \chi^{\bar{j}_q} \tag{288}$$

which bears a striking resemblance to the expression for the complex exterior derivative. In deriving the descent equations the identification of physical operators was made in the language differential forms so upon the identifications

$$\chi^i \cong d\phi^i, \qquad \chi^{\bar{j}} \cong d\phi^{\bar{i}} \tag{289}$$

$$Q_A \cong d = (\partial + \bar{\partial}) \tag{290}$$

we now have an isomorphism between the cohomology of Q_A corresponding to the set of physical operators and the de Rham cohomology of $d = (\partial + \overline{\partial})$:

$$H(Q_A) \cong \bigoplus_{p,q=0}^{n} H_d^{p,q}(M). \tag{291}$$

In fact we can impose further physical constraints. The highest weight states of our primary fields must be a unitary representation of the conformal group and so there is a unique decomposition into harmonic (p,q) forms since this corresponds to the (p,q) $U(1)$ charges of our theory. Now that we have identified our operators as differential forms we can determine their descent equations. For the A-model these read

$$\begin{aligned} \mathcal{O}^{(0)} &:= \phi \\ \mathcal{O}_A^{(1)} &= idz[\overline{Q}_-, \mathcal{O}_A^{(0)}] - id\bar{z}[\overline{O}_+, \mathcal{O}_A^{(0)}] \\ \mathcal{O}_A^{(2)} &= dzd\bar{z}\{Q_+, [\overline{Q}_-, \mathcal{O}_A^{(0)}]\}. \end{aligned} \tag{292}$$

We would now like to understand how we can utilize localization to turn these infinite dimensional path integrals into a finite number. From the calculation done in the appendix we know that the weight we must insert will have $2k$ more χ fields than ψ fields[20] but the calculation done does not explicitly state the exact number of each type of field. We will consider the generic case $dimH^1(\phi^*(TM)) = 0$ meaning there are no ψ insertions so we can simply consider inserting physical operators. The path integral localizes on the the minima of the action with respect to the bosonic fields just as in the case presented at the beginning of this section. Theses are simply the classical equations of motion which are given by

$$\partial_z \phi^{\bar{i}} = 0 \quad \partial_{\bar{z}} \phi^i = 0 \tag{293}$$

so the theory localizes and counts the number of holomorphic maps ϕ. Equivalently this is the condition that the bosonic part of the fermionic variations vanish. Stated otherwise, the world-sheet instantons of the theory correspond to holomorphic embeddings. The A-model moduli space $M_g(M,\beta)$, also referred to as the moduli of instantons of degree β [26], is the space

$$\mathcal{M}_g(M,\beta) := \{\phi : \Sigma \to M \mid \overline{\partial}\phi = 0,\ \phi_*[\Sigma_g]\beta \in H_2(M,\mathbb{Z})\}. \tag{294}$$

We would now like to write a general correlation function as an integral over this moduli space to better understand this counting scheme. It comes to no surprise that we must loot at the tangent space

[20] Recall that the insertion of fields into the correlator as weights is simply to absorb the fermionic zero modes and will vanish after the integration is done. Therefore there is no contradiction to not allowing the physical operators of the theory to contain terms with ψ.

to the modulispace $T\mathcal{M}_g(M,\beta)$ which characterizes deformations of the image of ϕ. This analysis was done by Witten in [28]. Since $\delta\phi^i = i\epsilon\chi^i$ we can impose the condition that this supersymmetric variation be holomorphic leaving $\overline{\partial}\chi = 0$ so by our previous identification we can think of χ as a variation in $T\mathcal{M}_g(M,\beta)$. To see this consider variations of the image of ϕ. These variations should also be holomorphic. Note that $d\phi^i = \partial\phi^i/\partial z^k dz^k$ (since ϕ is holomorphic) and since $\chi^i \in \phi^*(T^{1,0}M)$ then we see that varying the direction in the moduli space is associated to and element χ. We can now consider a measure on $\mathcal{M}_g(M,\beta)$ by pulling back forms from M.

Let $x_i \in \Sigma$ with $i \in \{1,\cdots,n\}$ and define the evaluation maps (one for each puncture) by

$$\begin{aligned} ev_i : \mathcal{M}_{g,n}(M,\beta) &\to M \\ \phi &\mapsto \phi(x_i). \end{aligned} \tag{295}$$

For $\xi \in H^\bullet(M)$ then we find the pullback of $ev_i^*\xi$ is a form in the moduli space and from our previous discussion these should be labeled by χ^i. Therefore we can view the vacuum expectation value of physical operators $O(x_i) = O_i$ as an integral over the moduli space. A correlation function of n physical operators is written as

$$\langle O_1 \cdots O_n \rangle = \sum_{\beta \in H_2(X,\mathbb{Z})} e^{-(\beta,\omega)} \int_{\mathcal{M}_{g,n}(M,\beta)} ev_1^*\xi_1 \wedge \cdots \wedge ev_n^*\xi_n =: \sum_{\beta \in H_2(X,\mathbb{Z})} e^{-(\beta,\omega)} N_{g,n}^\beta \tag{296}$$

where in the last equality we have defined the famous Gromov-Witten invariants $N_{g,n}^\beta$. As mentioned above these count the number of holomorphic maps from the n-puntured, genus g Riemann surface into the target space. Let us recast this expression to better understand how to compute these invariants. Consider the the cycles[21] D_i in M which are Poincare dual to the delta function representation of the Ω_{ξ_i}. That is, Ω has (delta function like) support in D_i. Let $C := \phi(\Sigma)$ be the image of the embedding and β its cohomology class. Then by our natural paring we expect

$$(\omega,\beta) = 2\pi \sum_{i=1}^{h^{1,1}} t_i d_i, \qquad d_i := C \cap D_i \tag{297}$$

where t_i is the (complexified) Kahler parameters defined by the special coordinates

$$t_i = \int_{S_i} \omega \tag{298}$$

for $span\{S_i\} = H_2(M)$. The parameters d_i are called intersection numbers and counts the number of times the two spaces intersect[22]. In the case that there are no interesctions $d_i = 0$ (or $\beta = 0$) the worldsheet is mapped to a single point so the moduli space is simply the entire target space

[21] In the literature these are known as divisors.

[22] There are a fair amount of subtleties assocaited with this and the reader if referred to **??** for the details.

$M_{g,n}(M,0) = M$. Since each physical operator has support in D_i we simply get the intersection number of the three operators Ω_{ξ_i}. Therefore the three point correlator can be expanded as the sum[18]

$$\langle O_1 \cdots O_n \rangle = D_i \cap D_j \cap D_k + \sum_{d_i \neq 0} N^{\beta}_{0,3} \int_{\beta} \xi_1 \int_{\beta} \xi_2 \int_{\beta} \xi_3 \prod_{i=1}^{h^{1,1}} e^{-2\pi i t_i d_i} \tag{299}$$

where $N^{\beta}_{0,0}$ is determined from the first terms and $N^{\beta}_{0,3}$ from the second. The second term can be thought of as a quantum correction which has brought many new insights into the study of 'stable maps' which the author admitly knows nothing about. See [14] for the gory details. For later use we define the Topological String prepotential

$$F_g(t) := \sum_{\beta} N^{\beta}_{g,n} \prod_{i=1}^{h^{1,1}} (e^{-2\pi t_i})^{d_i} \tag{300}$$

Deformations

Now, recall from our analysis of correlators of a 2d topological theory can be reduced to the product of order three in physical operator insertions $\langle O_{\xi_1} O_{\xi_2} O_{\xi_3} \rangle$. This notation make use fact that each operator is related to a differential form ξ. For the case of CY_3 by definition $c_1(TCY_3) = 0$ so by the results found in the appendix the difference of fermionic zero modes is

$$\ell_\chi - \ell_\psi = 2 \int_{\Sigma} \phi^* c_1(TCY_3) + 2m(1-g) \tag{301}$$

$$= 6. \tag{302}$$

To keep $\langle O_{\xi_1} O_{\xi_2} O_{\xi_3} \rangle$ from being anamolous ξ_i are (1,1) forms since both degrees must add to the (complex) dimension of CY_3. The genus zero correlator is therefore

$$C_{ijk}[t] = \langle O_{\xi_i} O_{\xi_j} O_{\xi_k} \rangle. \tag{303}$$

which is the Yukawa coupling defined by the prepotential of the special geometry. Its name comes from the fact that this term arises as a vertex interaction of vector multiplets. We therefore see that obtaining a prepotential of the theory amounts to finding a generating function of physical operators. The action of a cohomological theory was found to be invariant under deformations of the highest descendant state $O^{(2)}_{\xi_i}$ with respect to the Kahler moduli parameters t^i

$$\delta S = \sum_i t^i \int_{\Sigma} O^{(2)}_{\xi i} \tag{304}$$

so the insertion of such a term amounts to varying c_{ijk} by the kahler parameter

$$\frac{\partial}{\partial t^\alpha} C_{jkl}[t] = \frac{\partial}{\partial t^\alpha} \langle O_{\xi_j} O_{\xi_k} O_{\xi_l} \rangle = \langle O_{\xi_j} O_{\xi_k} O_{\xi_l} \int_\Sigma O^{(2)}_{\xi i} \rangle. \tag{305}$$

Since we are considering a surface isomorphic to S^2 one must fix at least three points which is often chosen to be the points $0, 1, \infty$.

8 B-model

The B-model turns out to be just as rich in structure as the A-model but much simpler. In the A-model localized on holomorphic maps while the B-model will localize on constant maps. Recall that the B-model was defined through a $(-,-)$ twist of the fields ψ_+, ψ_- which means they are now sections of the following bundles

$$\psi^i_+ := \psi^i_z \in \Gamma(\Sigma, K \otimes \phi^*(T^{1,0}M)) \qquad \psi^{\bar{i}}_+ \in \Gamma(\Sigma, \phi^*(T^{0,1}M)) \tag{306}$$

$$\psi^i_- := \psi^i_{\bar{z}} \in \Gamma(\Sigma, \overline{K} \otimes \phi^*(T^{1,0}M)) \qquad \psi^{\bar{i}}_- \in \Gamma(\Sigma, \phi^*(T^{0,1}M)). \tag{307}$$

As reasoned in the A-model, after twisting we see that the combination

$$Q_B := \overline{Q}_+ + \overline{Q}_- \tag{308}$$

transforms as a scalar under the embedded $U(1)'_E$ group. This combination is the BRST operator of the topological theory. We now define the following variables to make both the spin structure of these fields more apparent

$$\theta_i := g_{i\bar{j}}(\psi^{\bar{j}}_+ - \psi^{\bar{j}}_-). \qquad \eta^{\bar{i}} := \psi^{\bar{i}}_+ + \psi^{\bar{i}}_-. \tag{309}$$

and like the A-model this theory admits a slightly modified version of the conformal OPE's of the untwisted model

$$T(z)T(w) \sim \frac{2T(w)}{(z-w)^2} + \frac{\partial_w T(w)}{(z-w)} \tag{310}$$

$$T(z)G^\pm(w) \sim \frac{(3 \pm 1)G^\pm(w)}{2(z-w)^2} + \frac{\partial_w G^\pm(w)}{(z-w)} \tag{311}$$

$$T(z)J(w) \sim +\frac{J(w)}{(z-w)^2} + \frac{\partial_w J(z)}{(z-w)} + \frac{c}{3(z-w)^3} \tag{312}$$

$$G^+(z)G^-(w) \sim +\frac{(2c/3)}{(z-w)^3} + \frac{2T(w) + \partial_w J(w)}{(z-w)}. \tag{313}$$

Rewriting the $N = (2,2)$ action in terms of this field content we find the B-model action is given by

$$S_B = t\int_\Sigma \left(g_{i\bar{j}}\partial_z\phi^i\partial_{\bar{z}}\phi^{\bar{j}} + ig_{i\bar{j}}\eta^{\bar{j}}(D_{\bar{z}}\psi^i_z + D_z\psi^i_{\bar{z}}) + i\theta_i(D_{\bar{z}}\psi^i_z - D_z\psi^i_{\bar{z}}) + \frac{1}{2}R_{i\bar{j}k}{}^l\psi^i_z\psi^k_{\bar{z}}\eta^{\bar{j}}\theta_l \right) \tag{314}$$

the topological variations of this action are given by

$$\delta\phi^i = 0, \qquad \delta\phi^{\bar{i}} = i\epsilon\eta^{\bar{i}} \tag{315}$$

$$\delta\theta_i = 0, \qquad \delta\eta^{\bar{i}} = 0 \tag{316}$$

$$\delta\psi^i_\mu = -\epsilon\partial_\mu\phi^i, \qquad \delta\psi^{\bar{i}} = -\epsilon\partial_\mu\phi^{\bar{i}}. \tag{317}$$

In attempting to write this action as a Q_B exact term we similarly find that it is exact up to an extra term with is independent of the Kahler metric of the target space [30]

$$S = it\int_\Sigma d^2z\{Q,V\} + t\int_\Sigma d^2z\left(i\eta_i(D_{\bar{z}}\psi^i_z - D_z\psi^i_{\bar{z}}) + \frac{1}{2}R_{i\bar{j}k}{}^l\psi^i_z\psi^k_{\bar{z}}\eta^{\bar{j}}\theta_l \right) \tag{318}$$

where $V = -g_{i\bar{j}}(\psi^i_z\partial_{\bar{z}}\phi^{\bar{j}} + \partial_z\phi^{\bar{j}}\psi^i_z)$. The argument here is that a variation of the action by kahler class elements is Q_B exact and therefore do not depend on the Kahler moduli. This extra term is antisymmetric in z and $\bar{z}$ so this can be thought of as a (1,1)-form which after integration is naturally independent of the metric so it is also independent of the worldsheet metric. We do find however that the theory depends on the target space complex structure. From the variations of ϕ above it is easy to see that this is true. It is now clear that the B-model is a theory which depends on the complex structure moduli while the A-model was found to only depend on the Kahler class moduli. Conceptually it is now easy to see how Mirror symmetry is manifest in the language of topological sigma models: The A-model on a Calabi-Yau threefold M is the 'mirror' theory of the B-model on a mirror Calabi-Yau threefold W.

8.0.1 Localization

Since we impose the requirement that physical operators are Q_B closed we can only consider ϕ, $\eta^{\bar{j}}$ and θ_i. At the level of CFT this again means we consider fields with conformal weight $(0,0)$. The most general physical operator is therefore of the form

$$\mathcal{O}(z,\bar{z}) = \mathcal{O}_{\bar{i}_1\cdots\bar{i}_p}{}^{j_1\cdots j_q}\eta^{\bar{i}_1}\cdots\eta^{\bar{i}_p}\theta_{j_1}\cdots\theta_{j_q} \tag{319}$$

and since the only nonzero supersymmetry variation comes from $\{Q_B,\phi^{\bar{i}}\} = -\eta^{\bar{i}}$ we find

$$\{Q_B,\mathcal{O}\} = -\frac{\partial\mathcal{O}_{\bar{i}_1\cdots\bar{i}_p}{}^{j_1\cdots j_q}}{\partial\eta^{\bar{k}}}\eta^{\bar{k}}\eta^{\bar{i}_1}\cdots\eta^{\bar{i}_p}\theta_{j_1}\cdots\theta_{j_q} \tag{320}$$

which is exactly minus the (twisted) Doubeault operator. For η we therefore make the identification $\eta^{\bar{i}} \leftrightarrow d\bar{z}^i$ and since $\theta_i = g_{i\bar{j}}\eta^{\bar{j}} - 2g_{i\bar{j}}\psi_-^{\bar{j}}$ one concludes we identify θ_i with $\partial/\partial z^i$. Because the the latter operators were antisymmetric this identification should be made in $\bigwedge^q(T^{1,0}M)$. From all this there is the following isomorphism for the cohomology of physical operators $H_B(Q)$:

$$H(Q_B) \cong \bigoplus_{p,q=0}^{n} H_{\bar{\partial}}^{0,p}\left(M, \bigwedge^q T^{1,0}M\right) \tag{321}$$

$$Q_B \cong \bar{\partial} \qquad \eta^{\bar{i}} \cong d\bar{z}^i \qquad \theta_i \cong \frac{\partial}{\partial z^i}. \tag{322}$$

Now that we have identified the physical operators of the theory we would not how to determine what localaization tells us about the theory. Again we will work with the generic case in which we assume that the fields ψ do no contribute to the zero modes:ℓ_ψ. We will also assume that we are working on a Calabi-Yau manifold so that $c_1(TM) = 0$. To ensure the vector R-symmetry not become anomalous a physical operator must have an equal number of each field. From the appendix this means the number of zero mode contributions from each field is equal to the dimension of the target space $\ell = dim_{\mathbb{C}}(M)$. If we again argue that the action localizes on the classical equations of motion[23] for the embedding map or equally by setting the fermionic variations to zero we find

$$\partial_z\phi^i = \partial_z\phi^{\bar{i}} = \partial_{\bar{z}}\phi^i = \partial_{\bar{z}}\phi^{\bar{i}} = 0. \tag{323}$$

The only such solution to this is if ϕ is the constant map in which ϕ maps the entire world-sheet to a single point. Since there exists a map for each point of the target space the moduli space of constant maps contains the target space

$$M \subseteq \mathcal{M}_g(M). \tag{324}$$

One may have assumed that the moduli space is equal to the entire space and this happens to be true when we consider [14] a 3-punctured two sphere. The reason for this is that the three point correlator, as we mention in the A-model analysis, is fixed by specifying three points. Note that the operators as written cannot give a preoper measure of integration as they do not wedge to a top form. To obtain an integration measure we simply contract contract $\mathcal{O}$ with the (unique) holomorphic $(3,0)$ form Ω so that we obtain a top form:

$$\mathcal{O}_{\overline{i_1 i_2 i_3}}{}^{j_1 j_2 j_3} d\bar{z}^{i_1} \wedge d\bar{z}^{i_2} \wedge d\bar{z}^{i_3} \mapsto \mathcal{O}_{\overline{i_1 i_1 i_3}}{}^{j_1 j_2 j_3}\Omega_{j_1 j_2 j_3} d\bar{z}^{i_1} \wedge d\bar{z}^{i_2} \wedge d\bar{z}^{i_3} \wedge \Omega. \tag{325}$$

Which is exactly the Yukawa coupling term defined with respect to the prepotential of the complex structure moduli. This comes as no suprise as the B-model describes the complex moduli of the

[23] It is important to remember that these are the classical equations of motion with respect to exact part of the action

Calabi-Yau in consideration. The three point correlator on a Riemann surface of genus 0 is therefore

$$C_{ijk} = \partial_i \partial_j \partial_k F = \langle O_i O_j O_k \rangle = \int_M O_{\overline{i_1 i_1 i_3}}{}^{j_1 j_2 j_3} \Omega_{j_1 j_2 j_3} d\overline{z}^{i_1} \wedge d\overline{z}^{i_2} \wedge d\overline{z}^{i_3} \wedge \Omega. \tag{326}$$

Mirror Symmetry

The A and B models are related through mirror symmetry since each theory classifies half of the Calabi-Yau moduli. Generally calculations made in the A-model are much simplier than those in the B-model. Therefore the mirror isomorphism is an extrememly useful computational too to make calculate invariants in one theory and then translate them back to the other through mirror symmetry. In the mathematical literature this led to the study of Homological mirror symmetry. For the details of this see [14].

9 Topological String Theory

In tthis section we will couple the topological sigma models to gravity. This turns out to couple in exactly the same way that the bosonic string is coupled to gravity. We will then consider the D-branes which we can obtain from these string theories. We conclude with by stating how different topological invariants can be obtained from the topological string.

9.1 Lessons learned from the bosonic string

It turns out [8] the bosonic string is a topological theory if we take the target space to be a Calabi-Yau threefold. The reason for this is that this thoery has the same anomaly as our twisted topological sigma models on Calabi-Yau threefolds. Therefore we can take the analysis done for constructing a measure over the moduli of this system and do exactly the same for our models. Let us briefly state the basics of the Bosonic string theory.

In order to extend the A and B models to topological string theories we must couple the theory to gravity. In the path integral formalism this simply means that we integrate over all possible conformally invariant metrics of a Riemann surface of genus g and then sum over all possible surfaces. The partition function is then of the form

$$Z = \sum_{g=0}^{\infty} \int \mathcal{D}\phi \mathcal{D}h_g e^{-S[\phi,h]_g} \tag{327}$$

where ϕ is the the worldsheet embedding $\phi : \Sigma \to M$ and h is the worldsheet metric. The (classical) bosonic string is described by the Polyakov (sigma-model) action[24]

$$S[\phi, h] = \int_{\Sigma_g} d^2\sigma \sqrt{-h} h^{\alpha\beta} \partial_\alpha \phi^\mu \partial_\beta \phi_\mu \tag{328}$$

where σ are local coordinates on the world sheet σ. Our analysis is based on [19] [23]. Let $Diff(\Sigma)$ be the group of diffeomorphisms on Σ defined by $h \mapsto f^* h$ which also acts accordingly on the embedding maps $\phi^\mu \mapsto f^*\phi$ where $f : \sigma^\mu \mapsto \sigma'^\mu(\sigma)$, $f \in Diff(\Sigma)$ and let $Weyl(\Sigma_g)$ be the group of Weyl rescalings defined by $h \mapsto e^\theta h$. A full conformal variation of the metric is of the form

$$h_{\mu\nu} = e^\theta \frac{\partial \sigma^\alpha}{\partial \sigma'^\mu} \frac{\partial \sigma^\beta}{\partial \sigma'^\nu} h_{\alpha\beta}$$

Then it is easy to see that the Polyakov action is invariant under this combined symmetry and since

[24] Strictly speaking there is also an additional topological term which acts as a string coupling but we will ignore it here.

the pair (ϕ, h) characterizes the classical field of the theory we find this invariance is a classical symmetry of the action. In order to avoid infinities which arise from overcounting equivalent geometries when we quantizate the theory we gauge fix by taking equivalnce classes of the pair (ϕ, h) which are related via action of the group[25] $Diff \ltimes Wely$ and define our measure over this space. To formalize this let $\mathcal{E} := \{\phi : \Sigma \to M\}$ be the space of embeddings and $\mathcal{Y}_g$ the set of all metrics which are defined on Σ. The action of our measure is then defined on the space $(\mathcal{E} \times \mathcal{Y}_g)/\sim := (\mathcal{E} \times \mathcal{Y}_g)/Diff \ltimes Wely$ where the equivalence relation of the space is defined by $(\phi, h) \sim (f, e^\theta)(\phi, h) = (f^*\phi, e^\theta f^* h)$. The subspace $\mathcal{M}_g := \mathcal{Y}_g / Diff \ltimes Weyl$ is the Moduli space of Σ as it characterizes the solution space for the metric. Therefore the string partition function is given by

$$Z_g := \int_{(\mathcal{E}\times\mathcal{Y}_g)/\sim} [D\phi][Dh] e^{S[\phi,h]} = \int_{(\mathcal{E}\times\mathcal{Y}_g)} \frac{D\phi Dh}{Vol(Diff \ltimes Weyl)} e^{-S[\phi,h]} \tag{329}$$

where $vol(Diff \ltimes Weyl)$ is the formal volume of the group. Although the integration measure as currently written is infinite by virtue of the Feynmann path integral it turns out that it is actually finite dimensional. To see this we take a reference metric h and consider slight deformations δh. These deformations are elements of the tangent space of $(\mathcal{E} \times \mathcal{Y}_g)$ and are characterized by beltrami differentials $(\mu_i) = (\mu_i)^z_{\bar{z}} d\bar{z} \otimes \frac{\partial}{\partial z} \in T\mathcal{M}_g$, $i \in \{1, \cdots 3g-3\}$. The formalities associated with obtaining this result requires introducing a great deal of new ideas and although they are striaght forward we will omit them and refer to reader to [19] for the details.

In determining the string measure via the Faddeev-Popov method we obtain a ghost term

$$S_{gh} = \int d^2\xi b_{\alpha\beta} \partial^\alpha c^\beta \tag{330}$$

where b is a spin two ghost field. The energy momentum tensor of the theory can be written as $T(z) = \{Q_{BRST}, b_{zz}\}$ where Q_{BRST} is the BRST which arises from the introduction of ghost terms. The string amplitudes are of n-point functions are non-vanishing for $3g - 3 + n$ insertions of $b(z)$ [23]

$$\langle \phi_{i_1} \cdots \phi_{i_n} \rangle_g = \int_{\mathcal{M}_g} \left\langle \phi_{i_1} \cdots \phi_{i_n} \prod_{i=1}^{3g-3+n} dm^i d\overline{m}^i \int_\Sigma b_{zz} (\mu_i)^z_{\bar{z}} \int_\Sigma b_{\bar{z}\bar{z}} (\overline{\mu}_i)^{\bar{z}}_z \right\rangle \tag{331}$$

where m_i are elements of the cotangent bundle of the moduli space dual to μ_i. The lesson learned here is for twisted sigma models we can take take the nonphysical operator $G_{\alpha,\beta}$ from the expression $T_{\alpha\beta} = \{Q, G_{\alpha\beta}\}$ and use this to build a measure over the moduli space.

[25] Let $(G, *)$ and $(H, \circ)$ both be Lie groups and denote a left action of G on H by $\eta : G \times H \to H$. Then $G \ltimes H$ is semidirect product of G and H (with respect to η) defined by the group $(\{(g, \alpha) \in G \times H\}, \cdot)$ with the binary operation $(g, \alpha) \cdot (l, \beta) := (g * l, \alpha \circ \eta(g, \beta))$

9.2 Quantum Gravity

If we now couple our theories to graivity by including the metric as a spin 2 field called the graviton and its superpartner the gravitino Ψ_μ^α. The partition function will now take the form

$$Z \sim \sum_{g=0}^{\infty} \int_{(\mathcal{E}\times\mathcal{Y}_g)} \frac{Dh}{Vol(Diff \ltimes Weyl)} \int \mathcal{D}\phi\mathcal{D}[\psi] e^{-S[\phi,h,\psi]} \tag{332}$$

where ψ represents the measure for all fermionic fields in either model. The exact form is given defined by the *perturbative topological string amplitude*:

$$F = \sum_{g=0} \lambda^{2g-2} F_g \tag{333}$$

where λ is the topological string coupling constant and F_g is the topological free energy of the string defined by

$$F_g := \int_{\mathcal{M}_g} \left\langle \prod_{i=1}^{3g-3} dm^i d\overline{m}^i \int_\Sigma d^2z G_{zz}(\mu_i)^z_{\overline{z}} \int_\Sigma d^2z G_{\overline{zz}}(\overline{\mu}_i)^{\overline{z}}_z \right\rangle \tag{334}$$

where $\langle\ \rangle$ is taken for the respective A or B type theory and G is the operator related to the energy momentum tensor via $T = \{Q_{A/B}, G\}$. There are a couple of notes to be made about this definition. We have written the integral over conformally invariant metrics as $\mathcal{M}_g$ as it turns out the measure on $\mathcal{E} \times \mathcal{Y}_g$, after having factored out the group action, is completely determined by the complex structure. Therefore this moduli is simply the complex structure moduli. The expression for this partition function should also be thought of as a perturbative expansion in λ. This due to the fact that the free energy is simply the generating function for connected Feynmann diagrams and is related to the partition function through

$$Z[t] = exp(F[t]). \tag{335}$$

We can now utilize this measure to analysze the topological A and B- models as topological string theories. As before we will mainly focus on the A-model.

9.3 Topological branes

When we constructed the A and B-model we considered closed Riemannian surfaces. If we are to introduce branes we must generalize this discussion to worldsheets with boundaries. The introduction of boundaries will now lead to loss of symmetries since the worldsheet is no longer translationly invariant. By losing translational invariance we also loose the supersymmetry the original theory

had. In introducing branes we mentioned that half of the supersymmetry could be preserved if the 'right' boundary conditions were imposed. We will sketch how to obtain the boundary condtions that preserved this amount of supersymmetry my analyzing the conformal fields. We will follow the arguments in [20] for which the reader can find further details.

A-model branes (A-branes)

Let $\gamma = \phi(\partial\Sigma)$ be the sub-manifold which we map the boundary of the worldsheet onto. The A-model BRST operator was given by $Q_A = \oint dz G^+ + \oint d\bar{z}\overline{G}^+$ and so at $\mathfrak{R}(z)$ the super currents should satisfy

$$G^+ = \pm\overline{G}^+, \qquad G^- = \pm\overline{G}^- \tag{336}$$

which means the R currents should also be equal $J = \overline{J}$ at this boundary. From the supersymmetry transformations of the theory it is easy to see at the level of operators we have an expression of the form

$$\partial\phi^\mu = M^\mu_\nu(\phi)\bar{\partial}\phi^\nu, \quad \psi^\mu = \pm M^\mu_\nu(\phi)\psi^\nu_\mp \tag{337}$$

where M is some matrix as a function of ϕ and μ, ν now run over both holomorphic and anti-holomorphic indicies. This is simply stating that at the boundary we have an exact a relationship between left and right movers. We note that $G := G^+ + G^- = \pm\overline{G}^\pm$ and $T = \overline{T}$ on $\mathfrak{R}(z)$ being conserved is the statement that the $\mathcal{N} = 1$ subalgebra is preserved. These conditions together tell us that $M(\phi)$ is an orthogonal matrix

$$g_{\lambda\sigma}M^\lambda_\mu(\phi)M^\sigma_\nu(\phi) = g_{\mu\nu}. \tag{338}$$

If we now apply our topological variations to the first equation, for instance, then we find

$$M^i_j = M^{\bar{i}}_{\bar{j}} = 0. \tag{339}$$

Recall that $\partial\phi = +\bar{\partial}\phi$ led to Neumann boundary condition while $\partial\phi = -\bar{\partial}\phi$ led to Dirchlet boundary conditions. If we take a normalized eigenvector ϱ of R then the value of its eigenvalue determines whether or not the vector lies tangent or normal to the brane. In particular if $M\varrho = +\varrho$ defines Neumann boundary conditions so ϱ lies tangent to γ. However since the (almost) complex structure J is diagonal we find $M(J\varrho) = -\varrho$ so φ is also normal to γ. Since $J^2 = -1$ then $J^2\varrho = -\varrho$ and so the normal and tangent directs are not independent. Is means that the dimension of the sub-manifold is half of the dimension of γ. Because $J\varrho$ and ϱ are normal to one another we have

$$g(J\varrho, \varrho) = 0 \quad \Longrightarrow \quad \omega\big|_\gamma = 0 \tag{340}$$

and hence the Kahler form vanishes on γ. This defines a Lagrangian submanifold with the Kahler form the symplectic form on *gamma*. Therefore we conclude the following:

A D-brane wrapped on γ is a Lagrangian submanifold of (M, ω) which we call a topological A-brane.

When we introduced branes it was also noted that branes can be thought of as admitting a vector bundle structure over them. By introducing a B-field $B \in H^2(M, \mathbb{Z})$ on the target space we obtain a phase factor

$$exp\left(2\pi i t^{\alpha} \int_{\Sigma} \phi^* B\right). \tag{341}$$

This factor will change the boundary conditions imposed above. We will not give a thorough analysis of these conditions but will instead give an idea of how the theory changes. The deatils can be found in [14]. The B-field was well defined under $B \mapsto B + d\Lambda$ since the last term in

$$\int_{\sigma} \phi^* B \mapsto \int_{\Sigma} \phi^*(B + d\Lambda) = \int_{\Sigma} \phi^* B + \int_{\partial\Sigma} \phi^* \Lambda \tag{342}$$

vanished. We remedy this by constructing a vector bundle over γ which has a $U(1)$ connection $A \in \Sigma(\gamma)$ on it. This allows us to modify the action

$$S \mapsto S - 2\pi i t^{\alpha} \int_{\partial\Sigma} \phi^* A \tag{343}$$

and require the gauge condition $A \mapsto A + \Lambda$ as $B \mapsto B + d\Lambda$. In the present case it is easy to see that the B field is really the field strength of the connection $B = F = dA$. Moreover if we require the modified action to be BRST closed under Q_A then one must impose the condition $F = 0$. This means that a topological A-brane is a flat $U(1)$ gauge bundle over a symplectic manifold γ.

Suppose we have N D-branes which wrap M and let $\Sigma_{g,h}$ be a Riemannian surface of genus g with h boundaries D_i, for $i \in \{1, \cdots, h\}$

$$\partial\Sigma_{g,h} = \bigsqcup_{i}^{h} D_i \tag{344}$$

each mapped to some Lagrangian sub-manifold $\phi(D_i) \subset \gamma$. Then from the modification to the action the path integral will contain the following term

$$\prod_i \mathcal{P} exp\left(\oint_{D_i} \phi^* A\right) \tag{345}$$

where A is now a $U(N)$ gauge connection on γ pulled back to the brane.

This is not the first time that we have run into a symplectic manifold with a flat connection: the Chern-Simons moduli space was the space of flat connections which admitted the symplectic structure $\omega = \int \delta A \wedge \delta A$. In the next section we will obtain a duality relation between the Topological A-model and Chern-Simons on the resolved conifold. Lately we note that one requires a more sophisticated method to describe multiple branes which differ in dimension. In such cases it is much more natural to define these structures by sheaves. This abstraction can be described by the 'Fukaya' category. Again we refer the reader to [14] for this description.

B-model branes (B-branes)

We can follow the exact same analysis as was done for the *A*-model. The conditions we would like to impose are

$$G^+ = \pm\overline{G}^-, \qquad J = -\overline{J}. \tag{346}$$

If we proceed as in the *A*-model and define M the constraint on M which satisfies these boundary conditions are

$$M^{\bar{i}}_{j} = M^{i}_{\bar{j}} = 0 \tag{347}$$

and by the same reasoning the tangent and normal bundle are independent and therefore preserve the complex structure of the target space. Therefore

A D-brane wrapped on γ is a complex sub-manifold of (M, J) which we call a topological B-brane.

Adding a gauge field to the theory and requiring BRST invariance of Q_B we find [14] that B can only be a (1,1)-form field strength and so we can build a holomorphic vector bundle over it. Again if we consider multiple branes then the sheaf is described by the derived category of coherent sheaves.

Spacetime supersymmetry

In understanding the possible branes that the twisted topological string theories posses it is important to note that the above analysis was done to preserve supersymmetry on the worldsheet. When we compactify supergravity theories we would like to preserve spacetime supersymmetry as well. Let us give a brief outline of how this is done (see [20] for further details).

We start by bosonizing[26] the $U(1)$ currents

$$J = \sqrt{3}\partial\phi \qquad \overline{J} = \sqrt{3}\overline{\partial}\overline{\phi} \tag{348}$$

which satisfy the OPE

$$J(z)J(w) \sim \frac{3}{(z-w)^2}. \tag{349}$$

The bosonized supersymmetry generators will obtain a term $exp(i\sqrt{3}/2\phi)$ for left movers and the conjugate for right movers. For the A-model Imposing an equivalence on $\Re(z)$ geometrically this is equivalent to the condition

$$\Omega\Big|_\gamma = \overline{\Omega}\Big|_\gamma \tag{350}$$

where Ω is the $(3,0)$-form on the Calabi-Yau manifold. We know however that the condition $det(g) = \Omega \wedge \overline{\Omega}$ holds on any Calabi-Yau. Therefore Ω is a volume form on γ. A manifold γ with these properties is called a special lagrangian manifold and such spaces are required if would like to preserve spacetime supersymmetry. Similar conditions can be found for the B-model.

9.4 Gromov-Witten invariants

In our analysis of the A-model we mentioned that the free energy asymptotics in the string coupling λ is given by

$$F[t] = \sum_g \sum_{\beta \in H_2(M)} N_g^\beta q^\beta \lambda^{2g-2} \tag{351}$$

where the worldsheet instant amplitude q is given by $q^\beta = exp(2\pi i \beta_\alpha t^\alpha)$ and N_g^β are the Gromov-Witten invariants. These are topological invariants of M which are calculated by the topological A-model. Since we are considering constant maps ϕ into the Calabi-Yau M then $\beta = 0$ and we have the following genus expansion

$$F = \frac{1}{\lambda} N_0^{\beta=0} + N_1^{\beta=0} + \sum_{g\geq 2} \lambda^{2g-2} N_g^{\beta=0} + \sum_{g\geq 0} \sum_{\beta \neq 0} \lambda^{2g-2} F_g^\beta \tag{352}$$

[26] It turns out [23] if a free scalar field is defined on a circle of appropriate radius the theory is equivalent to the same theory replaced by fermionic fields. This is known as bosonization and is related to spectral flow between the RR and NS sectors mentioned earlier.

if we relate this to our previous analysis we find $N_0^{\beta=0} = \frac{1}{6}D_{ijk}t^i t^j t^k$. For $g \geq 2$ the constant map constirbution is determined by

$$N_{g,0} = \frac{(-1)^g \chi(M)}{2} \int_{\mathcal{M}_g(M,0)} c_{g-1}^3(T\mathcal{M}_g) \tag{353}$$

where $\chi(M)$ is the Euler characteristic of M and c_{g-1} the $(g-1)$th Chern-calss of the Hodge bundle over $\mathcal{M}_g$. In particular one can obtain a generating function as an asymptotic in λ which has the explicit expression [18]

$$F = -\frac{\chi(M)}{2}\left[\frac{\zeta(3)}{\lambda^2} + C - \sum_{g=2}^{\infty} \lambda^{2g-2} \frac{(-1)^g |B_{2g}||B_{2g-2}|}{2g(2g-2)(2g-2)!}\right] \tag{354}$$

where ζ is the zeta function,

$$C = \frac{1}{12} log\left(\frac{2\pi i}{\lambda}\right) - \frac{1}{2\pi^2}\zeta'(2) + \frac{1}{12}\gamma_E \tag{355}$$

B_n are Bernoulli numbers and γ_E the Euler-Masheroni constant. Therefore it is possible to calculate the Gromov-Witten invariants explicitly using perturbatative topological string theory. By considering the B-model as well as M-theory compactification more topological invariants can be obtained. See [14] and [18] for this discussion.

10 Open-Closed String Dualities

One of the most exciting and interesting apects of topological strings is the so called open-closed string dualities. These are dualities which relate open string theories to closed string theories. The most well known duality of this type is the AdS/CFT correspondence in which one finds $\mathcal{N} = 4$, $U(N)$ super Yang-Mills on $\mathbb{R}^{1,3}$ is dual to the closed type IIB superstring theory on $AdS \times S^5$. The former can be realized as the so called worldvolume limit of D-branes of type IIB open string on $\mathbb{R}^{1,9}$ with N D-branes wrapping $\mathbb{R}^{1,3}$. This correspondence is realized by taking the large N limit which reduces the theory to a matrix model calculation. What we learn from this is the notion of geometric transition: each theory is related to one another on different geometrical backgrounds. In this section we will focus on a single aspect of these beautiful dualities and as we shall see, doing so will combine every concept covered in this thesis thus far. The duality of interest is the duality between the Chern-Simons open string field theory and the closed A-model topological string theory.

10.1 Chern-Simons as an open string field theory

In [27] constructed a nonperturbative field theory of open strings. Although we will not require all the details of Witten's noncomuutative open string theory let us list some of the formal details that underpin it. We follow [25]. It is well known that the supersymmetry algebra splits up into two distinct types of elements. One for bosonic terms and one for fermionic terms. If we denote a bosonic term by b and a fermionic one by f then we know that the following multiplication rules follow: $bb = b$, $bf = f$, $ff = b$. This multiplication is formalized by graded algebras. In the case of the supersymmetry algebra we have a $\mathbb{Z}_2$ grading of the algebra. A ring which splits as $L = \bigoplus_{k=0}^{N-1} L_k$ with the product $\circ : (l_i, l_k) \mapsto l_i \circ l_k \in L_{(i+k)mod(N)}$ is called an N-graded algebra. The degree of an element $l \in L$ is given by $deg(l) = i$ if $l \in L_i$. We denote this degree by $g_l := deg(l)$. With this we can now begin to define the basic properties of Witten's open string theory.

Definition 18. *Let $\mathfrak{A}$ be a graded algebra with the non-commutative product $\star : \mathfrak{A} \star \mathfrak{A} \to \mathfrak{A}$ called the star product. Elements of $\mathfrak{A}$ are called string fields and the operator satisfies $deg(\Phi \star \Psi) = deg(\Phi) + deg(\Psi)$ for $\Phi, \Psi \in \mathfrak{A}$. We require this algebra to have additional structure:*

1. *Compatibility with a derivation (BRST operator) $Q : \mathfrak{A} \to \mathfrak{A}$ of degree 1 such that $deg(Q\Psi) = 1 + deg(\Psi)$*

2. *An 'integration' operand $\int : \mathfrak{A} \to \mathbb{C}$ which localizes on string states of degree 3*

$$\int \Psi = 0 \Leftrightarrow deg(\Psi) \neq 3. \tag{356}$$

We will call any such algebra an (open) string algebra.

The degree of an object in the algebra is referred to as the ghost number since in most instances the BRST operator Q of the open string theory is manifest from the addition of ghost fields.

Definition 19. *A string field theory is a collection* $(Q, \star, \int, \mathfrak{A})$ *such that*

1. *(Nilpotency)* $Q^2 = 0$
2. *(Inverse)* $\int Q\Psi = 0$ *for all* $\Psi \in \mathfrak{A}$
3. *(Cyclicity)* $\int \Psi \star \Phi = (-1)^{G_\Psi G_\Phi} \int \Phi \star \Psi$ *for all* $\Psi, \Phi \in \mathfrak{A}$
4. *(Associativity)* $(\Psi \star \Phi) \star \Xi = \Psi \star (\Phi \star \Xi)$ *for all* $\Psi, \Phi, \Xi \in \mathfrak{A}$

Let us know 'derive' the Chern-simons open string Lagrangian. Define the curvature $\mathcal{F} := Q\mathcal{A} + \mathcal{A} \star \mathcal{A}$ with $\delta\mathcal{F} = \mathcal{F} \star \Lambda - \Lambda \star \mathcal{F}$. By considering the self dual solution to an action similar to that of the Yang-Mills action $\int \mathcal{F} \star \mathcal{F}$ we find

$$\begin{aligned} \int \mathcal{F} \star \mathcal{F} &= \int (Q\mathcal{A} + \mathcal{A} \star \mathcal{A}) \star (Q\mathcal{A} + \mathcal{A} \star \mathcal{A}) \\ &= \int (Q\mathcal{A} \star Q\mathcal{A} + Q\mathcal{A} \star \mathcal{A} \star \mathcal{A} + \mathcal{A} \star \mathcal{A} \star Q\mathcal{A}) \\ &= \int Q\left(\mathcal{A} \star Q\mathcal{A} + \frac{2}{3}\mathcal{A} \star \mathcal{A} \star \mathcal{A}\right) \\ &= 0 \end{aligned} \tag{357}$$

where in going from line one to two we have used the fact that the integral vanishes for terms of which are not of degree 3. Property 3. for a string field theory is invoked in going from line two to three and the final line is from property 2. The Chern-Simons open string action is

$$S = \frac{1}{2\lambda}\left(\int \mathcal{A} \star Q\mathcal{A} + \frac{2}{3}\mathcal{A} \star \mathcal{A} \star \mathcal{A}\right) \tag{358}$$

for a degree 1 (ghost number 1) string field $\mathcal{A}$ and a coupling λ. Note that in order for this integral to be nonzero the the integration operator has ghost number -3 and so $G_\Phi = 1$. This action can also be understood in the context of CFT and vertex operators but we shall not do so here. A variation of the action $S \mapsto S + \delta S$ is invariant under the following 'gauge transformation' $\delta\mathcal{A} = Q\Lambda + \mathcal{A}\star\Lambda - \Lambda\star\mathcal{A}$ for $\Lambda \in \mathfrak{A}$ and $G_\Lambda = 0$. Let us now apply this abstraction to the open string fields. The string fields $\mathcal{A}$ are thought of as a functional of the embedding maps ϕ which are defined is defined at all points along the string. We can think of both the $\star$ operator and $\int$ as physical actions of the string and mathematically as functionals of the algebra of embedding functionals $\mathfrak{A}$. The integral operator simply takes the string, bends it in half, and glues the two pieces together. This action has the explicit form

$$\int \mathcal{A} := \int \mathcal{D}\phi \prod_{0 \le \sigma \le \frac{\pi}{2}} \delta[\phi(\sigma) - \phi(\pi - \sigma)]\mathcal{A}[\phi]. \tag{359}$$

The star product is defined similarly. It takes two strings, bends each one in half and glues the two bent halves together. For n string fields this concatenation is defined through

$$\int \mathcal{A}_1 \star \cdots \star \mathcal{A}_n := \int \prod_{i=1}^{n} \mathcal{D}\phi_i \prod_{i=1}^{n} \prod_{0 \leq \sigma_i \leq \frac{\pi}{2}} \delta[\phi_i(\sigma_i) - \phi_{i-1}(\pi - \sigma_{i-1})] \mathcal{A}_i[\phi_i] \tag{360}$$

The interpreation of $\mathcal{A}$ is that it is a a string field (functional) of the bosonic string fields ϕ, c and b

$$\mathcal{A} = \mathcal{A}[\phi(\sigma), c(\sigma), b(\sigma)]. \tag{361}$$

Because it is a functional of the zeromodes of the bosonic string it properly describes the midpoints which are used to define the above operators. In the limit that these zeromodes decouple the theory reduces to Chern-Simons and can further be related to the topological A-model string theory. This was was argued in [31] for the case in which the target space is the cotangent bundle T^*S^3 of the three shere. We now outline how this occurs following [31] and [18].

10.1.1 A-model on the Deformed Conifold

Let M be a general three manifold with $dim_{\mathbb{R}}(M) = 3$ and coordinates[27] q^α, $\alpha \in \{1, 2, 3\}$ and consider its cotangent bundle T^*M. Trivially M is a Lagrangian sub manifold so we may consider wrapping,say N, branes onto the base M. To understand the weak coupling limit we are interested in the classical limit $t \to \infty$ where t is the coupling parameter introduced in deriving the $N = (2, 2)$ sigma model. We now canonically quantize the theory so that we obtain a Hamiltonian picture. Because the A-model admits a flat connection we consider the case in which the world-sheet Σ is an infinite strip locally defined by coordinates $\sigma \in [0, \pi]$, $-\infty \leq \tau \leq \infty$ and metric $ds^2 = d\sigma^2 + d\tau^2$. The Grassmanian field ψ is one form valued and can be expanded as $\psi = \psi_\theta d\theta + \psi_\tau d\tau$. By self duality

$$(*\psi)_\alpha = \varepsilon_{\alpha\beta}\psi_\beta = \psi_\alpha \tag{362}$$

these two parameters are not independent. The commutation relations are easily read off from the A-model action

$$\left[\phi^I(\sigma'), \frac{d\phi^J}{d\tau}(\sigma)\right] = \frac{i}{\tau} g^{IJ} \delta(\sigma - \sigma') \tag{363}$$

$$\{\psi_\tau(\sigma), \chi(\sigma')\} = \frac{1}{t}\delta(\sigma - \sigma') \tag{364}$$

[27] Here we are making use of the formalism introduced for symplectic manifolds. Of course in the general case we replace q by (ϕ).

where I, J run over all coordinates. The Hamiltonian is given by

$$H = \int_0^\pi d\sigma T_{00} \tag{365}$$

and in writing $d\phi/d\tau$ as the canonical dual $\delta/\delta\phi$ we can express the Hamiltonian as

$$H = \frac{1}{2}\int_0^\pi \left(-\frac{1}{t}g^{IJ}\frac{\delta^2}{\delta\phi^I(\sigma)\delta\phi^J(\sigma)} + tg_{IJ}\frac{d\phi^I}{d\sigma}\frac{d\phi^J}{d\sigma}\right) + (fermions) \tag{366}$$

so in the limit $t \to \infty$ we require

$$\frac{d\phi^I}{d\sigma} = 0. \tag{367}$$

Therefore the embedding map $\phi : \Sigma \to T^*M$ is a constant map and as we reasoned in the B-model such maps, up to equivalences, determine a single point in M. The same analysis done for the fermions shows that these terms contribute to the Hamiltonian at order 1. At weak coupling we therefore determine that the functional $\mathcal{A}$ is described by the zero modes of the field content. This can be written as

$$\mathcal{A} = A^{(0)}(q) + \sum_{i=1}^{3}\chi^{k_1}\cdots\chi^{k_i}A^{(p)}_{k_1\cdots k_i} \tag{368}$$

and by the identifications made in our analysis of the A model this expression can be interpreted as a p-form on M. Since we are wrapping N D-branes onto M and we know that A-branes admit flat vector bundles $E \to M$ these forms take value in $H^\bullet(M, End(E))$ and are in the adjoint representation of the group action $U(N)$ on E. We now make the important observation: we found that $\mathcal{A}$ had ghost number one, therefore it must be expressed as

$$\mathcal{A} = \chi^i A_i(q) \tag{369}$$

and since A_i is a Hermitian matrix $\mathcal{A}$ reduces to a $U(N)$ gauge connection on $E \to M$. We conclude in the limit $t \to \infty$ open string field theory reduces as follows

$$\mathcal{A} \xrightarrow{t\to\infty} A, \qquad Q_A \xrightarrow{t\to\infty} d \tag{370}$$

$$\star \xrightarrow{t\to\infty} \wedge, \qquad \int \xrightarrow{t\to\infty} \int_M \otimes Tr \tag{371}$$

which means we can now write the action as

$$S = \frac{1}{2\lambda}\int_M Tr\left(A\wedge dA + \frac{2}{3}A\wedge A\wedge A\right) \tag{372}$$

and if we compare this to Chern-Simons theory with level k we have the following relationship with the string coupling

$$\lambda = \frac{2\pi}{k+N}. \tag{373}$$

What this tells us is that the topological string theory reduces to a point-like quantum field theory which is described by a *finite* number of fields. Furthermore since the manifold is symplectic there are no instanton corrections since

$$\int_\Sigma \phi^*(p_\mu \wedge q^\mu) = \int_{\partial\Sigma} \phi^*(p_\mu dq^\mu) = \int_{\phi(\partial\Sigma)} p_\mu dq^\mu = 0 \tag{374}$$

where in the last equality we have used the fact that the boundary (branes) $\partial\Sigma$ are mapped to the base M where p_μ vanish. The analysis done here can also be done on a general Calabi-Yau target space which can be found in [18].

The analysis done here was for any three manifold M but we will be interested in the case in which $M = S^3$ which we know describes the deformed conifold T^*S^3 and is therefore Calabi-Yau. With what has been done and said for general Chern-Simons gauge theories and their use in calculating knot invariants embedded in S^3 the natural next step in our analysis is to consider such cases.

10.1.2 B-model on general Calabi-Yau

The exact same analysis can be done to determine what the string field theory reduces to for the B-model. Here however let us assume that M is a general Calabi-Yau target space. We again find the condition $\phi^I/d\sigma = 0$ is required in the limit $t \to \infty$ and so ϕ is again a constant map. The string functional $\mathcal{A}$ becomes dependent on only the zero modes q^i and $\eta^{\bar{i}}$ and is given by

$$\mathcal{A} = A^{(0)}(q) + \sum_{p=1} \eta^{\bar{k}_1} \cdots \eta^{\bar{k}_p} A^{(p)}_{\bar{k}_1 \cdots \bar{k}_p}(q). \tag{375}$$

By our identifications this is a $(0,p)$-form on M and since we are wrapping N B-branes onto M these forms take value in $End(E)$ for a holomorphic vector bundle $E \to M$ in some irreducible representation $\mathcal{R}$ of G. Since we know it must reduce to a field with ghost number 1 we conclude the string functional must be a $(0,1)$ form which is valued in the endomorphism bundle $End(E)$

$$\mathcal{A} = \eta^{\bar{k}} A_{\bar{k}}(q) \in \Omega^{0,1}(M, End(E)) \cong \Omega^{0,1}(M) \otimes End(E). \tag{376}$$

Since we have made the identification $\mathcal{A} \to A$ we note that the Chern-Simons action will not be a top form on M. This too was a problem which was encountered on defining the Yukawa coupling

for the B-model and in spirit we make the following identifications

$$\mathcal{A} \overset{t\to\infty}{\longrightarrow} A, \qquad Q_B \overset{t\to\infty}{\longrightarrow} \overline{\partial} \tag{377}$$

$$\star \overset{t\to\infty}{\longrightarrow} \wedge, \qquad \int \overset{t\to\infty}{\longrightarrow} \int_M \Omega \wedge Tr_{\mathcal{R}} \tag{378}$$

where the integral of the string field theory now includes the holomorphic $(3,0)$ form. The string field action reduces to the holomorphic Chern-Simons action

$$S = \frac{1}{2\lambda}\int_M \Omega \wedge Tr_{\mathcal{R}}\left(A \wedge \overline{\partial} A + \frac{2}{3} A \wedge A \wedge A\right) \tag{379}$$

10.2 A-model, knots & Geometric transitions

Following [18] and [21] let us now consider the case in which we embed a knot $\mathcal{K}$ into the base S^3 of the deformed conifold. The introduction of knots as a one-dimensional cycle does not currently allow us to wrap branes around it as we require the submanifold of the target space to be Lagrangian in the case of the A-model. To resolve this issue we construct the conormal bundle $C_{\mathcal{K}}$ to the knot in S^3 which satisfies $\mathcal{K} = S^3 \cap C_{\mathcal{K}}$. This bundle is the sub-bundle of T^*S^3 defined over $\mathcal{K}$ as

$$C_{\mathcal{K}} = \left\{ (q(s), p) \in T^*S^3 \;\middle|\; \sum_i^3 p_i \dot{q}_i = 0,\ s \in [0, 2\pi] \right\} \tag{380}$$

where $\dot{q}_i := dq_i/ds$. Notice that the fibers of this bundle are *two*-dimensional subspace of $T^*_{q(s)}S^3$ so the co-normal bundle is topologically $S^1 \times \mathbb{R}^2$. Further, we can now wrap a different set of branes on $C_{\mathcal{K}}$. In introducing a new set of branes we have really introdcued two new sets of open strings.

If we wrap N D-branes onto S^3, M D-branes on $C_{\mathcal{K}}$ then the following possiblities are possible

(i) The ends of an open string end on two of the N D-branes giving rise to a $U(N)$ Chern-Simons gauge theory on S^3,

(ii) The ends of an open string end on two of the M D-branes giving rise to a $U(M)$ Chern-Simons gauge theory on $C_{\mathcal{K}}$,

(iii) One end of the open string ends on one of the N branes and the other on one of the M branes.

The third is of course the new case which we must consider. The analysis is similar to that which was done above for the A-model on the Deformed conifold since $C_{\mathcal{K}}$ is a sub-bundle of T^*S^3. Following this line of reasoning we must again impose the condition $d\phi^i/d\sigma = 0$ in the limit $t \to \infty$ meaning ϕ^i is constant on T^*S^3. However, because the string ends on two different sub-manifolds it must lie on their intersection $\mathcal{K}$. The Chern-Simons open string functional localize to the zero modes of the

coordinates q which parametrize $\mathcal{K}$ and the zero modes of the Grassmann odd variable χ. This is written as

$$\mathcal{A} = \varphi(q) + \chi\vartheta(q) \tag{381}$$

where φ is a complex scalar in the bi-fundamental representation of $SU(N) \otimes \overline{SU(M)}$ on $\mathcal{K}$ which we denote $\mathcal{R} = (N, \overline{M})$ and $\vartheta(q)$ some scalar function. The previous identification $Q_A \to d$ and $\chi \to d\phi$ still hold but since we are considering integration over the knot $\mathcal{K}$ integration reduces to

$$\int \to \oint_{\mathcal{K}} \tag{382}$$

The highest order differential form one can consider on $\mathcal{K}$ is a one form and so the Chern-Simons open field string action reduces to an integral

$$\oint_{\mathcal{K}} (\overline{\varphi} d\varphi + \varphi d\overline{\varphi}) = \oint_{\mathcal{K}} ds \frac{d}{ds}(\varphi\overline{\varphi}). \tag{383}$$

The scalar fields will couple to the gauge connection on the bundles that the D-branes admit. Let A be the $U(N)$ gauge connection on S^3 and $\tilde{A}$ the $U(M)$ connection on $C_{\mathcal{K}}$ then the coupling is given by

$$\oint_{\mathcal{K}} Tr_{\mathcal{R}} \left(\overline{\varphi} A \varphi - \varphi \tilde{A} \overline{\varphi} \right). \tag{384}$$

Since both of these expressions are Gaussian we can integrate them out of the partition function to obtain

$$Z_{CS} = exp\left[-log\ det\left(\frac{d}{ds} + (A_\mu - \tilde{A}_\mu)\frac{dq^\mu}{ds} \right) \right] \tag{385}$$

where we have expanded each connection as $A_\mu(dq^\mu/ds)ds$. Define the holonomies

$$U := \mathcal{P}exp\left(\oint_{\mathcal{K}} A \right) \in SU(N), \quad V := \mathcal{P}exp\left(\oint_{\mathcal{K}} \overline{A} \right) \in SU(M) \tag{386}$$

up to an irrelivant constant the partition function evaluates to [21]

$$Z(U, V) = exp\left(\sum_{n=1}^{\infty} \frac{1}{n} Tr(U^n) Tr(V^{-n}) \right). \tag{387}$$

We can obtain an effective theory of D-branes on $\mathcal{K}$ by integrating out the degrees of freedom coming from the D-branes on S^3. To do this we define the following generating function of Chern-

Simons observables (Wilson loops) on S^3 associated to $\mathcal{K}$ by

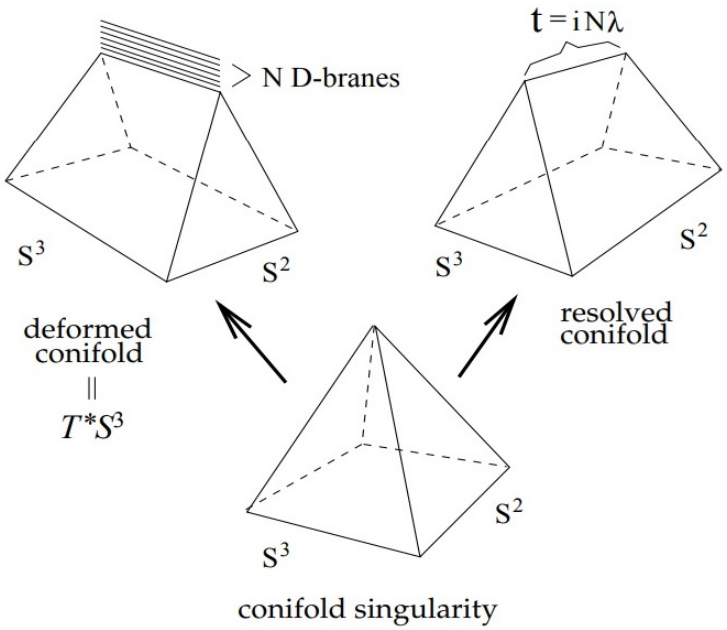
$$exp(-F(t,V)) := \frac{1}{Z_{CS}(A;S^3)} \int \mathcal{D}A exp\left(-S_{CS}(A,S^3) + \sum_{n=1}^{\infty} \frac{1}{n} Tr(U^n) Tr(V^{-n})\right). \tag{388}$$

here t is the 't Hooft coupling $t = iN\lambda$. The theory on $C_{\mathcal{K}}$ is now given by the effective action

$$S = S_{CS}(A;C_{\mathcal{K}}) + \sum_{n=1}^{\infty} \frac{1}{n} Tr(U^n) Tr(V^{-n}). \tag{389}$$

Now that we have an explicit description for the open string theory on T^*S^3 we would now like to understand the open-closed string duality by shrinking S^3 and 'blowing up' the S^2 fibration through the conifold singularity. This transition is depicted in Figure 1. below and is known as a geometrical transition. This 'gauge-gravity' conjecture was first proven in [11] by Gopakumar and Vafa and expanded upon in [22]. Let us briefly explain the closed string theory we obtain on a background geometry[28] of $\mathcal{O}(-1) \oplus \mathcal{O}(-1) \to \mathbb{CP}^1$. In this transition the N branes vanish and we are left with M branes which continue to wrap $C_{\mathcal{K}}$ as a sub Lagrangian manifold of resolved conifold.

[28] Recall $\mathbb{CP}^1 \cong S^2$.



A local representation of the geometric transition between the deformed and resolved conifold. Taken from [21].

The structure of this sub-bundle is not known for a general knot embedding of S^3. However certain cases have been done explicitly [18]. The gauge-gravity duality symbolically reads

$$F(t, V) = F_{CS}(V) \tag{390}$$

Where the left hand side is the free energy of the open topological string theory on the sub-bundle $C_{\mathcal{K}}$ of the resolved conifold while the right hand side is the free energy of the Chern-Simons gauge theory constructed above. This has been explicitly shown for the unknot in [21] with the topological string sector computation coming from considering the spectrum of $M2$ branes ending on $M5$ branes in M-theory.

11 Discussion

We have shown how the Chern-Simons open string field theory on the deformed conifold is dual to the topological A string on the resolved conifold. Further we showed how we can obtain knots invariants by considering a bundles over a knot embedded in the target space which we wrap branes on. It turns out that this construction can also be done form more general toric manifolds which contain T^*S^3. The reason one considers more general cases is due to the fact that in our analysis we did not learn too much about the gravity theory on the resolved conifold. More complicated geometries will give rise to more complex gravitational theories.
There are many different aspects of the topological string which we did not mention in this thesis

and we shall now name a few. One of the biggest aspect left out is the discussion on matrix models which play a crucial role in open-closed string dualities. In the appendix we give the very brief discussion on matrix models which is sufficient to understand literature which discusses it. Another aspect which was not discussed was the notion of the topological vertex. This is an approach to constructing dulaities by 'gluing' together local Calabi-Yau manifolds. Also much of our analysis was focused on the A-model and of course there is a lot to be learned from considering the B-model in greater depth. In particular an important aspect one must consider in the topological B-model string theory is the holomorphic anomaly. Finally, the topological string is related to $\mathcal{N} = 2$ supersymmetric gauge theories and the Seiberg-Witten prepotential which is used to calculate topological invariants. There is also an interesting relationship between BPS states of black holes and the topological string. Needless to say the topological string is an extremely useful tool for understanding the deeper meaning behind string theory.

Part IV

Appendix:

11.1 Sheafs

Because of their use in string geometry we give give the basics of sheafs. These are classes of objects that are defined over the manifold which give us information about the structure of the manifold. For instance we will be interested in the set of zeros of holomorphic sections of a complex bundle which defines a sheaf of functions on M (which we define below). These turn out to relate to topological invariants of the bundle.

Definition 20. *Let M be a topological space. A presheaf $\mathcal{F}$ consists of a group $\mathcal{F}(U)$ for all open subsets $U \subset M$ and for every inclusion $V \subseteq U$ of open subsets a restriction homomorphism $r_{VU} : \mathcal{F}(U) \to \mathcal{F}(V)$. We require $\mathcal{F}(\emptyset) = 0$, $r_{UU} = id_U$ and if $W \subseteq V \subseteq U$ then $r_{UW} = r_{VW} \circ r_{UV}$.*

Elements of $\mathcal{F}(U)$ are referred to as sections of $\mathcal{F}$ over U. Intuitively we can think of $\mathcal{F}(U)$ as some class of functions on U. For $s \in \mathcal{F}(U)$ we define $s|_V := r_{VU}(s)$.

Definition 21. *A presheaf $\mathcal{F}$ is a sheaf if*

(i) For every $s \in \mathcal{F}(U)$, if $U = \cup_i U_i$ is an open cover and $s|_{U_i} = 0$ for all i, then $s = 0$.
(ii) If $U = \cup U_i$ is an open cover and $s_i \in \mathcal{F}(U_i)$ is such that $s_i|_{U_i \cap U_j} = s_i|_{U_i \cap U_j}$ for all i, j, then there is a section $s \in \mathcal{F}(U)$ such that $S|_{U_i} = s_i$ for all i.

If M is a complex manifold and $E \to M$ a holomorphic vector bundle, then define the following sheafs on it

- $\mathscr{O}(U)$: holomorphic function on U
- $\mathscr{O}(U)$: nowhere vanishing holomorphic function on U
- $\Omega^p(U)$: holomorphic p-forms on U
- $\mathscr{O}(E)(U)$: holomorphic sections of E over U

A sequence $\cdots A_1 \xrightarrow{\alpha_1} A_2 \xrightarrow{\alpha_2} A_3 \xrightarrow{\alpha_3} \cdots$ of spaces and maps satisfying $Im(\alpha_i) = Ker(\alpha_{i+1})$ at each level of i is called an exact sequence. If $0 \to A \xrightarrow{\alpha} B$ is exact then α is injective and if $B \xrightarrow{\beta} C \to 0$ is exact then β is surjective. The sequence $0 \to A \xrightarrow{\alpha} B \xrightarrow{\beta} C \to 0$ is called a short exact sequence. Note this means $A \subseteq B$ and $C = B/A$. A map $\alpha : \mathcal{F} \to \mathcal{G}$ between sheaves on M is called a morphism of sheaves if for $\alpha_U : \mathcal{F}(U) \to \mathcal{G}(U)$ for all open subsets $U \subset V \subset M$ such that

$r_{VU} \circ \alpha_U = \alpha_V \circ \tilde{r}_{VU}$ where r is the restriction map on $\mathcal{F}$ and $\tilde{r}$ is the restriction map on $\mathcal{G}$. The sequence $0 \to \mathcal{F} \xrightarrow{\alpha} \mathcal{G} \xrightarrow{\beta} \mathcal{H} \to 0$ is exact if for all $U \subset M$, and $0 \to \mathcal{F}(U) \xrightarrow{\alpha_U} \mathcal{G}(U) \xrightarrow{\beta_U} \mathcal{H}(U) \to 0$ is exact if for a section $s \in \mathcal{H}(U)$ and $p \in U$ there exists a neighborhood of $p \in V \subset u$ and a section $t \in G(V)$ such that $\beta_V(t) = s|_V$. The complex of sheaves $\cdots \to \mathcal{F}_n \xrightarrow{\alpha_n} \mathcal{F}_{n+1} \xrightarrow{\alpha_{n+1}} \cdots$ is exact if $\alpha_{n+1} \circ \alpha_n = 0$ and $0 \to ker(\alpha_n) \to \mathcal{F}_n \to ker(\alpha_{n+1}) \to 0$ is a short exact sequence for all n. Although we will not define them here we denote the Cech cohomology classes by u

12 Gauge theories as Matrix Models

It turns out to be extremely useful to look at 'zero-dimensional' quantum gauge theories and the perturbative expansions of their partition functions. These models can be used to find further open-closed string dulaities which were not mentioned in the main text. With this in mind we now introduce some basics of matrix models. Here we follow [18] closely.

Let $M \in GL(\mathbb{C}, N)$ be a Hermitean matrix and define a potential term $W : GL(\mathbb{C}, N) \to \mathbb{P}(M)$ as an adjoint invariant polynomial

$$W(M) = \frac{1}{2} Tr(M^2) + \sum_{p \geq 3} \frac{g_p}{p} Tr(M^3) \tag{391}$$

where the trace is taken over the defining representation of M and g_p is a coupling constant. The name comes from the fact that under the action of the adjoint representation of the group $U(N)$ we find $M \mapsto UMU^\dagger$ we find $W(UMU^\dagger) = W(M)$ for $U \in U(N)$. When we take a path integral over all possible M we will be overcounting and must mod out these terms. This is of course done by diving the path integral by $vol(U(N))$. We then have the gauged matrix model partition function

$$Z_0(M) := \frac{1}{Vol(U(N))} \int dM e^{-\frac{1}{\lambda} W(M)} \tag{392}$$

for some coupling constant λ. The measure for this integral is given by the Haar measure

$$dM = \prod_{i=1}^{N} dM_{ii} \prod_{1 < i < j \leq N} d\mathcal{R}e(M_{ij}) \mathcal{I}m(M_{ij}). \tag{393}$$

Using a generalization of the isomorphism between $SU(2) \cong S^3$ the volume of the unitary group $U(N)$ is given by

$$Vol(U(N)) = \frac{(2\pi)^{N(N+1)/2}}{G_2(N+1)} \tag{394}$$

where $G_2(z)$ is the Barnes function which generalizes the gamma function. The analytically continued gamma function satisfies $\Gamma(z+1) = z\Gamma(z)$, $\Gamma(1) = 1$ and similarly the Barnes function is defined recursively through $G_2(z+1) = \Gamma(z) G_2(z)$, $G_2(1) = 1$. Therefore we can express the volume in more familiar terms

$$Vol(U(N)) = \frac{(2\pi)^{N(N+1)/2}}{\Gamma(N)\Gamma(N-1)\cdots\Gamma(1)} \tag{395}$$

which is the product of the volume of unit N spheres. The correlation function is defined as one would expect

$$\langle f(x) \rangle = \frac{\int dM f(x) e^{-\frac{1}{\lambda} W(M)}}{\int dM e^{-\frac{1}{\lambda} W(M)}}. \tag{396}$$

The utility of matrix models comes from interpreting them as relating the gauge theory that the

$U(N)$ group underpins. To get a better idea of how to work with the theory we now give the simplest possible example.

A preliminary example

Consider the case $N = 1$. In particular this means $M = x \in \mathbb{C}$, $a := 1/\lambda$ and $\mathrm{W(M)} = 1_{\frac{}{2Tr(x^2)=\frac{1}{2}x^2}}$. The partition function is a simple Gaussian integral

$$Z_0 := \int_{-\infty}^{\infty} dx e^{-\frac{a}{2}x^2} = \sqrt{\frac{2\pi}{a}}. \tag{397}$$

Consider the correlator of an even degree polynomial[29] $\langle x^{2k} \rangle$ for $k \in \mathbb{Z}$

$$\langle x^{2k} \rangle = \frac{1}{Z_0} \int_{-\infty}^{\infty} dx x^{2k} e^{-\frac{a}{2}x^2} = \frac{1}{Z_0} \int_{-\infty}^{\infty} \left(-2\frac{\partial}{\partial a}\right)^k e^{-\frac{a}{2}x^2} = \frac{1}{Z_0} \left(-2\frac{\partial}{\partial a}\right)^k Z_0 = \frac{(2k-1)!!}{a^k}$$

here $(2k-1)!! = (2k-1)(2k-3)\cdots$. Using this expression we can construct higher order polynomial terms in the integral density of the partition function. For instance we may consider adding a 'potential' term $\frac{g}{4!}x^4$ for some coupling g

$$Z(x,g) := \int_{-\infty}^{\infty} dx e^{-\frac{a}{2}x^2 - \frac{g}{4!}x^4} = \int_{-\infty}^{\infty} dx e^{-\frac{a}{2}x^2} \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{-g}{4!}\right)^n x^{4n} \tag{398}$$

in general we cannot swap the integral and summation because the sum is not convergent. Introducing a asymptotic limit we can do so without worry. (For a quick review of asymptotic expansions and Gaussian integrals see [2]). Thus we can consider standard perturbation theory and we obtain

$$Z(x,g) = \sum_{n=0}^{\infty} \left[\frac{(4n-1)!!}{n!(4!)^n}\right] (-g\lambda)^n Z_0 \tag{399}$$

This generates all standard Feynmann diagrams associated to the standard 'ϕ^4' theory. In fact the numerical prefactor in the summation tells us the following information

$$\frac{(4n-1)!!}{n!(4!)^n} = (\text{Number of Feynmann Diagrams with n-verticies}). \tag{400}$$

We see that this zero dimensional matrix model mimics the analysis of a scalar quantum field theory. Of course to get rid of the redundancies associated to counting disconnected diagrams we introduce

[29] Odd degrees vanish. This can be though of as the zero dimensional case of normal ordering.

the free energy formally as

$$F = log(Z) \tag{401}$$

which contains only connected diagrams. Going back to the general case we let M_{ij} be in the adjoint rep $N \otimes \overline{N}$ of $U(N)$ and consider only the 'kinetic' square part of the potential $W(M)$: $\langle M_{ij} M_{kl} \rangle = \lambda \delta_{il} \delta_{kj}$ which we express graphically as double line propagators. These are known as a fat or ribbon graph and are the propagators of the matrix model. This double line notation was introduced by 't Hooft. Note $\langle M_{ij} \rangle = 0$ as one would expect. Vacuum diagrams are obtained by tracing over the expression for the propagator

$$Tr(M^2)\rangle = \sum_{i,j}^{N} \langle M_{ij} M_{ji} \rangle = \sum_{ij}^{N} \lambda \delta_{ij} \delta_{ji} = \lambda N^2 \tag{402}$$

the interpretation of taking the trace amounts to connecting the ends of the double line propagator. An important expansion is that of $\langle Tr(M^3)^2 \rangle$ which generates Feynmann diagrams associated to cubic verticies. We find [18]

$$\langle Tr(M^3)^2 \rangle = \lambda^3 (12N^3 + 3N) \tag{403}$$

which has contributions from three topologically distinct ribbon graphs. Each ribbon graph is described by the following topological information

- E = Number of edges of a fat graph (propogators)
- V_p = Number of verticies with p legs
- h = Number of closed loops
- $V = \sum_p V_p$ (Total number of Verticies)

and we find that each graph contributes a factor of

$$\lambda^{E-V} N^h \prod_p g_p^{V_p}. \tag{404}$$

We now make an important identification: recall that the Euler charactersitic χ of a Riemann surface Σ of genus g (once it has been triangulated [19]) is given by

$$\chi(\Sigma) = 2g - 2 = E - V - h \tag{405}$$

so now can now think of each topologically distinct ribbon graph as defining a Rieman surface of genus g which has h holes (boundaries). The contributing topological factor $\lambda^{E-V} N^h \prod_p g_p^{V_p}$. can be

written in a more suggestive way

$$\lambda^{2g-2} t^h \prod_p g_p^{V_p}, \qquad t := N\lambda \tag{406}$$

where t is called the 't Hooft parameter which will play an important role in our analysis. All the information just presented is encoded in the free energy expansion

$$F = \sum_{g=0}^{\infty} \sum_{h=1}^{\infty} F_{g,h} \lambda^{2g-2} t^h = \sum_{g=0}^{\infty} F_g(t) \lambda^{2g-2} \tag{407}$$

$$F_g(t) := \sum_{h=1}^{\infty} F_{g,h} t^h \tag{408}$$

where in the second line we have introduced the free energy at genus g factor F_g. This expression is exactly as the one found for perturbative expansion of the topological string. By condiering matrix model limits of certain theories one can find dualities between them and the topological string theories discussed. See [18].

13 Sigma model Anomalies

We present the proof for the difference in zero modes of the fermionic fields which lead to an axial anomoaly. The utility of our presentation is that we do not require the use of index theorems. Although the index theorem is a beautiful branch of mathematics showing an intamite relationship between analaysis and gemoetry it would require a long digression. For an introduction to index theorems see [19] chapter 12.

$N = (2,2)$ **Sigma Model**

Since $\psi_+ \in \Gamma(K^{1/2} \otimes \phi^* T^{1,0}M)$ then for this bundle the Riemann-Roch theorem reads

$$\sum_{i=0}^{m} dim[H^i(K^{1/2} \otimes \phi^*(T^{1,0}M))] = \int_\Sigma ch(K^{1/2} \otimes \phi^*(T^{1,0}M)) td(T\Sigma) \tag{409}$$

where $H^i(K^{1/2} \otimes \phi^*(T^{1,0}M))$ is the i-th sheaf cohomology. Given some bundle E Serre duality states $H^i(E) = H^{m-i}(K \otimes \overline{E})^*$ where $*$ indicates the dual vector space. By invoking Serre duality we find $H^1(K^{1/2} \otimes \phi^*(T^{1,0}M)) = H^0(K^{1/2} \otimes \phi^*(T^{0,1}M))^*$ and since the (finite) dimension of the dual of the vector space is equal to the dimension of the vector space itself we conclude the value of $dim(H^1)$ is simply ℓ_-. This means that the difference in zero modes is given by

$$\ell_+ - \ell_- = dim(ker(D_{\bar{z}})) - dim(ker(D_z)) = \int_\Sigma ch(K^{1/2} \otimes \phi^*(T^{1,0}M)) td(\Sigma). \tag{410}$$

Invoking the splitting principle for Chern characters the left hand side can be written, up to volume forms on Σ, as

$$\begin{aligned}
\ell_+ - \ell_- &= \int_\Sigma ch(K^{1/2} \otimes \phi^*(T^{1,0}M)) td(T\Sigma) = \int_\Sigma \sqrt{ch(K)} ch(\phi^*(T^{1,0}M)) td(T\Sigma) \\
&= \int_\Sigma \left(1 - \frac{1}{2} c_1(T^{1,0}\Sigma)\right)\left(m + \phi^* c_1(T^{1,0}M\right) td(T^{1,0}\Sigma) \\
&= \int_\Sigma \left(1 - \frac{1}{2} c_1(T^{1,0}\Sigma)\right)\left(m + \phi^* c_1(T^{1,0}M) td(T^{1,0}\Sigma)\right) \\
&= \int_\Sigma \left(1 + \phi^* c_1(T^{1,0}M) - \frac{m}{2} c_1(T^{1,0}\Sigma)\right)\left(1 + \frac{1}{2} c_1(T^{1,0}\Sigma)\right) \\
&= \int_\Sigma \phi^* c_1(T^{1,0}M) \\
&= \int_{\phi(\Sigma)} c_1(T^{1,0}M) \quad \Leftrightarrow \quad c_1(T^{1,0}M) \stackrel{!}{=} 0
\end{aligned} \tag{411}$$

A-model zero-modes

In a similar manner let ℓ_χ and ℓ_ψ be the zero modes of χ and ψ respectively. Then if we invoke both Serre duality on $H^1(\phi^* T^{1,0}(M))$ and the Riemann-Roch theorem we find

$$2k := \ell_\chi - \ell_\psi = 2\int_\Sigma (m + \phi^* c_1(T^{1,0}M))(1 + \frac{1}{2}c_1) \tag{412}$$

$$= 2\int_\Sigma \phi^* c_1(TM) + 2\int_\Sigma \frac{m}{2} c_1(T\Sigma) \tag{413}$$

$$= 2\int_\Sigma \phi^* c_1(TM) + 2m(1-g) \tag{414}$$

where $\int_\Sigma c_1(T\Sigma) = 2 - 2g$ was used.

B-model zero-modes

By the exact same analysis of the A-model we find

$$(\ell_\eta + \ell_\theta) - \ell_\psi = 2\int_\Sigma \phi^* c_1(TM) + 2m(1-g). \tag{415}$$

14 Categorification of Topological Quantum Field Theories:

A beautiful formulation of TQFT's has been formulated by Atiyah in terms of category theory. Here we state only the basics of this formulation in order to connect the ideas that we develop in the main text. The utility of this formalism is that when we analyze string amplitudes as sums over topologies of the underlying manifold. In principle it is entirely possible to analyze Topological String theory without this formalism but in an effort to understand every aspect of TQFTs I have added this section. The reader interested in a mathematically rigoruous exposition see [**?**] [**?**]. Here we follow [**?**]

Definition 22. *A category $\mathcal{K}$ consists of the following information*

1. *A class of objects $Ob(\mathcal{K})$ of C*
2. *A set of morphisms $Mor(A,B) = \{f : A \to B\}$ for every pair $A, B \in Ob(\mathcal{K})$*
3. *A binary operator which defines composition $Mor(A,B) \times Mor(B,C) \to Mor(A,C)$ satisfying*
   (a) *(associativity) $f \circ (g \circ h) = (f \circ g) \circ h$ for $f \in Mor(C,D)$, $g \in Mor(B,C)$, $h \in Mor(A,B)$*
   (b) *(neutral elements) For every $A \in Ob(\mathcal{K})$ $\exists!\ 1_A \in Mor(A,A)$ such that $1_A \circ f = f$, $g \circ 1_A = g$ for all $f \in Mor(B,A), g \in Mor(A,B)$*

Let $\mathcal{K}_1$ and $\mathcal{K}_2$ be two cateogries. If there exists a map $F : \mathcal{K}_1 \to \mathcal{K}_2$ such that $F(X) \in Ob(\mathcal{K}_2)$ for every $X \in Ob(\mathcal{K}_1)$ and for every morphism $f \in Mor(X,Y)$ of $\mathcal{K}_1$ we have the corresponding morphism $F(f) \in Mor(F(X), F(Y))$ of $\mathcal{K}_2$ then F is said to be a functor.[30]

In physics we often work with linear categories which correspond to categories with linear morphisms. The most familiar is the category **Vect** of vector spaces where Ob(**Vect**) are vector spaces and Mor is the space of linear maps between vector spaces. The category of d-dimensional manifolds **Man**(d) has smooth, compact, oriented manifolds (X) as its objects and its morphisms $M : X \to Y$ are (d+1) dimensional smooth, oriented manifolds with two boundaries that are isomorphic to X and Y, each with opposite orientations. Pictorially this can be represented as.

[30] We also note that $F(1_X) = 1_{F(X)}$. If $F(f \circ g) = F(f) \hat{\circ} F(g)$ then F is said to be a covariant functor and if $F(f \circ g) = F(g) \hat{\circ} F(f)$ then F is said to be a contravariant functor.

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